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Reinforcement Learning for Residential Energy Management System Control

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Declaration of Authorship

I, Paulo Ricardo Lissa, declare that the research described in this Thesis is all my own work, which I undertook at the National University of Ireland, Galway during the years 2017 - 2021. This work has not previously been presented for an award in this or another university.

Abstract

Modern solutions for residential energy management systems control are emerging and helping to improve the way energy is consumed, bringing enhanced comfort and savings to end-users through smart and autonomous actions. This has been facilitated by the introduction of new Internet-of-Things devices for home applications and the advances of the communication infrastructure. In addition to that, the increasing necessity of reducing greenhouse emissions has leveraged the use of renewable sources, hence on-site energy generation such as PV is becoming a reality in a number of buildings. However, this creates a more complex environment and extracting the best it can offer is not an easy task for the operator. For instance, users may not know the best time to turn on or off some specific load according to the cost of energy, or sometimes not be at home to get benefits from the PV generation. One of the possible solutions is the utilisation of machine learning-based adaptative methods, such as reinforcement learning, which can learn how to operate the system in a near-optimal manner by directly interacting with the environment. This thesis aims to investigate and demonstrate how reinforcement learning can improve residential energy management systems control. Several simulated experiments are conducted based on real demand response projects, applying multiple reinforcement learning algorithms, such as Q-learning and deep Q-networks, thus proving the efficacy of such techniques in this domain.

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List of Abbreviations

AI	Artificial Intelligence
AMI	Advanced Metering Infrastructure
ANN	Artificial Neural Network
A3C	Asynchronous Advantage Actor-Critic
BEM	Building Energy Model
BS	Building Simulator
CPP	Critical Peak Pricing
DDPG	Deep Deterministic Policy Gradient
DHW	Domestic Hot Water
DLC	Direct-Load Control
DQN	Deep Q-Network
DR	Demand Response
DRL	Deep Reinforcement Learning
EMS	Energy Management System
EU	European Union
EV	Electric Vehicle
HEMS	Home Energy Management System
HP	Heat Pump
HVAC	Heating, Ventilation, and Air Conditioning
IC	Interruptible/Curtailable

ICT	Information and Communications Technology
IESVE	Integrated Environmental Solution Virtual Environment
IoT	Internet-of-Things
MDP	Markov Decision Process
ML	Machine Learning
MLP	Multi-Layer Perceptron
MPC	Model Predictive Control
NN	Neural Network
PV	Photovoltaic
RES	Renewable Energy System
RL	Reinforcement Learning
RTP	Real-Time Pricing
SAC	Soft Actor-Critic
SARSA	State–Action–Reward–State–Action
SVM	Support Vector Machine
TD	Temporal Difference
TF	Transfer Learning
ToC	Time out Comfort
ToU	Time-of-Use

Chapter 1

Introduction

1.1 Motivation

Energy has always been an important topic for human beings, as it impacts every aspect of our daily lives, from work and industry production activities to leisure and well-being matters. A lot has been achieved to improve technologies for generation, transmission and distribution of energy to facilitate the access to this important resource, aiming to increase capacity and avoid power interruptions. However, traditional power plants based on fossil fuels are costly to deploy and increase greenhouse gas production, which is harmful to the environment. The use of renewable energy resources is emerging as a sustainable option, and its application is increasing every year. It is expected to represent 32% of the total energy production in Europe by 2030 [1], affecting directly the energy matrix. Renewable energy systems (RES), such as solar or wind generation, have an intermittent capacity, as their energy production relies on weather conditions. This makes matching demand and production a complex management task.

New Internet-of-Things (IoT) technologies, along with the advances and rollout of communication infrastructure and smart metering for residences, allow for real-time remote verification and control of smart devices. For instance, users can check and manage energy on-site production, energy consumption at the appliance level, and also comfort parameters, such as indoor temperature and CO₂ concentration. This brings new possibilities for energy management, where residential consumers can play an important role in helping to balance the energy grid. As an example, they can change their usual consumption behaviour and take advantage of dynamic tariffs

or periods where PV production is higher, hence reducing energy demand in peak hours. However, deciding when to perform a control action or being available to do so at the best time may be challenging, thus increasing the need for reducing human’s manual intervention, which can be done by leveraging smarter control solutions. Considering this new and modern environment, how to properly operate its components autonomously with the highest possible savings, without affecting the user’s comfort?

A number of solutions for residential energy management have arisen recently to improve the way energy is consumed. Prior to the introduction of adaptive solutions, the most common methods in terms of control were manual and programmable schedule, which allowed the operator to choose the period to turn the heating, ventilation, and air conditioning (HVAC) system or some specific appliance on/off. However, the schedule or manual operation of certain equipment may not be performed considering other elements that could bring savings, such as the PV (photovoltaic) energy produced in the residence or the cost of energy in that time. In this regard, machine learning-based adaptive methods may be a solution to achieve even greater cost reductions whilst maintaining equivalent occupant comfort levels. Reinforcement learning (RL) algorithms, a sub-field of machine learning (ML), can learn the optimal policy by interacting directly with their environment, choosing actions based on their previous experiences within the domain. After large numbers of trials, the algorithm can learn optimal or near-optimal control policies for operating the equipment, with no prior knowledge. That raises the question: How can reinforcement learning algorithms improve residential energy management and control?

1.2 Hypothesis

This research aim is to investigate and demonstrate how reinforcement learning-based control algorithms can improve residential energy management systems, bringing cost savings and enhanced comfort to end-users and improving the smart grid reliability.

1.3 Contributions

The main contributions of this thesis are summarised as follows:

- The development of an RL (Q-learning) algorithm that utilises transfer learning (TF) for HVAC control, allowing for much faster learning rates in response to building spatial changes or different geographic locations. (Chapter 3)
- The development of a deep reinforcement learning (DRL) algorithm for indoor and domestic hot water (DHW) tank temperature control, aiming to reduce energy consumption by optimising the usage of PV energy production. The algorithm also provides load shifting, helping the energy grid balance. (Chapter 4)
- The development of a DRL algorithm with transfer learning capabilities for domestic hot water tank temperature control of a number of houses in a microgrid, aiming to reduce energy consumption from the grid by optimising the usage of PV production, and enhancing financial savings through a time-of-use (ToU) tariff scheme. (Chapter 5)
- A thorough investigation of the impact of user comfort levels and energy savings is conducted, with performance comparison against conventional control methods. (Chapters 3, 4 and 5)

1.4 Publications

The main contents of this thesis have been published in journals and conference proceedings. The list of publications is found below, where the first author is the primary author.

1.4.1 Journal Papers

1. **Lissa, P.**, Schukat, M. and Barrett, E. Transfer Learning Applied to Reinforcement Learning-Based HVAC Control. SN COMPUT. SCI. 1, 127 (2020). <https://doi.org/10.1007/s42979-020-00146-7>.
2. **Lissa, P.**, Deane, C., Schukat, M., Seri, F., Keane, M. and Barrett, E. Deep Reinforcement Learning for Home Energy Management System Control, Energy and AI, Volume 3, 2021, 100043, ISSN 2666-5468. <https://doi.org/10.1016/j.egyai.2020.100043>.

3. **Lissa, P.**, Schukat, M., Keane, M. and Barrett, E. Transfer Learning Applied to DRL-Based Heat Pump Control to Leverage Micro-grid Energy Efficiency. Smart Energy, 2021, 100044, ISSN 2666-9552, <https://doi.org/10.1016/j.segy.2021.100044>.

1.4.2 Conference Proceedings

1. **Lissa, P.**, Seri, F. and Keane, M. Demand Response for Residential Buildings: Case Studies and DR Program Design by the RESPOND Project. In Proceedings of the 4th Annual APEEN Conference 2019 - Energy Demand-Side Management and Electricity Markets.
2. **Lissa, P.**, Peretti, D., Schukat, M., Barrett, E., Seri, F. and Keane, M. Machine Learning Methods Applied to Building Energy Production and Consumption Prediction. In Proceedings of the 27th AIAI Irish Conference on Artificial Intelligence and Cognitive Science. AICS 2019. <http://ceur-ws.org/Vol-2563/>.

1.5 Thesis Overview

This thesis is composed of six chapters and an appendix, as depicted in the diagram in Figure 1.1. Chapters 3 to 5 are the main content of the research, each of them has a different set of experiments considering the algorithms applied and EMS features. Chapters 1, 2, 6 and Appendix are designed to provide a better understanding of the domain, helping to link the objectives with the results achieved in the experiments.

The overall content of each chapter is described as follows:

- Chapter 1: Presents the motivations and main contributions of the thesis.
- Chapter 2: Provides a literature review about the main concepts utilised in the experiments.
- Chapter 3: In this chapter, the first experiment is presented. It is about transfer learning applied to a reinforcement learning-based (Q-learning) HVAC control, considering buildings in different locations and with different dimensions.

- Chapter 4: This chapter presents the second experiment, which is an upgraded version of the previous one. New EMS features were added to the environment (PV generation and dynamic setpoint) and the tabular Q-learning was replaced by a deep Q-network.
- Chapter 5: The third and last experiment is presented in Chapter 5. It explores transfer learning applied to DRL-based heat pump control in a number of houses, aiming to speed up the learning process.
- Chapter 6: This chapter shows the main conclusions linked with the research contributions, presents the limitations and provides ideas for future works.
- Appendix: The appendix comprises two conference proceedings, which complements part of the literature presented in Chapter 2 about demand response and machine learning.

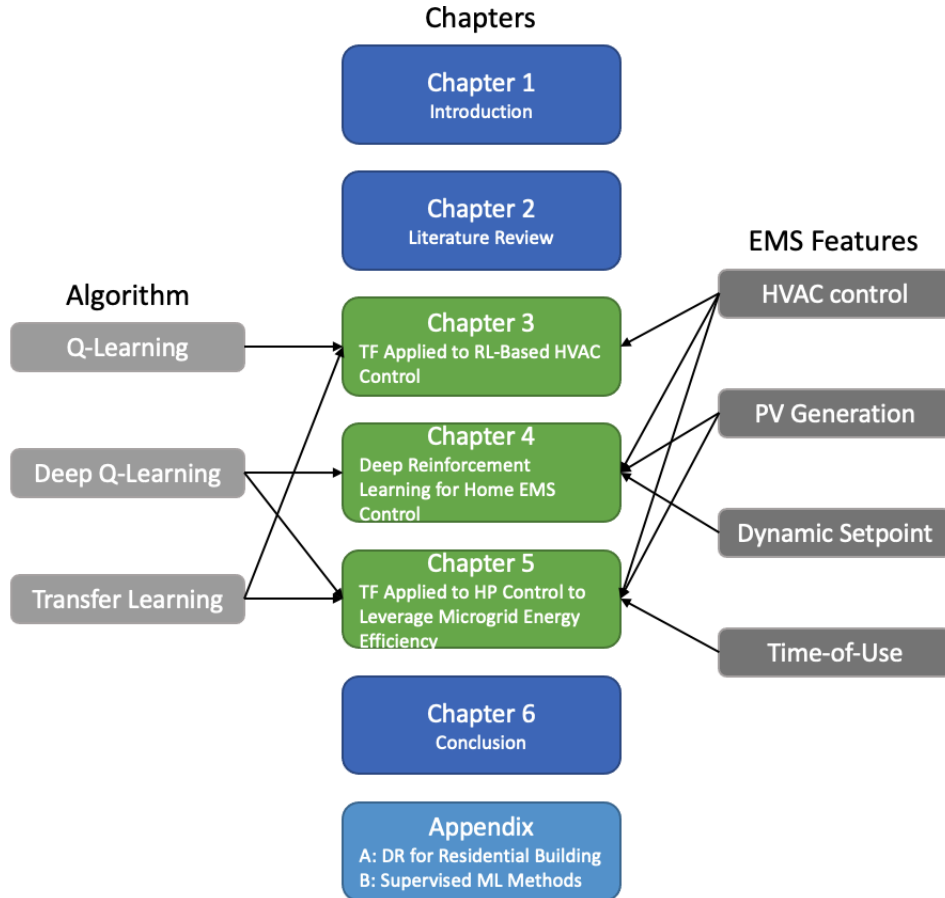


Figure 1.1: Thesis structure.

Chapter 2

Literature Review

This literature review aims to introduce the key concepts utilised in this research, to support the understanding of the experiments in the next chapters. It is divided into two main topics: energy management systems and machine learning. The first explains the components of this domain, showing the main challenges and introducing the principles of smart control and demand response (DR) as a solution. The second topic explores the application of machine learning algorithms to enhance the way the user consumes energy. It shows the different ML approaches and their application in energy and other domains, with an additional focus on reinforcement learning, the object of this research.

The content outlined in this chapter was partly published in the two conference proceedings described in subsection 1.4.2, which are also available in the Appendix section.

2.1 Smart Grid and Energy Management Systems

Smart grid is a very broad concept with differing views on its scope but at its broadest, it can be seen as an evolved electrical infrastructure that uses information and communications technology to transform the way that electrical energy is generated, transported and consumed. It becomes more relevant every year and its application can be seen in areas such as energy distribution and transmission, distributed generation and electric vehicles, all facilitated by an advanced information and communication technology (ICT) infrastructure. The objectives are to increase the efficiency, reliability, security, economy and sustainability of electrical infrastructure. To achieve

that, a smart energy management system (EMS) is necessary to process and control all the smart grid components. In general, an EMS can be described as a computer-based system composed of tools that enables better monitoring and control of the grid energy production and consumption assets, bringing optimisation capabilities for both electric utilities and end-users, which can be applied on a large or small scale, from power plants to individual dwellings. Figure 2.1 shows the general components and application of the smart grid.

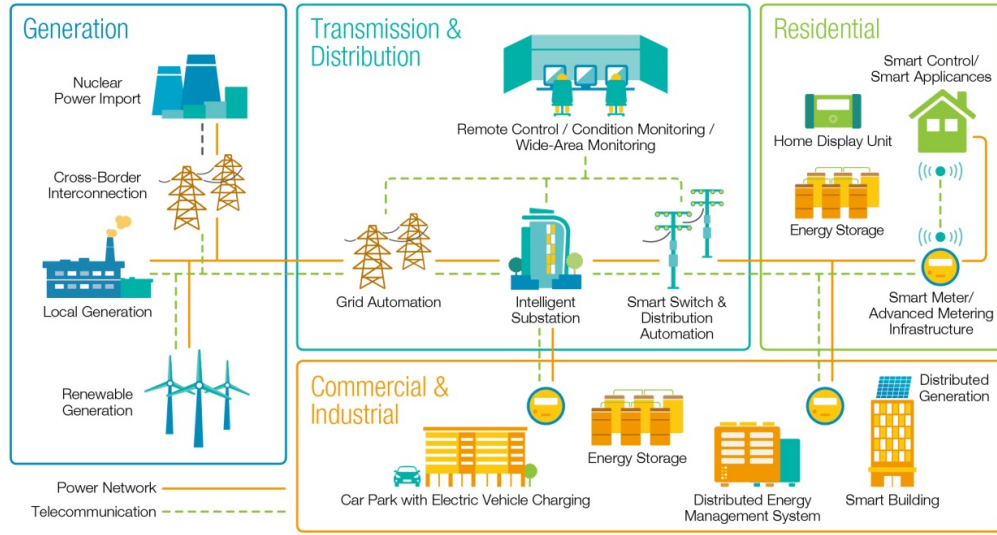


Figure 2.1: Smart grid components. Source: CLP Power [2].

The focus of this research is the residential domain, in which energy consumption accounted for 26.3% of the total energy consumption in the European Union (EU) in 2019, according to the Eurostat report [3]. Furthermore, the report also states that most of the energy consumption by households is used for heating purposes (63.6%) and water heating (14.8%), followed by lighting and general electrical appliances (14.1%). From the energy production perspective, studies from the International Energy Agency [4] show that energy production from renewable is increasing over the years, representing almost 20% of the total generation in the EU, compared with just 9.6% in 2004. Also, it is expected to expand to 32% by 2030 as stated by the European Commission [1], which also includes generation systems owned by end-users.

Renewable sources, such as PV and wind generators, rely on weather conditions, and coordinating demand and production is one of the chal-

allenges arising from this change in the energy matrix. Furthermore, users' consumption patterns contribute directly to sporadic peaks of demand and, to support this variation, utility companies need to increase their generation to avoid interruptions in power supply. However, backup power plants are a costly solution to match demand during peak events, and they also increase greenhouse gas production [5]. An alternative method to match energy demand and production can be found in demand response programs, which aim to stimulate users to use energy in a more distributed manner over the day, thus reducing energy peaks.

2.1.1 Demand Response

Demand response is defined as *“changes in electric usage by end-use customers from their normal consumption patterns in response to changes in the price of electricity over time, or to incentive payments designed to induce lower electricity use at times of high wholesale market prices or when system reliability is jeopardized”* [6]. Demand response programs help both consumers and utilities to reduce peak demand and price volatility. Siano [7], in his survey about demand response and smart grids, identified three different ways in which customers can participate in DR: reducing their energy consumption through load curtailment strategies, moving energy consumption to a different time period, or using on-site standby generated energy, thus limiting their dependence on the main grid. Moreover, the potential benefits of DR include participant and other customers bills savings, grid reliability, market performance, improved choice (customers have options for electricity costs management) and system security, as system operators have the flexibility to meet contingencies.

Research carried out by Sia Partners [8] estimated that at the European level there was a total of 800 TWh of energy consumption for processes with DR potential in 2012, which represented 29% of the total electricity consumption in these countries, similarly distributed across the industrial, the tertiary and the residential sectors. Around 42% of the potential was identified in refrigerators and freezers (12%), heating systems and boilers (10%), space and water heating (10%) and ventilation (10%). After a further analysis considering the installed capacity, the capacity guaranteed, and reduction potential per process, [8] estimated a DR potential in Europe of 52,35 GW, where residential applications corresponded to 42% of this total,

accounting for 22.11 GW. The distribution across the different sectors and subgroups can be found in Figure 2.2.

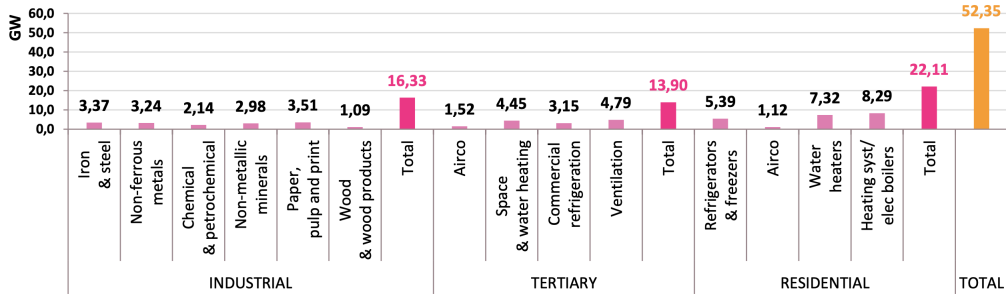


Figure 2.2: Demand response potential. Source: Sia Partners [8].

There are different types of demand response, the US Department of Energy [6] proposed that DR programs can be categorised into two main groups: incentive-based and price-based. Starting with incentive-based programs, end-users get a recompense to reduce their electric loads on request or for giving permissions to regulate their electric loads, as can be seen in RESPOND [9] and REACT [10] EU projects. These kinds of programs were initiated with industrial and large commercial customers, but residential customers have recently been included. There are six different subcategories of incentive-based programs, grouped into classical and market-based. Classical ones are direct-load control (DLC) and interruptible/curtailable services (IC). Demand bidding, emergency DR, capacity market and ancillary service belong to the market-based programs. Moving to price-based demand response, they are programs where the electricity price is not fixed; it varies in time. This category is divided into three major groups: time-of-use rates, critical peak pricing (CPP) and real-time pricing (RTP). The common goal is to reduce electricity consumption over a certain period. An illustrative chart with the DR classification can be seen in Appendix A (Figure A.1). There is also additional information on demand response concepts, including drivers and barriers for its application and an analysis of three case studies countries: Ireland, Spain and Denmark, as part of the RESPOND project [9].

2.1.2 Residential EMS Components

The advancements in metering and control technologies have been important enablers for energy management systems and demand response programs. Smart meters are the key point of connection between the end-users and the utility. They can electronically measure customer energy consumption and other parameters depending on the model and transmit the information in a certain time interval to the utility or the responsible for data processing and management [7]. In 2020, the European Commission [12] stated in their final report about smart metering deployment that the member states committed to rolling out close to 200 million smart meters for electricity by 2020, which represents that almost 72% of European consumers would have a smart meter for electricity. To support such rollout, the whole metering infrastructure is also being upgraded with communication capabilities, now called advanced metering infrastructure (AMI). An AMI comprises a number of integrated technologies and applications, including smart meters, communication networks, meter data management systems, gateways and systems to support data integration and software application. Furthermore, it supports two-way communication between customers and utilities, expanding energy management and control possibilities.

Smart meters provide information regarding the aggregated energy consumption and, sometimes production, of a household. However, if detailed and individual metering is required for a specific purpose, a new set of in-home devices should be deployed. The popularisation and increasing number of Internet-of-Things devices are leveraging behind-the-meter applications, adding even more possibilities of energy management and control and creating the smart home concept. IoT refers to a network of dedicated physical objects (things) that contain embedded technology to communicate and sense or interact with their internal states or the external environment. According to IoT Analytics [13], the number of IoT-connected devices globally reached 11.7 billion in 2020, and it is expected to achieve over 30 billion IoT connections by 2025. In another research, Gartner [14] pointed out that IoT for building automation would represent the segment with the largest growth rate in 2020 (42%), followed by automotive (31%) and health care (29%). There are different devices for smart home applications, such as temperature sensors, thermostats and smart plugs, that can be used as part

of DR control loop and energy management, thus improving user's comfort, reducing energy bills, and improving the network's reliability.

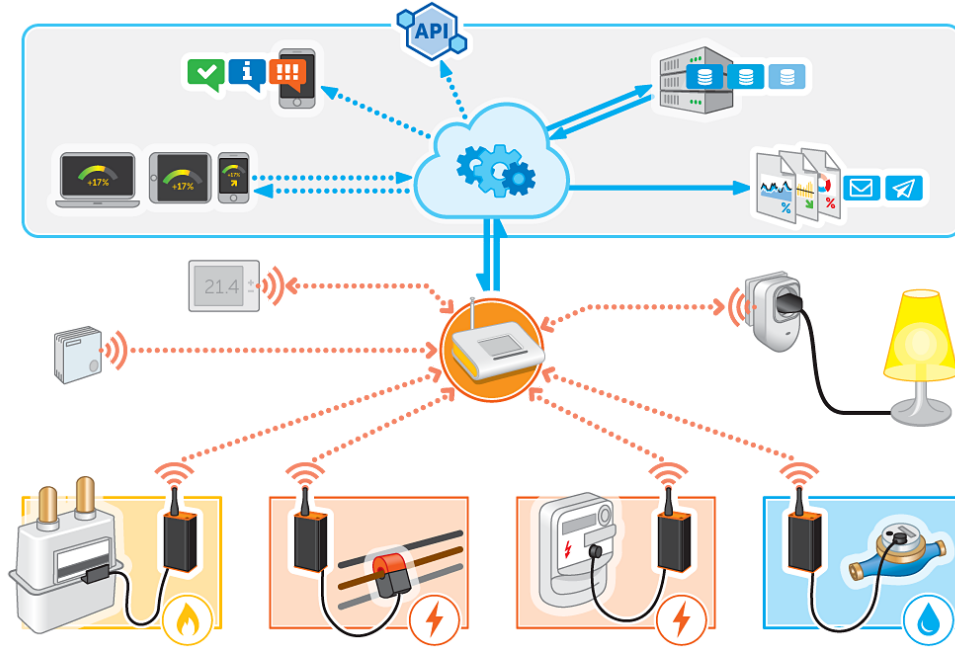


Figure 2.3: Residential EMS architecture. Source: Energomonitor [11].

Figure 2.3 shows an example of smart home architecture. The centre point of the solution is the gateway, which is responsible for collecting the data from other devices and send it to the database. The EMS can read this information, perform analysis and usage optimisation, and send signals back to the house, providing suggestions for turning on or off some appliances or explicit send a control action to some equipment, such as HVAC systems. If this kind of system is available in several houses, the utility or EMS operator can manage the energy consumption of those customers considering the energy peak in that specific grid, offering incentives if they reduce their consumption at a certain time. Proper energy management of microgrids can improve the energy system in isolated communities, such as small islands, by leveraging on-site production and reducing their dependency on the energy that comes from the integrated energy system through submarine transmission cables, which are expensive if an upgrade is needed [10].

Some IoT devices are designed to improve the user's comfort and knowledge about their consumption behaviour. For example, temperature, humidity and CO₂ sensors can provide real-time information about the air quality of the environment. Adding smart thermostats to HVAC systems can en-

able remote and autonomous control, helping to keep comfort standards and reducing the energy bill if some optimisation is applied, for instance, activating the system when the electricity price is lower. Similarly, smart plugs can be connected to appliances such as washing machines, tumble dryers, and dishwashers and the loads can be managed and scheduled according to the user's needs. Bigger loads, such as electric vehicle charging stations, can also have specific relays and sensors and be part of the residential EMS.

Finally, if a house has its own energy generation, such as a PV system, sensors can be installed to gather real-time information about the production. Battery systems can also be part of the solution, storing energy when there is excess in production and being available when necessary. At the same time that new technologies bring benefits for user's and utilities, they also increase the complexity of the grid. Users are not always aware of the technology or sometimes are not available to perform control actions when necessary, which may impact the DR program success. Also, the PV systems peak of production usually happens during the afternoons, and the higher consumption patterns of a specific dwelling may happen in the morning and evening. To exploit the best of the energy management system a number of smart control techniques have emerged, aiming to perform actions and achieve the highest benefits, with the minimum of manual intervention necessary. For instance, the smart control can find a near-optimal operation mode for an HVAC system, considering the PV production and ToU tariffs as inputs, targeting keeping the user's comfort with the lowest possible cost.

2.1.3 Smart Control

To exploit as much as possible the benefits from the new devices and energy management system, a proper control strategy has to be deployed to replace manual or basic rule-based models. Manual actions are not effective because they rely on the user's availability to perform some intervention, such as turning on/off an appliance. Basic rule-based control or scheduled by the user methods, such as setting up temperature setpoints for a building, can act autonomously. Still, they may not perform the actions considering other elements that could bring savings, such as the PV production or energy cost in that time.

Overall, choosing the best time for using an appliance is a load scheduling problem. Leitão et al. [15] classified the methods for load scheduling into

five categories, described as follows:

1. Mathematical optimisation: There are different techniques based on deterministic formulations in this approach: Linear, Non-Linear, Convex and Dynamic programming. The objective function and constraints are expressed in different ways for each technique, such as linear or non-linear relationships, sequential scheduling decisions or geometric optimisation.
2. Heuristic and meta-heuristic methods: They are an alternative to the mathematical formulation, which are computationally expensive for large problems. They follow high-level procedures to search for possible solutions. Genetic algorithms and differential evolution are examples of this technique.
3. Model-based predictive control (MPC): This is a control algorithm based on a predictive model of the process. The model predicts the future output based on historical information. MPC methods [37, 59] are usually costly in terms of modelling, as they require a detailed plant model and need data acquisition.
4. Game theory approaches: They are usually applied in a multi-agent framework, where each agent chooses a strategy to maximise an individual utility function, which considers the operation objectives. There are two main categories, cooperative and non-cooperative. The first one communicates with other agents to find the best solution, while the second focus on self-interest and only communicates if there will be a benefit involved.
5. Machine learning: As an option to the previous techniques, machine learning models are becoming popular, mainly due to the new capabilities derived from the technology advances presented in subsection 2.1.3. For instance, considering that now a huge amount of data started to be available, it can be used to build prediction models using regression models or artificial neural networks (ANN), which will help to define the scheduling process (an example of application of this method can be seen in Appendix B). Some adaptative models are not data-driven, such as reinforcement learning, which can learn the op-

timal policy by interacting directly with their environment, choosing actions based on their previous experiences within the domain.

Machine learning-based models are the key control strategy applied in this research. More detailed information about how it works and the difference of other machine learning models can be found in the next subsections and also depicted in the experiments in Chapters 2, 3 and 4.

2.2 Machine Learning

Machine learning can be defined as programming computer algorithms to optimise a certain task by using example data or past experience. Different from traditional methods, it is commonly used when there is a lack of knowledge about the model or if one may be interested in finding another model that may be more efficient [16]. For instance, machine learning algorithms can identify correlation patterns in big datasets, which may not be possible to be done manually due to the complexity of information. The ML model may not represent the complete task or process, but it will try to approximate as much as possible for the majority of the cases. This is important in terms of computational resources, as estimating values can be faster than a detailed calculation. Nowadays, the application of ML is found in a vast range of fields, such as robotics, speech recognition, stock market, medicine and building control and optimisation, to name a few. Machine learning can be divided into three main groups: supervised learning, unsupervised learning, and reinforcement learning, as presented in Figure 2.4 along with some applications.

Supervised learning has two main streams, classification and regression, where there is a set of inputs and outputs, and the task is to learn the mapping that connects both [16]. An example of regression can be found in Appendix B, in which the machine learning task is to predict energy consumption and production of a building, given a number of inputs, such as weather conditions and timestamps. There, it is also presented different types of regression methods: artificial neural network, decision forest, boosted decision tree, and linear regression. Regarding classification, a common example of application is image classification. The user can use a dataset with images and labels as input training for the machine model that will process the information. Once the model is trained, new images can be used as input and

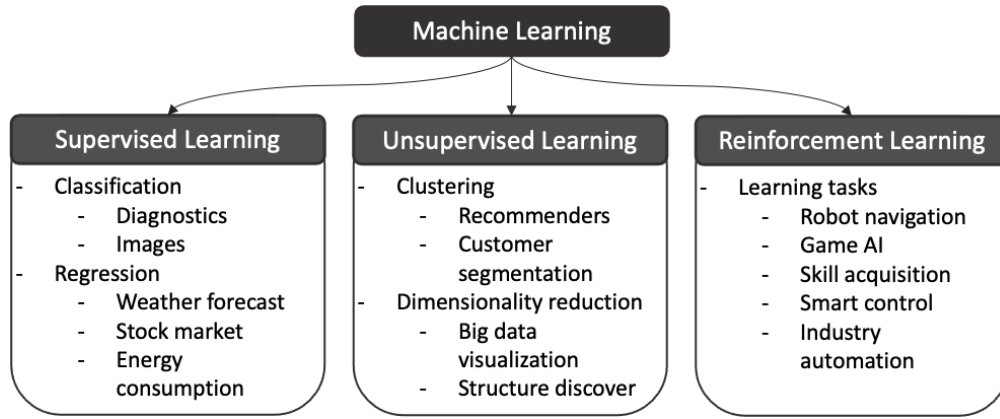


Figure 2.4: Machine learning classification.

the model will find the correlations and return the label with the highest percentage match.

As opposed to supervised learning, unsupervised learning does not have the supervisor role. In other words, the dataset has only input data and the machine learning task is to find the regularities in the entries, such as patterns that happen more than others [16]. An example of unsupervised learning is clustering in image compression, where the algorithm groups pixel with similar colours that are frequently occurring in the image. In market research, clustering can be used to identify customer buying patterns that may be related to demographics, attitudes and preferences.

Reinforcement learning [17] can be a solution when there is a lack of information about the environment and problem to be solved. In this model, the agent can learn the optimal policy by interacting directly with their environment, choosing actions based on their previous experiences within the domain. RL algorithms can adjust dynamically in changing environments, such as energy management in buildings, without the need to re-train the models as in supervised learning. For each action taken at a determined state, the agent moves to the next state and receives a reward that rules how good or bad was that decision. After large numbers of trials, the algorithm can learn optimal or near-optimal policies, with no prior knowledge. The agent should be able to evaluate policies and learn from past good actions to generate a policy [16]. The application of RL can be seen in many areas, such as robot navigation [39], games [24,25], industry automation, and smart control [20,53].

2.2.1 Reinforcement Learning

Reinforcement learning is the algorithm chosen as a base for the experiments of this research and, due to its importance, has its main concepts further explored in this subsection. Additional information on the algorithms and examples of their applications are also found in Chapters 3, 4 and 5, aiming to enhance the understanding of each specific experiment.

Markov Decision Process

The primary objective of any reinforcement learning algorithm is the solution of the Markov Decision Process (MDP), in which the main property (Markov property) states that the “*future is independent of past given the present*”. In other words, it means that the current state succinctly captures the entirety of the state history into the current state, thus one does not need to maintain a historical record. A common example is a chequers board, whereby the current confirmation of the pieces on the board is sufficient for a player to make a move, the entire history of all states or actions leading to this current configuration is not needed. A MDP model is generally composed of four components: Environment state space (S), Action space (A), Probability distributions of state transitions ($p(s_{t+1}|s_t, a_t)$) and Probability distribution governing the rewards received ($q(s_{t+1}|s_t, a_t)$). Figure 2.5 shows a graphical representation of the MDP.

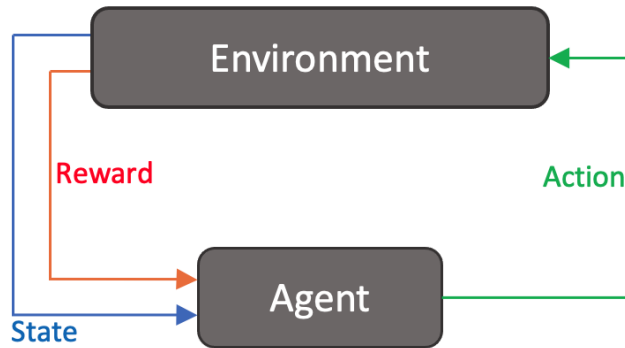


Figure 2.5: Markov decision process.

Overall, an MDP task can be discretized into time periods, where at each period t the agent occupies a state $s_t \in S$, and then chooses an action a_t from the set of all possible actions within the current state. This action

then results in a state transition to s_{t+1} and an immediate numerical reward $R(s_t, a_t)$. The state transition probability $p(s_{t+1}|s_t, a_t)$ governs the probability that the agent will transition to state s_{t+1} as a result of choosing a_t in s_t . The numerical reward received upon arrival at the next state is governed by $q(s_{t+1}|s_t, a_t)$ and represents the benefit of choosing a_t whilst in s_t . Solving an MDP results in the output of a policy, which is a mapping from states to actions guiding the agent’s decisions over the entire learning period.

Algorithms

In cases where an MDP model is complete and fully observable, the problem can be solved using traditional techniques such as dynamic programming, such as value iteration. However, the vast majority of real-world problems cannot be fully observed, as part of the environment may be unknown. To solve this issue, there are two main categories of RL that can be used as a possible solution, which are to estimate the model by using statistical methods (model-based reinforcement learning) or by directly estimating its value function or policy (model-free reinforcement learning). Wang and Hong [18], in their work about reinforcement learning for building control, identified 73 articles in this field where the most popular method is the model-free RL, with 70 studies. One of the reasons is that model-based can be time-consuming to deploy, as a model for the environment is needed, for instance a data-driven model (e.g. deep neural network), or a physics-based model (e.g. thermal resistance–thermal capacity model). In this regard, this thesis considers a model-free RL, where learners attempt to directly approximate a control policy through environmental interactions [53, 68–73, 75–77].

Authors in [18] and [52], state that the three main approaches to find the optimal control policy without learning the model are value-based, policy gradient, and actor-critic. The first approach starts with a random value function and updates to an enhanced value function in an iterative process until reaching the optimal value function. An example is the off-policy Q-learning algorithm, where the optimal policy is created by selecting the action that optimises the value function at a certain state. On-policy methods, such as SARSA, can take advantage of learning from different control strategy, if an MPC or rule-based model is available for reference is available, for instance. The second approach, policy gradient, uses optimisation techniques to directly search for an optimal policy, instead of the value function.

The third approach is an enhanced version of the second, the actor-critic methods estimate both policy and the value function, and update both. The actor makes an action when it observes a state, then the critic marks the actor's performance. There are also other algorithms that are variations of the two last approaches, such as DDPG (deep deterministic policy gradient), A3C (asynchronous advantage actor-critic) and SAC (soft actor-critic).

Overall, on-policy methods are more stable but off-policy methods are more likely to achieve global optima, as they will have more freedom by not following some predetermined controllers. Moreover, actor-critic and policy gradient methods are usually applied to continuous state and actions state and, in this research, both are discrete. Therefore, it is preferable to formulate the estimations tabular Q-learning, which has a straightforward deployment with a guaranteed convergence after a number of trials [17]. Tabular Q-learning methods can be very precise, but they need to interact with the environment and revisit states a number of times in order to learn a good policy, which may be a problem for large state-spaces. As an option, the Q-table values can be approximated instead by using a deep Q-network, for instance. Both chosen methods, tabular Q-learning and deep Q-network are depicted below.

Q-Learning

MDP models can be applied to energy management systems, in case of some missing information about the environment a model-free technique can be used to generate the optimal policies, for example as in Q-learning [43]. This method is part of the temporal difference methods and is capable of making predictions incrementally and in an online way. For each time where a state is non-terminal, the update rule in Equation (2.1) is calculated.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)] \quad (2.1)$$

Actions are selected according to the policy π and approximations of $Q^\pi(s_t, a_t)$ are calculated after each time interval. To balance exploration and exploitation and achieve the best results, an action selection policy, for instance ϵ -greedy, can be selected. The ϵ -greedy strategy chooses the best

action from the policy most of the time, however the agent explores a certain amount of the time ruled by ϵ . After a number of trials, the agent should converge to an optimal or near-optimal operation, which means it will choose the actions that lead to the highest reward. The importance of future rewards is governed by the discount factor γ , situated in a range from 0 to 1. Values close to 1 assigns a greater weight to it, while a value close to 0 considers only the most recent rewards. Finally, α is a value lower than 1 that represents the learning rate of value estimates over the learning process.

Deep Q-Network

Tabular Q-learning methods can be very precise, but they need to interact with the environment and revisit states a number of times in order to learn a good policy. The learning time to converge to an optimal policy depends on the state-space size, where for each additional state or action included, the problem size can increase exponentially, which is known as the curse of dimensionality. This issue can be mitigated by using a deep reinforcement learning approach, which replaces the Q-table with a function approximator, using for example a deep Q-network (DQN) to estimate the Q-values [31, 100]. Figure 2.6 illustrates the comparison of a Q-table and DQN. The size of the Q-table will depend on the dimension of the number of actions (n) and the number of possible states (m). The DQN can be designed according to the user needs, and its general architecture is composed of an input layer (neural network nodes that will receive the states), hidden layers (where the calculations and transformations happen) and an output layer, which will have the size of the action space (n). This model does not rely on the amount of possible states (m).

As DQN has a built-in back-propagating optimiser for the learning process, the learning rate α is no longer needed in Equation (2.1). Hence, both $Q(s_t, a_t)$ terms cancel each other, resulting in the simplified Equation (2.2).

$$Q(s_t, a_t) \leftarrow r_{t+1} + \gamma \max_a Q(s_{t+1}, a) \quad (2.2)$$

In summary, both Q-learning and DQN models have advantages and disadvantages and when designing an RL experiment, one should decide on which algorithm will be applied, considering the accuracy and learning time expected. As stated by the authors in [29], the RL algorithms to be used

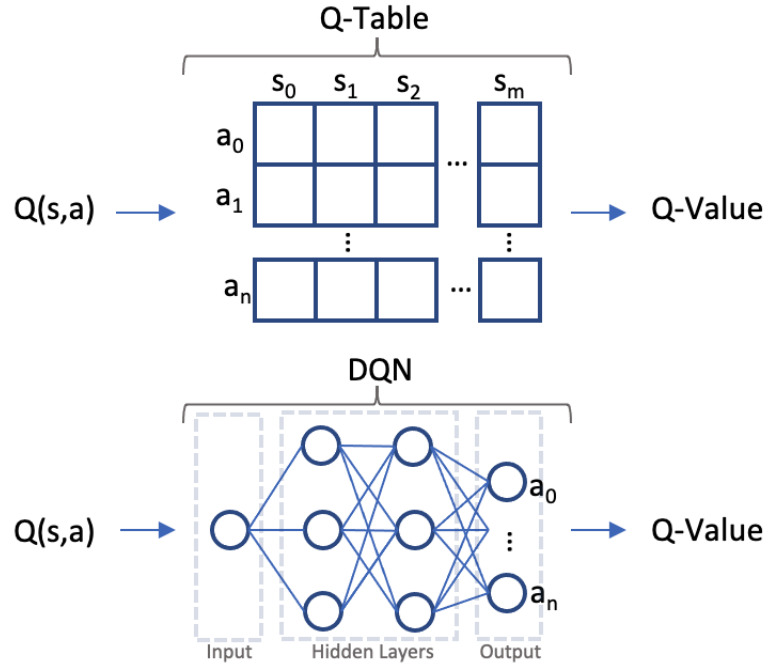


Figure 2.6: Q-table vs DQN.

should be selected according to the complexity of the problem being tackled. In some cases, a simpler algorithm may be effective enough to solve a task.

Transfer Learning

Transfer learning refers to the use of experience learnt by performing one task to improve another related, but different, task [79]. The objective is to increase convergence speed. Instead of learning from the beginning without prior knowledge, the agent integrates value function estimates from other agents to perform the tasks, achieving better results faster as a result of the knowledge sharing. Taylor and Stone [79] describes in their review five different types of transfer learning approaches, as follows:

1. Allowed task differences: Transfer learning methods can transfer between MDPs that have different transition functions, state spaces, targets, state variables, reward functions, and/or a set of actions. The agent's representation of the world (sensors and actuators) remains the same in both models, while the true state variables and actions can change.
2. Source task selection: In the simplest approach, it is assumed that only a single task has been learned and that the human has selected

it, ensuring that the agent should use it for transfer. Some models allow learning multiple source tasks and use them all, including the creation of libraries.

3. Transferred knowledge: This technique states that the type of knowledge transferred can be primarily characterised by its specificity. To improve the target task learning process, a policy, task model, prior distributions or action-value function can be used.
4. Task mappings: This approach is applied when the state variables and actions have the same semantic meanings but a different label. For instance, the mapping would be needed if the knowledge was acquired from a robot that executes a task on the right side of a manufacturing process, and the user needs to install another robot in the opposite direction. In that case, actions from "right" should be mapped to "left" to ensure the proper task execution in a different installation.
5. Allowed learners: The type of knowledge transferred directly affects the type of learner that is applicable [79]. For instance, a transfer learning algorithm that transfers an action-value function would need the target task agent to use some technique to exploit the transferred knowledge, such as temporal difference (TD).

Silva and Costa [80], in their survey about transfer learning for multi-agent systems, proposed two main types of transfer: Intra-Agent Transfer and Inter-Agent Transfer. The first relates to agents without explicit communication for accessing knowledge of the agents. For instance, the knowledge obtained from an environment with a set of sensors and actuators is used in another, but similar, environment. The second type considers that the information shared from another agent can be merged with the agent who receives it.

Transfer learning applied to RL can be seen in works from different domains, such as computer vision [21–23], games [24, 25] and 3D environments [26, 27]. In the energy and buildings domain, Xu et al. [41] applied TF to a DRL-based HVAC control, transferring knowledge from different zones, equipment and locations, thus reducing the training time, energy cost, and temperature violations. Finally, authors in [41, 80] also explored TF, but

a with focus on building predictive models, predicting energy consumption, temperature and humidity.

2.2.2 Energy Systems and Reinforcement Learning

Although reinforcement learning applied to energy-related fields has been object of research since the late 90s, it has been observed an increasing number the activities in this research area in the last decade. Vázquez-Cantelly and Nagy [29], in their review about RL for demand response, found more than one hundred works from 1997 to 2018, where 90 of these were published after 2012. From the energy application point of view, the authors classified the articles into four groups: HVAC and DHW, electric vehicles (EV) and hybrid vehicles, distributed generation and electrical storage, and appliances. The first two accounted for the majority of the publications, followed by DG and appliances. Across all the groups, Q-learning is the algorithm most used, representing around 50% of the total. Other algorithms found in their research include batch reinforcement learning, SARSA, actor critic, Monte Carlo and others less used.

Mason and Grijalva [49] explored reinforcement learning for autonomous building energy management in their review, in which the main applications identified were autonomous control systems, IoT, batteries and smart grid. The popular algorithms identified by the authors are Q-learning, SARSA and Actor critic. Mason and Grijalva [49], conclude that RL significantly improves the energy efficiency of dwellings, where savings of around 10% for HVAC applications, 20% for DHW and more than 20% for complete building EMS have been reported in the articles part of their review.

In a third review, Han et al. [52] studied reinforcement learning methodologies for controlling occupant comfort in buildings. The comfort types analysed were lighting, thermal, indoor air quality (CO₂, humidity and ventilation). Although RL applications are growing over the years in this field, comparing to other control types (i.e. rule-based, occupant-based, hard and soft control), RL models accounted for just 13% of the papers investigated. Regarding the distribution of algorithms for indoor comfort control, once again Q-learning was the most popular.

Wang and Hong [18] carried out recent research about the opportunities and challenges of reinforcement learning for building controls. Their survey found that about 42% of the models uses a table to record the value asso-

ciated with each action-state pair. However, using a deep neural network as a value function approximation represented more than 50% of studies in this field since 2018. Mason and Grijalva [49] and Vázquez-Cantelly and Nagy [29] also observed a trend towards the utilisation of ANNs or deep learning techniques in the RL algorithms. HVAC was the main source of control across the publications, followed by battery, appliances and DHW. Wang and Hong [18] identified significant barriers limiting the application of RL for real building controls. Starting with the training process, it can be time-consuming and/or data-demanding, which may impact the building comfort and energy consumption during the learning process. Moreover, the studies and implementations of transfer learning are small in this field, so it is unclear how controllers can be generalised and used in other buildings. Finally, it is not easy to compare models as the environments across studies vary. In this regard, Vázquez-Cantelly et al. [19] launched an initiative called “CityLearn Challenge 2020”, aiming to standardise research in multi-agent RL for demand response and urban energy management. The environment was made available for other researchers, who had the task to develop agents to control and coordinate components of 9 buildings in a microgrid: domestic hot water storage, chilled water storage, and electrical storage.

This subsection aimed to give an overview of the recent developments of reinforcement learning applied to energy systems. Additional works on this area are found in the next chapters, where an individual assessment is provided according to each specific experiment, also including their main achievements in terms of savings and comfort performance.

2.3 Summary and Experiments Methodology

The literature review guided the decisions on the algorithms chosen and the set of experiments performed. As presented in subsection 2.2.1, model-free RL approaches using tabular Q-learning and deep Q-network have been chosen as opposed to other ML models. The main reason is that the model is not fully observable, and there was no previous data to be used as a reference for supervised and unsupervised RL, MPC, or dynamic programming algorithms. Mathematical optimisation is computationally expensive; heuristic and meta-heuristic methods could be a solution in this regard, but they are

not flexible enough and will tend not to find an optimal policy if the environment changes constantly. In the energy management domain, for instance, the PV production can vary from day to day, and it can change even more when comparing different seasons. Finally, rule-based or scheduled methods rely on the user’s knowledge about the equipment and willingness to perform a different action if the environment dynamics change and, similarly to the previous method, it is not autonomously flexible. The key argument for the selection of the model-free RL is its adaptability, as buildings can sometimes be unpredictable and suffer changes, retrofits, or human intervention that affect their dynamics or energy consumption patterns.

After reviewing the control methods, the first experiment was designed considering the replication of a recognised work in the field about reinforcement learning applied to HVAC control, from Barrett and Linder [20]. HVAC systems are one of the most relevant loads in a building, and also has the necessary flexibility in terms of periods to be turned on/off. As can be seen in Figure 2.7, the inputs are the indoor temperature and weather forecast, and the outputs are the control actions (on/off), targeting to keep the comfort level adequate for the user. As the complete building model was not available and due to the size of the state-space and actions, which were discrete, a model-free tabular Q-learning has been applied. If the actions were continuous, such as fan speed of the HVAC system or temperature setpoints, a different algorithm would be needed, for instance, actor-critic. Next, in order to extend upon [20], transfer learning capabilities have been added to the model (red marked in the agent block of Figure 2.7), aiming to speed up the learning process. The details and results of the first experiment can be found in Chapter 3.

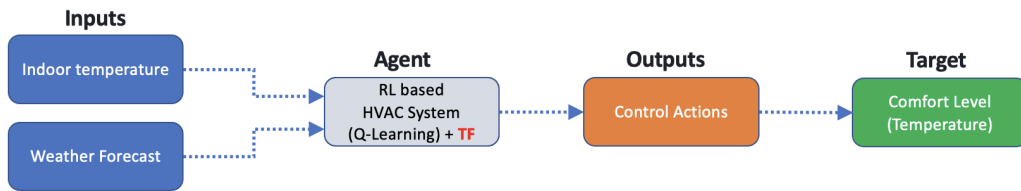


Figure 2.7: Experiment 1.

For the second experiment, the previous model has been upgraded considering DHW temperature and the PV production as new inputs. The control actions are still discrete, but the possibilities now are to turn on/off indoor heating or DHW. New targets have been added, which are energy

savings and load shifting. Due to all these new features, this new model is now called EMS. However, this is a more complex scenario with a much bigger state-space and, in order to tackle the curse of dimensionality, the Q-table has been replaced by a Q-network. Figure 2.8 shows the diagram of the second experiment, the details and results can be seen in Chapter 4.

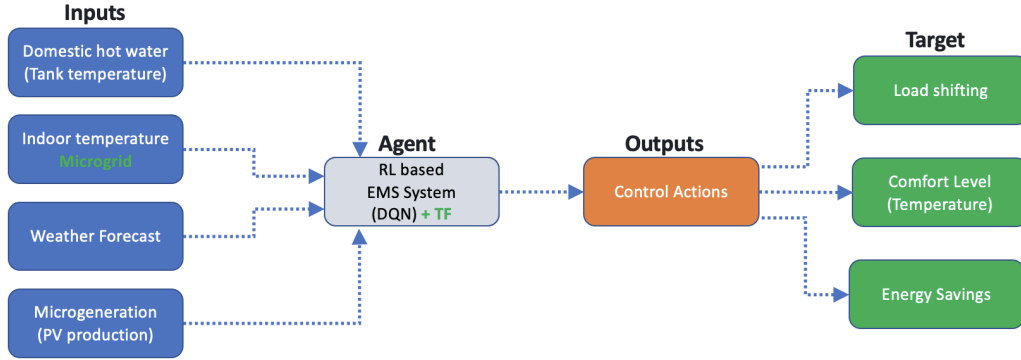


Figure 2.8: Experiments 2 and 3.

The third and last experiment is a natural evolution of the previous two. It uses the DQN algorithm from experiment 2 as a base, and adds the transfer learning capabilities of experiment 1, considering a microgrid environment. The modifications are green marked in the agent and inputs blocks of Figure 2.8. The main load considered in this experiment is DHW, as it can be used along with the PV production to create a buffer of hot water in periods of high energy production, to be used later on when the cost of energy is higher. The details and results of this experiments can be found in Chapter 5.

Chapter 3

Transfer Learning Applied to Reinforcement Learning-Based HVAC Control

The work outlined in this chapter was published in:

Lissa, P., Schukat, M. and Barrett, E. Transfer Learning Applied to Reinforcement Learning-Based HVAC Control. SN COMPUT. SCI. 1, 127 (2020). <https://doi.org/10.1007/s42979-020-00146-7>.

Abstract. Modern control solutions for HVAC have demonstrated excellent cost and energy savings through the utilisation of machine learning techniques. However, a challenging problem faced by most machine learning tasks is the amount of time and data required to train effective policies in the absence of prior knowledge. Considering that buildings from a specific geographical location share common environmental and structural features, this chapter investigates the impact of spatial changes on performance accuracy through the use of transfer learning applied to reinforcement learning based HVAC control. We propose the development of an adapted RL (Q-learning) algorithm which can transfer HVAC control policies, adjusting themselves according to spatial changes. We examine the performance of our approach across multiple different locations. Moreover, an analysis of the user's time out comfort has been made, comparing models with and without transfer learning. The results from different cases show that after applying transfer learning the learning time to train optimal or near-optimal control policies

was reduced by more than a factor of 6 when comparing to experiments without it. Furthermore, the test case where the spatial variation was lower than 50% achieved a similar performance for both dynamic and static HVAC control, presenting an average time out comfort error of around 2.5%. From the user’s perspective, it means they will not feel any additional discomfort, as the number of minutes out of the comfort zone for the static version is approximately the same for a 1-day interval. Finally, when examining the effect of transfer learning on geographical changes, the proposed method demonstrated higher performance in countries where the temperature variation is lower, reducing time out comfort by one-third. If an agent receives a policy from a place where the environmental conditions are very different the proposed method will still work and find the best policy, but not as fast as receiving it from a similar place.

3.1 Introduction

Heating, ventilation, and air conditioning systems, commonly known as HVAC, have been applied in a large variety of buildings to ensure occupant comfort and to provide a healthy environment for users. However, its application can consume large amounts of energy if the management and control of the equipment is not operated in an optimal manner. A number of solutions, based on smart-thermostats, and learning user occupancy patterns have arisen recently and are replacing manual control configurations [20,31,32]. Prior to the introduction of adaptive control solutions, the most common method was the programmable schedule, which allowed the operator to choose the period to turn the HVAC system on/off given the most likely occupancy patterns and specify the temperature set-point of the given room. This achieved energy and cost reductions when compared to leaving the HVAC always on, but recent work [20] has proven that machine learning-based adaptive methods can achieve even greater cost reductions whilst maintaining equivalent occupant comfort levels. Whilst achieving a sustainable solution with a better efficiency using machine learning techniques has demonstrated efficacy in the domain, one problem is the amount of time and data required to train effective policies in the absence of prior knowledge. In this work, we propose the use of transfer learning to speed-up the training times required for a reinforcement learning agent to determine

an optimal policy for HVAC control.

Smart-thermostats based on reinforcement learning, a sub-field of ML, can learn the optimal policy by interacting directly with their environment, choosing actions based on its previous experiences within the domain. After large numbers of trials, the algorithm can learn optimal or near-optimal control policies for operating the equipment, with no prior knowledge. This is one of the primary benefits of reinforcement Learning, as it allows for a control strategy to be approximated simply by interacting with the environment. Given enough time and experience and a reward signal indicating the benefit as to which actions to take whilst in a particular state, a learning agent can determine what actions or sequence of actions should be taken for a given state. For example, in HVAC control problems, an agent would be penalised for maintaining the temperature set-point when the room is vacant and not doing so when occupied. Concurrently, it would be rewarded for not switching on the device in a room which is not in use and thus achieving a cost saving with respect to energy consumption. The goal of the reinforcement learning agent is to ensure that a cost minimisation is achieved whilst concurrently ensuring occupant comfort is not compromised.

Achieving accurate control policies requires a substantial learning time, and often many thousands of learning trials to achieve a good approximation of the thermal dynamics of the space. One method which has shown promise in reducing the learning time and data required in more mainstream machine learning is known as transfer learning [23]. Transfer learning allows for the transfer of information between one agent to another, usually at the beginning of the learning process, in an effort to reduce the time taken by a learner to learn an optimal control policy. The hypothesis from an reinforcement learning perspective is that whilst a policy trained on one environment is not optimal for a separate disparate environment, many similar low-level states and actions are potentially shared between environments and thus the transfer of experience should avoid the need to visit all states during learning, which will result in a significant speed-up in learning times. Outside of the HVAC control domain, transfer learning has been successfully applied to a range of domains including computer vision [21–23], games [24, 25] and 3D environments [26, 27], demonstrating its proven efficacy. In a reinforcement learning context, it makes sense to employ it as the agent no longer needs to learn without any prior knowledge but instead initialises its value estimates

based on other agents’ observations. This raises the questions: can learning be transferred between reinforcement learning-based HVAC agents with different spatial characteristics in a similar location; and what is the impact when evaluating it in different geographic locations?

Office buildings, apartment blocks, homes in a housing estate and buildings from a specific geographical location all share similarity regarding their environment and structural features and even the materials used during the construction. In particular, each unit in an apartment complex or home in a housing estate will likely contain the same construction materials, insulation panels, have similar heat transfer rates and r -values. Thus, logically it makes sense that a control strategy learned for a smart thermostat in one apartment should theoretically be transferable to an apartment down the corridor. The thermal constants of the room may vary slightly, one apartment may be north facing, another more southerly but in general transfer learning should speed up the agent learning time when learning without any prior knowledge (which is common for reinforcement learners). The external factors such as weather conditions, temperature and air humidity, tend also not to vary dramatically between two nearby houses, which will also help the agent make decisions based on another’s observations. By transferring a basic coarse-grained policy, a new agent can focus on fine tuning the policy to their specific environment possibly achieving even higher accuracy than when learning from scratch. This will represent, in the HVAC and smart-thermostat domain, an immediate energy saving and less time out of comfort for occupants as this usually occurs during the early training phases.

In summary, the main contributions of this chapter include:

- The development of an RL (Q-learning) algorithm which utilises transfer learning for HVAC control, allowing for much faster learning rates in response to building spatial changes or different geographic locations.
- Thorough empirical analysis of a number of different spatial environments and different geographical locations is conducted, with analysis of models both with and without transfer learning presented.
- Thorough investigation of the impact of user comfort levels when training models using transfer learning.

The rest of this chapter is organised as follows: *Related Work*, which shows the related works regarding HVAC smart control and relevant learning approaches from other domains. *Learning Method and Initial Environment Setup* provides information about the principles of ML applied in this research. *Methodology*, describes how the environment was structured and explains the variables and system dynamics for the cases tested. The *Results* section presents all the relevant outputs of our experiments, showing the performance of the algorithm and comparing the methods. Finally, *Conclusions* recaps the main points of the chapter.

3.2 Related Work

This section aims to provide an overview about HVAC systems and current control methods, detailing recent works related to RL methods applied to HVAC control. Furthermore, some examples from other domains about the application of RL with transfer learning are discussed to illustrate the benefits that this method can bring to the HVAC domain.

3.2.1 HVAC Control Systems

HVAC systems are designed to satisfy user’s thermal comfort. Many options are available depending on the energy and environmental design, being categorised by the functional criteria (e.g., heating, ventilating and/or air conditioning) and the control method [28]. In a generic way, they consist of a thermostat, that senses the environmental temperature and controls the other heating and/or cooling components based on the thermostat values. From the user’s perspective, the HVAC control can be controlled by setting a temperature setpoint and/or scheduling the period of when the device is on/off. More recent control systems, which use ML algorithms, have been studied and implemented aimed at providing an autonomous optimised control. The key element for this is the smart-thermostat, which can provide additional sensing information, such as building/room occupancy, time to temperature estimates based on observations, in addition to complex algorithms for controlling the actuators.

Vázquez-Canteli and Nagy [29], in their work reinforcement learning for demand response: a review of algorithms and modelling techniques, provide an extensive review of RL for demand response and have found 105 articles

over the past 20 years, in which 24 are related to the HVAC domain. There have been a variety of algorithmic approaches, some of them are model based (e.g. Batch-RL) and others are model free or online (e.g. Q-learning), which are recommended when there is no complete model available, a common situation when dealing with real world challenges [30].

Barrett and Linder [20] proposed a model free (Q-learning) RL architecture which can support occupancy prediction and HVAC control, optimising user comfort and energy costs. Furthermore, their research revealed that although many patents has been found in the last years, only a few specific projects have been identified in the research community. The results from their simulations showed the efficiency of the model proposed, achieving 55% and 10% of cost reductions against “always on” and “Programmable” methods, respectively. Chen et al. [46] also applied a model-free Q-learning for HVAC control and window systems for natural ventilation, achieving up to 23% of energy consumption and 80% less discomfort degree hours in a case study deployed in Los Angeles.

In recent works, Wei et al. [31] and Nagy et al. [32] demonstrated the application of deep reinforcement learning with a data-driven approach for HVAC operation. The first one has achieved cost reductions from 20 to 70% comparing their results with scheduling methods. The second one, besides the model-based solution, created an additional RL model-free method for comparisons, which performed slightly worse. This was expected because online methods do not have previous knowledge and then need to explore before finding the optimal policy. Reference [33] used a model-free RL method applying recurrent neural networks for control and management of a central HVAC of a building. The results showed the RL controller has improved the thermal comfort by 15.5% and reduced energy consumption by 5.03% when compared to a variable control method, which consisted of an adjusting formula that tracks the external temperature and controls the HVAC accordingly. Finally, Gao et al. [34] proposed a deep RL-based framework for energy optimisation and thermal comfort control, working with continuous variables in the deep deterministic policy gradients model, comparing it against DQN, Q-learning and SARSA algorithms. Their study did not consider transfer learning techniques.

Additional HVAC control methods for energy efficiency and comfort, which are not from the field of RL, can be found in Refs. [35, 36] - Fuzzy

control techniques; and [37] - Model predictive control.

3.2.2 Transfer Learning for Reinforcement Learning

Transfer learning involves transferring previous experience from an agent to perform another related, but different, task [38]. In other words, it means taking value function estimates learned through interactions with one environment and transferring them to initialise the value functions of another model. This practice can lead to convergence speed ups, as the agent does not need to learn the policy from the beginning without prior knowledge. Taylor and Stone [38] provide a survey about TF for RL domains, where they classified the methods according to its capabilities and goals. Our research belongs to the first group, where the same state variables and actions are shared between agents. Other groups include methods with different states and actions or methods that learn a mapping between tasks.

The effectiveness of transfer learning can be seen across different domains. Barreto et al. [39] proposed a transfer framework where the reward function changes amongst navigation tasks of a simulated robot arm, working in the generalisation of Bellman’s classic policy improvement theorem, which extends the original result from one to multiple decision policies, and the Dayan’s successor representation for a continuous space. The simulated robot arm was allowed to train in only 4 of 12 tasks to achieve a target location. A quick performance improvement was observed in each of the 12 tasks, even for those without a specialised policy. Killian et al. [40] presented a transfer learning approach applied to patient diagnosis, considering that they may have a similar response to treatments.

In a related domain, Mocanu et al. [41] demonstrated an unsupervised energy prediction method using reinforcement cross-building transfer learning, having studied the Q-learning, SARSA and deep belief networks algorithms. Simulations included the use of a residential with non-electric heater data for prediction of a residence with electric heater, having shown RMSE from 0.02 to 2.67 across different combinations of models and scenarios.

Whilst transfer learning has shown its effectiveness across a number of domains, to the best of our knowledge this is the first time it has been applied to HVAC control, addressing spatial and geographical changes, while ensuring user’s comfort.

3.3 Learning Method and Initial Environment Setup

This chapter aims to examine the use of transfer learning applied to reinforcement learning in the HVAC domain when presented with different spatial environments and different geographical locations. To do so, we have set up a standard environment considering the utilisation of the proof methods from [20], which consists of a RL model free (Q-learning) agent. The theory behind the model and the parameterisation of the agent is described in this section. The transfer learning method and the new experimental scenarios are presented in *Methodology*.

3.3.1 Markov Decision Process

The primary objective of any reinforcement learning algorithm is the solution of the Markov Decision Process, in which the main property (Markov property) states that the “*future is independent of the past given the present*”. In other words, it means that the current state succinctly captures the entirety of the state history into the current state, thus one does not need to maintain a historical record. A common example is a chequers board, whereby the current confirmation of the pieces on the board is sufficient for a player to make a move, the entire history of all states or actions leading to this current configuration is not needed.

A MDP model is generally composed of four components: the state space (S), the set of possible actions (A), the probability distributions regarding state transitions ($p(s_{t+1}|s_t, a_t)$) and the probability distribution governing the rewards received ($q(s_{t+1}|s_t, a_t)$).

In most MDP tasks the problem can be discretized into time periods. At the end of each time period t the agent occupies state $s_t \in S$. The agent chooses an action $a_t \in A(s_t)$, where $A(s_t)$ is the set of all possible actions within state s_t . The execution of the chosen action, results in a state transition to s_{t+1} and an immediate numerical reward $R(s_t, a_t)$. The state transition probability $p(s_{t+1}|s_t, a_t)$ governs the likelihood that the agent will transition to state s_{t+1} as a result of choosing a_t in s_t .

The numerical reward received upon arrival at the next state is governed by $q(s_{t+1}|s_t, a_t)$ and is indicative as to the benefit of choosing a_t whilst in s_t . The solution of a MDP results in the output of a policy π , denoting a mapping from states to actions, guiding the agent’s decisions over the

entire learning period. In the specific case where a complete environmental model is known, i.e. (S, A, p, q) are fully observable, the problem reduces to a planning problem [42] and can be solved using traditional dynamic programming techniques such as value iteration.

However, if there is no complete model available, which is often common with real world problems, then one must either attempt to approximate the missing model (model based reinforcement learning) or directly estimate the value function or policy (model free reinforcement learning). Model-based methods use statistical techniques to approximate the missing model [30], whereas model free learners attempt to directly approximate a control policy through environmental interactions. The approach has been widely applied to real-world problems, which allows for stricter comparisons with previous work and is capable of finding an optimal or near optimal control policy in a reasonable time.

3.3.2 Reinforcement Learning

MDP models can resolve uncertainty issues, such as the environmental effects on HVAC systems. However, due to the lack of prior knowledge about the complete model, the distribution of rewards and/or transitions probabilities, techniques from Dynamic Programming cannot be used to find an optimal policy. In that case, a model-free reinforcement learning method, such as Q-learning [43], can be applied to generate optimal policies in case of an incomplete environmental model.

Q-learning belongs to a collection of algorithms called temporal difference methods. Not requiring a complete model of the environment, TD methods possess a significant advantage and have the capability of being able to make predictions incrementally and in an online fashion. The update rule for Q-learning is defined as:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)] \quad (3.1)$$

and calculated each time a state is reached which is non terminal. Approximations of $Q^\pi(s_t, a_t)$, which are indicative as to the benefit of taking action a while in state s , are calculated after each time interval. Actions are chosen based on π , the policy being followed. A number of action selec-

tion policies can be used to decide what action to select whilst occupying a particular state. Examples include ϵ -greedy, softmax and unbiased sampling [44]. The goal of these selection strategies is often to carefully balance exploration and exploitation to yield the best possible results in the shortest possible time frame. Over time the actions selected should converge to the optimal where the agents consistently choose actions which present it with the greatest amount of cumulative reward over the course of the interaction. In the case of ϵ -greedy, the goal is to choose the best action most of the time except for a certain amount of time governed by ϵ when the agent chooses an exploratory action. Let $A'(s) \subseteq A(s)$, be the set of all non-greedy actions. The probability of selection for each non-greedy action is reduced to $\frac{\epsilon}{|A'(s)|}$, resulting in a probability of $1 - \epsilon$ for the greedy strategy.

Estimated action values for each state action pair $Q^\pi(s_t, a_t)$ can be represented in tabular form or as part of a generalised function approximator. The goal of the learning agent is to maximise its returns in the long run, often forgoing short term gains in place of long-term benefits. By introducing a discount factor γ , ($0 < \gamma < 1$), an agent's degree of myopia can be controlled. A value close to 1 for γ assigns a greater weight to future rewards, whilst a value close to 0 considers only the most recent rewards. Reinforcement learning-based approaches are capable of reasoning over multiple actions, choosing only those which yield the greatest cumulative reward over the entire duration of the episode. The steps involved in Q-learning are depicted by Algorithm 3.1.

ALGORITHM 3.1: RL Algorithm (Q-learning)

Initialise $Q(s, a)$ arbitrarily

Repeat (for each episode)

Initialise s

repeat

Choose a from s using policy from $Q(\epsilon$ -greedy)

Take action a and observe r, s_{t+1}

$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$

$s \leftarrow s_{t+1}$

until s is terminal

end

Q-learning can often require significant experience within a given environment to learn a good policy. This is largely determined by the size of the state and action space. In particular, tabular Q-learning methods require continuous updating of the value estimates through repeatedly revisiting the states and choosing actions in the environment. As the size of the state action space grows, this problem can become more pronounced (often referred to as the curse of dimensionality), where each additional state or action variable added, increases the problem size exponentially. For the purposes of this work, we utilise tabular Q-learning methods, by applying transfer learning we demonstrate how convergence speed ups are achievable.

3.3.3 HVAC Control and Occupancy Predictor

This section describes the reinforcement learning HVAC control specifics and the occupancy predictor algorithm proposed by Barrett and Linder [20]. Both have an important role serving as a base for the spatial and transfer learning tests in later stages of this work.

This work extends upon the previous work by Barrett and Linder [20], examining the effects of transfer learning on varying the spatial dimensions or geographical locations when learning policies for HVAC control. Thus, this work adopts the same Bayesian approach to occupancy estimation, where all that is required to compute the posterior probability is an initial prior probability and subsequent environmental experience. It requires an occupancy sensor to provide it with the necessary evaluative feedback to update the model over time. Every minute the learning agent queries the sensor which returns a Boolean result (true or false) depending on whether or not the room was occupied at that time.

In addition, this research also employs Q-learning by framing the environment as a MDP, similar to our previous work [20]. This allows for a strict comparison between this chapter and the previous paper and to examine in detail the benefits of transfer learning. The set of states S , actions A and rewards have slightly changed in order to consider cooling actions (which were not considered in the previous article) and are as follows:

- rt : is the room temperature (source: temperature sensor, unit: °C);
- tto : is the time to occupancy (source: occupancy predictor, unit: minutes);

- ot : is the outside temperature (source: weather station, unit: °C);
- $Heat_{on}$: turns on heating;
- $Cool_{on}$: turns on cooling;
- $HVAC_{off}$: turns off heating and cooling.

Reference [20] HVAC controller actions are executed at discrete time intervals of 1 min, at the end of each minute the learning agent observes the current state of the environment and chooses whether or not to execute an automated HVAC action (turn on or off heat or cooling). Temperatures ot and it range from 0 to 40°C.

These are approximately the minimum and maximum temperature values of real data obtained from Weatherbit.io [47], considering 213 days dating from 1 April 2019 to 31 October 2019. The rewards achievable by the learning agent are distributed in accordance with certain scenarios that arise and are scalar in value. A setpoint variable sp specifies the user defined objective temperature setting. The new rewards rules are calculated as follows:

1. $(Room_{occupied} == false) \& (Action == Heat_{on} || Action == Cool_{on}) \& (rt > sp || rt < sp); R = -1.$
2. $(Room_{occupied} == true) \& ((Action == Heat_{on}) \& (rt > sp) || (Action == Cool_{on}) \& (rt < sp)); R = -3.$
3. $(Room_{occupied} == false) \& (Action == Heat_{on} || Action == Cool_{on}) \& (rt == sp); R = -1.$
4. $(Room_{occupied} == true) \& (Action == Heat_{on} || Action == Cool_{on}) \& (rt == sp); R = -1.$
5. $(Room_{occupied} == false) \& (Action == HVAC_{off}) \& (rt < sp || rt > sp); R = 0.$
6. $(Action == HVAC_{off}) \& (rt == sp); R = 0.$
7. $Room_{occupied} == true) \& (Action == HVAC_{off}) \& (rt < sp || rt > sp); R = -3.$

A threshold around the setpoint of plus or minus 1°C has been assumed by [20]. If the setpoint temperature of 23°C is specified, the variable sp will range from $22^{\circ}\text{C} - 24^{\circ}\text{C}$ (which is the comfort parameter in the next sections). More detailed discussion about the HVAC control and occupancy predictor can be found in [20].

3.4 Methodology

3.4.1 Spatial Analysis

After replicating the results found in Ref. [20], where Q-learning method achieved 55% and 10% of cost reduction compared to “Always on” and “Programmable control” method, respectively, we examined the effects of varying the spatial dimensions by creating a new testing environment, which consists of three new rooms with different dimensions. Our hypothesis is that the rooms are in the same building, and hence share the same weather conditions, building construction materials and thus have the same heat transfer characteristics. This is analogous to having multiple apartments in the same complex. Room number 2 presents the highest change, having more than twice of the total area when compared to the original room 1. Rooms 3 and 4 have a smaller variation, being approximately 50% bigger or smaller compared to Room 1, respectively. For simulation purposes, the rooms have the same average u -value of 0.49 and no internal loads, as the aim is to identify the agent behaviour considering heating losses by the surface. The weather dataset in this experiment is from Galway (city in Ireland). The full information and comparison of the rooms can be found in Table 3.1.

Table 3.1: Test cases characteristics.

	Room 1	Room 2	Room 3	Room 4
Total Area	16 m ²	36 m ²	25 m ²	9 m ²
Height	2.6 m	2.6 m	2.6 m	2.6 m
Width	4 m	6 m	5 m	3 m
Length	4 m	6 m	5 m	3 m
Volume	41.6 m ³	93.6 m ³	65 m ³	23.4 m ³
Surface Area	57.6 m ²	98.4 m ²	77 m ²	40.2 m ²
% of Room 1	100%	225%	157%	56%

Experiment 1: Independent Learners' Study

The aim of the independent learners' study is to establish a baseline for algorithmic comparative purposes. It consists of training each learner from rooms 1 to 4 independently of each other with no prior knowledge or knowledge transfer over 10 episodes of learning. Q-learning was configured with a learning rate α of 0.5 for all experiments in this chapter, it observes the consequences of the transition probabilities T and adjusts accordingly. This number of episodes has been chosen because experiments have showed that after less than five episodes the Q-learning algorithm has already achieved stability and convergence. Thus, having ten episodes represents a hypothetical case where the sensor has been installed at the same place for a long period. This experiment follows the generic Algorithm 3.1 presented previously and it represents, for instance, the installation of a new HVAC unit in a suite of offices, without having previous knowledge of the HVAC systems of the same building.

Experiment 2: Transfer Learning with Static HVAC Power

The second experiment follows the same structure of the independent learners study. However, instead of learning from the beginning, the agents from rooms 2 to 4 will use the knowledge obtained from Room 1, without any policy restart. Furthermore, it is important to notice that all environmental conditions, as well the HVAC power output, are fixed. The generic representation of this experiment can be seen in Algorithm 3.2. In the real world, it is equivalent to suddenly changing the building space without ad-

justing or upgrading the current HVAC system, such as integrating/merging offices or even changing the air volume and exposed surface area due to new furniture.

ALGORITHM 3.2: TF with static HVAC power

Initialise standard Environment

Initialise standard HVAC power

Initialise Room 1 Dynamics

Initialise $Q(s, a)$ arbitrarily

Repeat (for 10 episodes)

Initialise s

repeat

Choose a from s using policy from $Q(\epsilon\text{-greedy})$

Take action a and observe r, s_{t+1}

$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$

$s \leftarrow s_{t+1}$

until s is terminal

update Room Dynamics

Repeat (for 10 episodes)

Initialise s

repeat

Choose a from s using policy from $Q(\epsilon\text{-greedy})$

Take action a and observe r, s_{t+1}

$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$

$s \leftarrow s_{t+1}$

until s is terminal

end

Experiment 3: Transfer Learning with Dynamic HVAC Power

Finally, experiment 3 is an upgraded version of experiment 2. It uses the same previous principle of transfer learning, but now the new algorithm aims to achieve the same best level of accuracy from room 1 episodes with a short-time return. After the observations from experiment 2 it was noticed that the configuration of the HVAC power has impacted in how quickly the system achieves the optimal policy, so we proposed a dynamic HVAC control to improve the performance.

This method is more complex when compared to experiment 2 and is fully automated, thus it does not need any additional inputs from the user, because the needed adjusts are made by the algorithm. After the first ten episodes learnt from room 1, the algorithm updates the room dynamics and before the next episode it executes a partial episode, taking the actions according to the previous policy and comparing the average rewards (Rwd) with the new expected future rewards. If the difference surpasses 20%, the HVAC power is increased or decreased by 10% and the partial episode starts again, until the difference is lower than the threshold.

In summary, the output of this experiment is the dynamic control of the HVAC system. It is important to note that the power limit specification of the HVAC system should be respected to avoid degradation of the equipment. Algorithm 3.3 shows how the decisions are made.

ALGORITHM 3.3: TF with dynamic HVAC power

Initialise standard Environment

Initialise standard HVAC power

Initialise Room 1 Dynamics

Initialise $Q(s, a)$ arbitrarily

Repeat (for 10 episodes)

 Initialise s

repeat

 Choose a from s using policy from $Q(\epsilon\text{-greedy})$

 Take action a and observe r, s_{t+1}

$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$

$s \leftarrow s_{t+1}$

until s is terminal

update Room Dynamics

repeat

 update $HVAC_{power} \pm 10\%$

until $|\sum Rwd_{episode}| / |\sum Rwd_{episode-1}| < 20\%$

Repeat (for 10 episodes)

 Initialise s

repeat

 Choose a from s using policy from $Q(\epsilon\text{-greedy})$

 Take action a and observe r, s_{t+1}

```


$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$$


$$s \leftarrow s_{t+1}$$

until  $s$  is terminal
end

```

3.4.2 Geographical Change Analysis

This second set of experiments aims to analyse the performance of transfer learning when using a trained agent at one specific place and then applying it to a different geographic location. For comparison purposes, we selected room 1 from Table 3.1 and the Galway dataset as a base, the same considerations about internal load and building characteristics are consistent as detailed at the beginning of “Spatial Analysis”. After ten training episodes, the policy is transferred to another environment in a different geographic location with different weather conditions. Algorithm 3.4 presents a generic representation of the experiment.

ALGORITHM 3.4: TF applied to different environment

```

Initialise standard Environment
Initialise standard HVAC power
Initialise Room 1 Dynamics
Initialise  $Q(s, a)$  arbitrarily
  Repeat (for 10 episodes)
    Initialise  $s$ 
    repeat
      Choose  $a$  from  $s$  using policy from  $Q(\epsilon\text{-greedy})$ 
      Take action  $a$  and observe  $r, s_{t+1}$ 
       $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$ 
       $s \leftarrow s_{t+1}$ 
    until  $s$  is terminal
  update Weather Dataset
  Repeat (for 10 episodes)
    Initialise  $s$ 
    repeat
      Choose  $a$  from  $s$  using policy from  $Q(\epsilon\text{-greedy})$ 
      Take action  $a$  and observe  $r, s_{t+1}$ 
       $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$ 

```

```

     $s \leftarrow s_{t+1}$ 
    until  $s$  is terminal
end

```

To have more than one view about transfer learning capabilities, cities from two distinct countries have been selected: Aarhus (Denmark), where weather conditions are slightly similar to Galway (Ireland); and Madrid (Spain), which presents extreme temperatures. The period of analysis is the same for all the datasets, 213 days from 1 April 2019 to 31 October 2019, Table 3.2 presents the main characteristics of each one of them.

Table 3.2: Temperature characteristics.

	Galway (Ireland)	Aarhus (Denmark)	Madrid (Spain)
Minimum °C	5.30	-0.15	0.06
Maximum °C	20.51	27.90	39.96
Average °C	13.27	13.26	20.09

3.5 Results

3.5.1 Performance Analysis of Spatial Changes

After the simulations it was observed that the same behaviour happened to the three new rooms across the experiments mentioned in the previous section. Generally speaking, experiment 1 simulations show that after the 5th episode all the agents achieve stabilisation. This can be seen by analysing the average rewards for each episode, represented by the blue lines in the four charts of Fig. 3.1. This behaviour is expected as each agent had to learn the policy without any previous knowledge.

The application of the algorithm defined in experiment 2 has improved the results by 17% on average. Here, transfer learning helps to reduce the average rewards in the initial episodes, as can be seen in the dashed orange lines of Fig. 3.1. Both rooms 2 and 3 present a stabilised curve since the 1st episode. This means that the information acquired from room 1 is enough to achieve the same performance as learning by themselves after a number of episodes, hence speeding-up the learning process. Room 4 also achieves

a faster convergence, but due to its dynamics the stabilisation occurs only in later stages. However, the agents do not achieve the original lowest average rewards of room 1, and that was the motivation of the deployment of experiment 3.

The HVAC system of this experiment was dimensioned to Room 1 and it considers a static output power of 1000 W for both heating and cooling, which may be too much or too little for the new rooms. Updating the HVAC power dynamically allows the rewards, and consequently the errors, to achieve a performance close to the average of room 1, successfully completing the objective of transfer learning. Figure 3.1 shows that there is no significant difference between the average rewards of Room 1 (blue line of first chart) and the average rewards after experiment 3 applied to the new rooms (dotted green lines). Note that this happens since the first episodes. Experiment 3 has improved the results by 45% on average for rooms 2 and 3; room 4 achieves 92%, mainly because the initial average rewards in experiment 1 for this case are much lower when compared to the others.

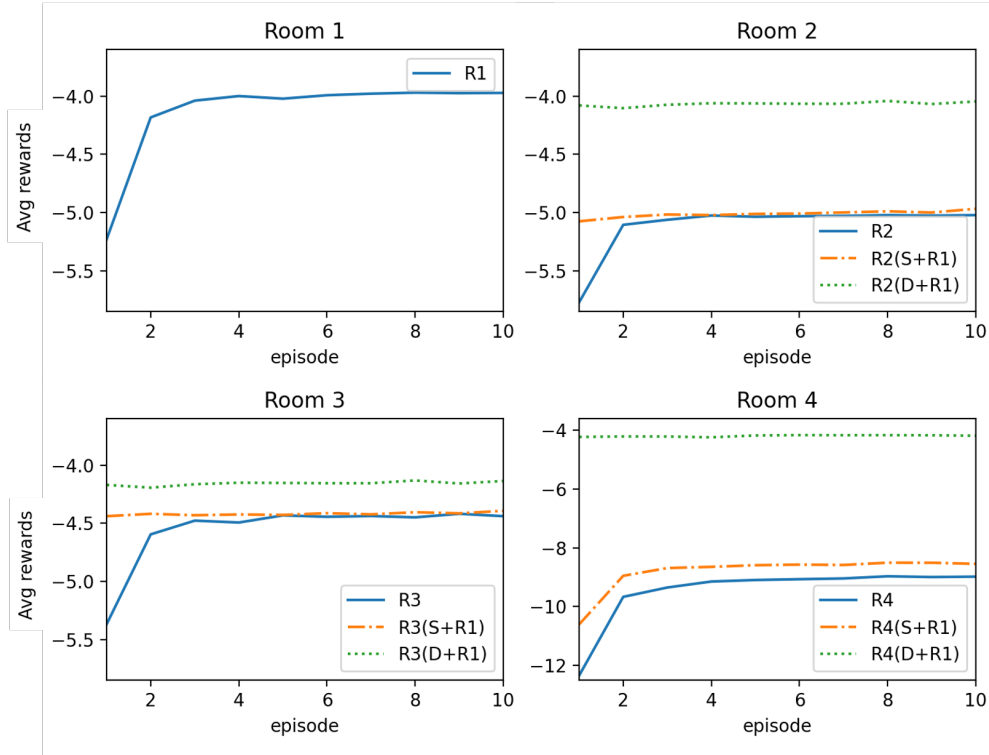


Figure 3.1: Spatial Changes - Cumulative average rewards per episode (S+R1: Static HVAC and TF from room 1. D+R1: Dynamic HVAC and TF from room 1).

To show the effect of each one of the experiments for the different rooms, an individual analysis has been carried out. Starting with room 2, the one with higher spatial variation, it is possible to verify that without using previous learning the average rewards after experiment 1 are lower than for the 1st room. Using transfer learning (experiment 2), the convergence is faster as the value never achieves the minimum reward average of room 1. Applying the HVAC dynamic control yields almost no variation between the minimum reward average of room 1 and room 2, but the HVAC power output has to reach 1700 W to keep the performance.

Room 3, which has an area size between room 1 and 2, shows quite similar results when compared to room 1. It can be observed that if the new room is not greater than 50% of the reference room, the static HVAC (experiment 2) can achieve a reasonable result. To achieve the minimum average rewards in this room the HVAC power reached 1300 W. Finally, the smallest room (number 4) had a poor performance and experiment 3 was necessary to achieve a low rewards average. In that case, the HVAC power was higher than necessary, thus the algorithm reduced it to 700 W.

The results show that even if the total area is reduced, one has to adjust the HVAC power. The main reason for this is that the agent learns how to manage the inertia and heat transfer of the room, that means having high-power HVAC in a small room or a low-power HVAC in a big room can be challenging from a control perspective. For instance, turning the high-power HVAC on in the small room could result in getting lower temperatures than expected, or low-power HVAC could never achieve the expected temperature due to the room's size.

The system modelled by Barrett and Linder [20], the basis of this work, utilises a HVAC power of 1000 W, operating with 100% efficiency. In this study, we applied Equation (3.2) to determine the temperature increase or decrease per timestep when the HVAC is turned on.

$$\Delta T = \frac{((HVAC_{power} - Heat_{transf}) \times Air_{dens} \times Room_{vol} \times StepSize)}{SpecificHeat} \quad (3.2)$$

where $HVAC_{power}$ is positive for heating and negative for cooling and the heat transfer model to determine the changes in the environment follows Equation (3.3).

$$Heat_{transf} = uValue \times surfaceArea \times (rt - ot) \quad (3.3)$$

3.5.2 Performance Analysis of Geographical Changes

To test the transfer learning across the three different countries, the following simulations were performed:

1. Baseline without transfer learning: Galway (IE), Aarhus (DK), Madrid (SP).
2. Transfer learning from Galway (IE) to: Aarhus (DK(IE)), Madrid (SP(IE)).
3. Transfer learning from Aarhus (DK) to: Galway (IE(DK)), Madrid (SP(DK)).
4. Transfer learning from Spain (SP) to: Galway (IE(SP)), Aarhus (DK(SP)).

Over time, the Spanish case study achieves the lowest average reward per episode, followed by the Danish and Irish ones. This can be seen in the blue lines of the charts from Fig. 3.2, which represents the initial tests without transfer learning for each country.

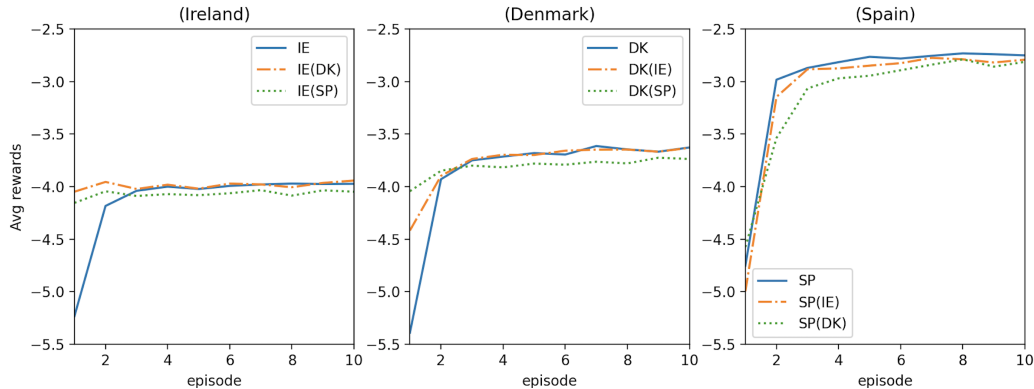


Figure 3.2: Geographical changes - Cumulative average rewards per episode.

Although Galway presents the lowest average reward, it is the one who converges faster after transfer learning from Aarhus and Madrid. The reward

curves for both transfer cases are stable since the first episode, represented by the orange dashed and green dotted lines, respectively, in the first chart of Fig. 3.2. It represents an average improvement of 27% and is going to impact directly the user’s comfort analysis, to be presented in the next subsection.

It was observed that the Danish experiment was able to reduce the average rewards by 33% when using transfer learning from Ireland, but only 22% when it comes from Spain. Finally, despite of presenting the best average rewards over time, the Spanish learning convergence is only improved by 4% using learning from Denmark and performs 4% worse when the data comes from Ireland. The charts 2 and 3 from Fig. 3.2 show the results for Denmark and Spain, respectively.

3.5.3 User’s Comfort Analysis

The result of the user’s comfort analysis is strongly correlated to the performance from the previous section. Nonetheless, it brings a better view of the real impact on the user’s life and the suitability of applying transfer learning to tackle spatial changes in the HVAC domain. The approach chosen was to count the daily average time out of the comfort (ToC) zone of 22°C and 24°C, which is $\pm 1^\circ\text{C}$ of the setpoint of 23°C. ASHRAE [45] recommends an indoor temperature range between approximately 19.4 to 27.7°C for thermal comfort purposes, which depends on the season, humidity and other factors. The chosen comfort temperature ranging from 22 to 24°C considers a more critical hypothetical case, where the user specifies the setpoint and the HVAC system automatically adjusts itself to meet the requirements.

Room number 1 has been included only for comparison purposes, its experiment 1 represents the ToC of the first episode. Experiment 2 and 3 present the same low values because they are the HVAC power reference for the other rooms in both situations. Rooms 2 and 3, both with higher dimensions compared to 1, present a similar trend of reductions across the experiments. Despite having a slightly reduced surface area, room 4 presents the worst results during experiments 2 and 3, but after experiment 3 it has reached almost the same performance as the other rooms.

The number of minutes out of the comfort zone ranges from approximately 100–630 in the first experiment and from 36 to 519 in the second one. In the last experiment, all rooms have reached around 40 min on aver-

age. Room number 3, as mentioned in the performance analysis, achieves a good average of 36 minutes for experiment 2, quite close to its 38 min result for experiment 3. Table 3.3 details the number of minutes out of comfort per room and experiments, as well as bringing information about its daily representative percentage.

Table 3.3: Time out comfort (ToC) per room.

Experiment:	1	2	3
Room 1			
Avg ToC (min)	121	43	43
Avg ToC (%)	8.42%	3.04%	3.04%
Room 2			
Avg ToC (min)	99	46	35
Avg ToC (%)	6.90%	3.21%	2.46%
Room 3			
Avg ToC (min)	101	36	38
Avg ToC (%)	7.00%	2.52%	2.66%
Room 4			
Avg ToC (min)	630	519	52
Avg ToC (%)	43.74%	36.02%	3.65%

Overall, an average ToC below 3% represents 40 min or less of discomfort in an entire day. The results presented show only the values regarding the first episode, it means that after that, as the number of trials progresses, the ToC will tend to decrease and in theory drop to 0, if the policy turns to completely greedy.

Moving to ToC analysis per country, Ireland presents a reduction from 8.42% to around 3% after applying TF from both Denmark and Spain, which represents 80 min less out of comfort. Denmark reached around 90 and 70 min less ToC with transfer learning from Spain and Ireland, respectively. On the other hand, as mentioned in the performance section, Spain was not able to get the same level of results achieved in the other two countries, reaching only 26 min reduction using TF from Denmark and almost no improvement from Ireland. The summary of results can be seen in Table 3.4.

Table 3.4: Time out comfort (ToC) per country.

Average ToC:	Minutes	(%)
Ireland - (IE) – without TF	121	8.42%
with TF from Denmark - IE(DK)	46	3.20%
with TF from Spain - IE(SP)	47	3.04%
Denmark - (DK) – without TF	154	10.69%
with TF from Ireland - DK(IE)	80	5.59%
with TF from Spain - DK(SP)	60	3.21%
Spain - (SP) – without TF	163	11.33%
with TF from Ireland - SP(IE)	157	10.88%
with TF from Denmark - SP(DK)	137	9.49%

Results could be improved for Spain or even other places that face a higher range of temperatures over the year if the agent is exposed to a new set of environmental conditions. However, increasing the state action space size will raise the problem exponentially and decrease policy convergence speed, because tabular Q-learning methods require continuous updating of the value estimates through repeatedly revisiting the states and choosing actions in the environment [20].

3.6 Conclusions

This work has demonstrated the benefits of applying transfer learning to handle spatial and geographical changes in autonomous HVAC control systems. The results of the three spatial test cases show that the algorithms proposed present a similar behaviour in all situations, having reduced the error rate and average rewards after experiment 2 and achieving the optimal policy after the dynamic HVAC control experiment 3. That means transfer learning has been achieved across the episodes even after spatial changes. Furthermore, if the total area of the new room is just slightly different from the original room, the dynamic control is not needed, as can be seen in the results of room 3.

Through this new approach, the time out of comfort has been drastically reduced by more than 6 times in some of the spatial tests. Consequently, the user would experience only 3% of inadequate temperatures during an

entire day, which ranges from 7 to 43% without transfer learning. Due to the HVAC power limitation, if the total area variation is very high, the agent may never achieve an optimal policy. In that case, the installation of a new HVAC system may be necessary to meet the requirements.

The geographical changes analysis also shows the benefits of applying transfer learning to achieve a faster optimal convergence. In some cases, it reduces by one-third the time out comfort, but the algorithm performs better where the temperature over time does not have major variations. The Irish case for instance presents a stable curve since the first episode, after receiving the policy from Spain or Denmark.

Chapter 4

Deep Reinforcement Learning for Home Energy Management System Control

The work outlined in this chapter was published in:

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<https://doi.org/10.1016/j.egyai.2020.100043>.

Abstract. The use of machine learning techniques has been proven to be a viable solution for smart home energy management. These techniques autonomously control heating and domestic hot water systems, which are the most relevant loads in a dwelling, helping consumers to reduce energy consumption and also improving their comfort. Moreover, the number of houses equipped with renewable energy resources is increasing, and this is a key element for energy usage optimisation, where coordinating loads and production can bring additional savings and reduce peak loads. In this regard, we propose the development of a deep reinforcement learning algorithm for indoor and domestic hot water temperature control, aiming to reduce energy consumption by optimising the usage of PV energy production. Furthermore, a methodology for a new dynamic indoor temperature setpoint definition is presented, thus allowing greater flexibility and savings. The results show that the proposed DRL algorithm combined with the dynamic

setpoint achieved on average 8% of energy savings compared to a rule-based algorithm, reaching up to 16% of savings over the summer period. Moreover, the users' comfort has not been compromised, as the algorithm is calibrated to not exceed more than 1% of the time out the specified temperature set-points. Additional analysis shows that further savings could be achieved if the time out of comfort is increased, which could be agreed according to users' needs. Regarding demand side management, the DRL control shows efficiency by anticipating and delaying actions for a PV self-consumption optimisation, performing over 10% of load shifting. Finally, the renewable energy consumption is 9.5% higher for the DRL-based model compared to the rule-based, which means less energy consumed from the grid.

4.1 Introduction

A report from Eurostat [3] shows that energy consumption in the residential sector accounted for approximately 26.1% of total energy consumption in the EU in 2018, where the main consumption was in heating systems and water heating, consuming 63.6% and 14.8% of the total, respectively. Moreover, most of the residential energy consumption is covered mainly by natural gas (32.1%) and electricity (24.7%), while renewables account for just 19.5%. There is a global trend of renewable energy asset expansion, as they represented almost two-thirds of the new net world electricity capacity additions in 2016 and it is expected to increase by 43% between 2017 and 2022, according to studies from the International Energy Agency [4]. To reach the key target set by the EU of at least 32% share for renewable energy this growth has to continue [48]. However, as solar and wind generation rely on weather conditions, challenges due to intermittent generation have to be solved, and solutions for energy management such as demand response and photovoltaic (PV) self-consumption optimisation can play a key role in this regard.

Residences with automated actuators and monitored sensors, known as smart homes, can get benefits from advanced machine learning techniques for energy management and achieve better results in terms of energy savings when compared with scheduling or manual control methods, which in some cases depend on the user's behaviour or do not consider all the variables for optimal operation. This can be seen in Barrett and Linder's [20]

work, where they have proven that machine learning-based adaptive methods, such as reinforcement learning, can achieve even greater cost reductions in the HVAC domain, the most relevant load in European residences. Reinforcement learning is a sub-field of ML; it can learn the optimal policy by interacting directly with their environment, choosing actions based on its previous experiences within the domain, with no prior knowledge. After a number of trials and a reward signal indicating the benefit linked to the actions in a particular state, the agent can decide the best action to be taken for a given state. In the context of energy management, the rewards could be based on performing control actions when the cost of energy is low, or the production of energy in the house is high. The target would be, for example, saving energy as much as possible, but also respecting the user's needs.

Reinforcement Learning can be applied through different approaches. Vázquez-Canteli and Nagy [29] presented an extensive review of algorithms and modelling techniques in the demand response domain, where they classified articles according to their application and ability to address the problems of speed of convergence, and curse of dimensionality. Mason and Grijalva [49], in their review about reinforcement learning for autonomous building energy management, also explored different types of RL, concluding that applications of deep reinforcement learning algorithms are expected to keep growing due to their increased effectiveness over traditional approaches.

In this work, we propose the use of reinforcement learning to manage and control the heating system and domestic hot water, with PV self-consumption optimisation. The main contributions of this chapter include:

- The development of a deep reinforcement learning algorithm for indoor and domestic hot water tank temperature control, aiming to reduce energy consumption by optimising the usage of PV energy production. The algorithm also provides load shifting, helping the energy grid balance.
- A thorough investigation of the impact of user comfort levels and energy savings is conducted, with performance comparison against conventional control methods.
- A methodology is proposed for a flexible indoor temperature comfort threshold definition, considering a real case study in Ireland.

The rest of this chapter is organised as follows: *Related Work* shows the related works regarding reinforcement learning applied to home energy management systems. *Environment Setup* provides information about the simulated environment and the temperature threshold definition. *DRL-Based Home Energy Management System*, describes the algorithm and tests performed. The *Results* section presents all the relevant outputs of our experiments, showing the performance of the algorithm and comparing the methods. Finally, *Conclusions* recaps the main points of the chapter.

4.2 Related Work

A home energy management system (HEMS) is a system deployed in the home containing both hardware and software that allows users to manage their energy consumption and production, thus enabling participation in the electricity market. These systems are becoming increasingly popular due to the recent advancement in residential metering, monitoring and controlling, which allows a more reliable energy management and new business schemes, such as demand response programs. Shareef et al. [50] presented the evolution of HEMS from works since 1970, showing control methods such as rule-based and artificial intelligence, where the common targets were energy savings, CO₂ emission reductions, and user's comfort. They also pointed out a future trend of self-learning AI techniques for HEMS. Lee and Cheng [51] found that 34% of the papers about energy management systems in the same period belong to the residential sector. Vázquez-Canteli and Nagy [29] categorised papers about reinforcement learning for demand response across four different groups: HVAC and DHW, electrical vehicles, smart appliances, and distributed generation with storage; the first two groups represented more than two-thirds of the recent publications. Q-learning is the most popular learning method [29, 49, 52], with deep Q-learning gaining popularity since 2015 due to its capability to handle learning rate issues from large state-action space, known as the curse of dimensionality. After analysing a number of studies, authors in [49] showed that RL algorithms can improve residential energy efficiency, and although projects can vary significantly, typical savings of approximately 10% and 20% can be achieved for HVAC applications and DHW, respectively.

Regarding the HVAC application, Barrett and Linder [20] proposed a

tabular Q-learning RL architecture for occupancy prediction and HVAC control, optimising user comfort and energy costs achieving 55% and 10% of cost reductions compared to "always on" and "programmable" methods, respectively. Extending upon this research Lissa et al. [53] developed a transfer learning technique which successfully sped up the learning time taken to learn optimal policies in the domain. Chen et al. [46] applied model-free Q-learning for HVAC control and window systems for natural ventilation, achieving up to 23% of energy savings. Wei et al. [31] applied a data driven DRL, with cost reductions from 20% to 70% when comparing their results with scheduling methods. Nagy et al. [32] used a data-driven approach, outperforming rule-based control by between 5% and 10%. Wang et al. [33] reduced energy consumption of a central HVAC by 5% through a model-free RL method applying recurrent neural networks. Gao et al. [34], Zhang and Lam [54] and Valladares et al. [55] applied DRL for energy optimisation and thermal comfort control.

Moving to DRL applied to other home applications, Wan et al. [56] proposed an algorithm to minimise the energy cost of a house with battery energy storage. Liang et al. [57] also worked with energy storage systems, but in coordination with HVAC control. One of the most recent studies in this field belongs to Sangyoon and Dae-Hyun [58], where they proposed a hierarchical DRL for scheduling appliance usage in a single home, including an air conditioner, a washing machine, a PV system, an energy storage device, and an electric vehicle, under a time-of-use pricing model.

The aforementioned model-free methods have proposed solutions for different home energy system architectures. However, to the best of our knowledge, this is the first time that DRL is proposed as a solution for indoor heating and DHW control, aligned with PV self-consumption optimisation and a variable setpoint strategy. There are also other control methods for comfort and energy management that are not from the RL domain, such as fuzzy control [35, 36], or model predictive control [37, 59], where building an accurate model for a house can be time-consuming.

4.3 Environment Setup

4.3.1 Building Simulator

The first step of the experiment involved the development of the Building Simulator (BS) application which enabled the DRL control algorithms to be applied. The case study is a residential dwelling located on the Island of Inis Mór, Co. Galway, Ireland. The dwelling was built in the 1970's and has a total floor area of 110 m². In recent years the dwelling has been upgraded, including additional insulation to the walls and roof and installation of an 8.5 kW Mitsubishi heat pump along with a PV panel array consisting of 8 panels, with a total nominal power of 2 kWp. The construction characteristics of the building can be seen in Table 4.1.

Table 4.1: Construction characteristics.

Construction	U-Value (W/m ² K)
Wall	0.27
Roof	1.50
Windows	3.00
Ground floor	0.45
Internal partitions	1.80
Doors	1.35

The heating system installed in the dwelling is an 8.5 kW Mitsubishi Electric Ecodan heat pump. The heat pump connects to a 170 L hot water cylinder which is used to store hot water for both space heating and DHW. The DHW mode operates automatically based on the upper and lower limits set by the system installer as seen in Fig. 4.1. This is influenced by the DHW max temperature and the DHW maximum temperature drop. The hot water is automatically heated once the tank temperature exceeds the DHW max temp drop. For example, if the DHW max temperature is set to 50°C and the lower threshold is 40°C, there is a delta temperature of 10 degrees [60].

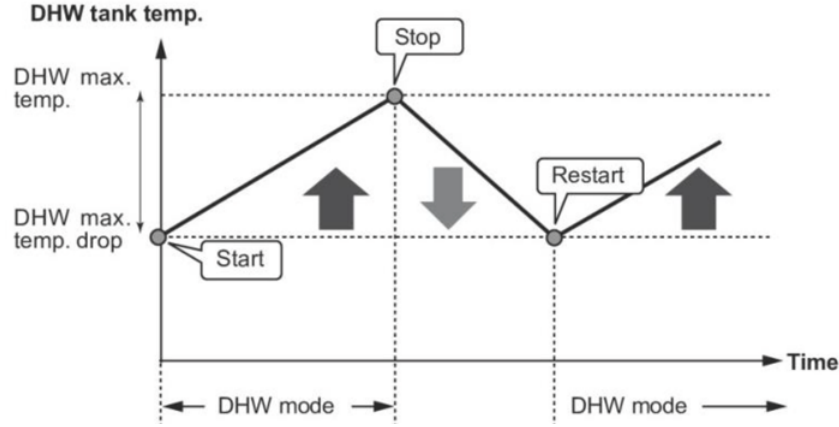


Figure 4.1: DHW operation [60].

The BS was formed by developing a building energy model (BEM) extracting key data from the case study dwelling. Sensors were installed in the dwelling which measured indoor temperature $^{\circ}\text{C}$, total electricity consumption (kWh), total PV production (kWh) and heat pump electricity consumption (kWh). The characteristics obtained from the site survey, along with data collected from each of the sensors, were used to develop a detailed and calibrated white-box model, using the Integrated Environmental Solution Virtual Environment (IESVE) software [61]. Simulations have been carried out to identify the main parameters and heating transfer dynamics necessary to build a reduced grey-box model. The parameters extracted from the white-box model were indoor air temperature increase and decrease rates for both, DHW and indoor temperatures, considering their behaviour during stationary conditions (system off) and when actions are performed.

Considering the heat pump off, the indoor temperature drop ranges from 0.0057°C to 0.0230°C every 5 min, depending on how high the difference between them is, if the indoor temperature is higher than the outdoor temperature. DHW temperature only relies on tank insulation, so the temperature drop is around 0.075°C per 5-min interval for the operational range of $40\text{--}55^{\circ}\text{C}$, slightly reducing this value until reaching the indoor temperature level. Regarding heat pump on, the temperature increases on average 0.2°C and 2.75°C every 5 min for space heating and DHW function, respectively. The new building simulator grey-box model reads the current environment state every 5 min and estimates the next DHW and indoor temperature values, calculated from the rates established previously. The benefit is that the new simulator can run the environment step-by-step, which allows us to

implement our proposed DRL control approach.

4.3.2 Indoor Thermal Comfort Threshold Definition

Demand response programs can take advantage of occupant behaviour changes, which can increase flexibility during the control decision process, hence allowing higher energy savings. However, authors in [62,63] stated that thresholds must be identified prior to deploying a strategy. In order to assess the thermal comfort of the occupant during the analysis, an indoor temperature threshold was developed based on a statistical analysis on the measured indoor temperature. The approach stems from Sweetnam et al. [62], who calculate the mean internal temperature (MIT) and upper and lower thresholds for 15-min intervals for each home in a 31-dwelling case-study, calculating the effect of a demand response control system on the heat pump. Upper and lower thresholds are based on calculating the upper and lower deciles of temperature. The outcome of the data analysis is a unified temperature profile bandwidth over a 4-month period which is used to compare temperatures pre and post instalment of HEMS.

Different approaches can be used to determine the best threshold to handle savings and comfort. For instance, a popular method for assessing building comfort is the predicted mean vote, which was first proposed by Fanger [64] and is an index that predicts the mean value of the thermal sensation votes of a group of people according to a seven-point sensation scale [65]. Comfort compliance is achieved if the PMV is within a range of $-0.5 < x < 0.5$ as per ASHRAE Standard 55-2017 guidelines [65]. This method is more suitable for application in controlled office environments but less suitable for homes where residents may have other individual preferences for temperature conditions, and the assessment of temperature may depend on other criteria than in an office environment.

To identify the individual preferences for the case study, the homeowner was interviewed on the preferred indoor temperature and occupancy hours, where two key assumptions were identified. Firstly, it is assumed that over the previous heating season months the occupant is satisfied with the indoor temperature when the heating is on. Secondly, through validation with the homeowner, the temperature can be altered for the analysis. Data was gathered at 30-min intervals for each day over three months. In the case of significant outliers or anomalies, data was cleaned by interpolating the

temperature based on previous and proceeding timeslot.

For each of the months, the cumulative indoor temperature was calculated for each hour of the day. For example, the indoor temperature value at 00:00 h summed for each day (92 days in total) and the mean temperature of that hour was calculated by Equation (4.1):

$$MIT = \frac{\sum_{n=1}^N T_{indoor}}{N} \quad (4.1)$$

Where N is the total number of days for which the data is gathered. The MIT is calculated for every 30 minutes and is outlined in Fig. 3.2. To measure the upper and lower limits of MIT , the standard deviation (Equation (4.2)) is calculated from the mean.

$$\sigma = \sqrt{\frac{\sum |x - MIT|^2}{N}} \quad (4.2)$$

Thus, two times the standard deviation was formulated by multiplying the standard deviation by 2.

The approach outputted the MIT for each hour of the selected months and the associated lower and upper threshold limits for analysis. It is assumed that changes in indoor temperature which are within this temperature bandwidth are acceptable and can be utilised for the demand response strategy. The MIT and associated upper and lower limits are outlined in Fig. 4.2.

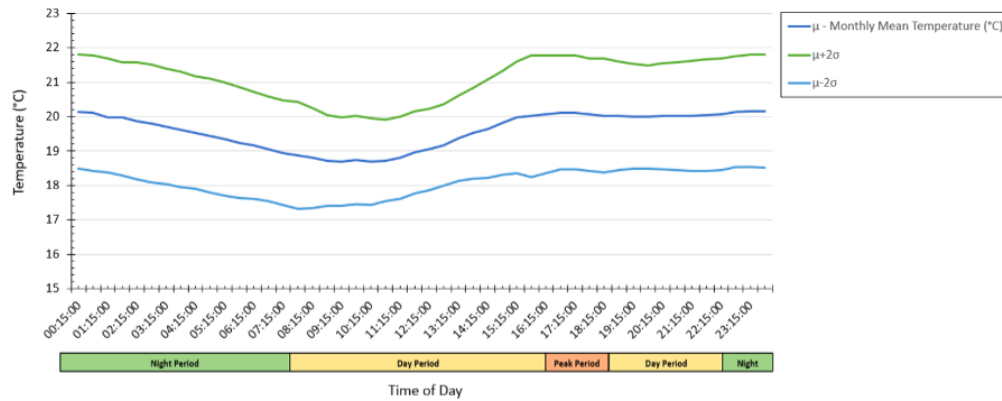


Figure 4.2: Temperature threshold.

The proposed threshold is utilised as part of our DRL environment and

control optimisation in later stages, helping to achieve additional savings without impacting user’s comfort.

4.4 DRL-Based Home Energy Management System

4.4.1 Markov Decision Process and Reinforcement Learning

The Markov Decision Process is basically composed of 4 components: the state space (S), the set of possible actions (A), the probability distributions regarding state transitions ($p(s_{t+1}|s_t, a_t)$) and the probability distribution governing the rewards received ($q(s_{t+1}|s_t, a_t)$). Its main property states that the *“future is independent of the past given the present”*, which means that the current state incorporates all the information regarding past states, hence there is no need of keeping a historical record of actions performed or states.

An MDP task usually can be discretized into time periods, where at each period t the agent occupies a state $s_t \in S$, and then chooses an action a_t from the set of all possible actions within the current state. Performing the chosen action results in a state transition to s_{t+1} and an immediate numerical reward $R(s_t, a_t)$. The state transition probability $p(s_{t+1}|s_t, a_t)$ governs the likelihood that the agent will transition to state s_{t+1} as a result of choosing a_t in s_t . The numerical reward received upon arrival at the next state is governed by $q(s_{t+1}|s_t, a_t)$ and is indicative as to the benefit of choosing a_t whilst in s_t . Solving an MDP leads to an output policy, which contains information about actions versus states, helping the agent’s decisions over the entire learning period.

If the complete environment model is known, the problem can be solved through traditional dynamic programming techniques such as value iteration. However, most real-world problems are not fully observed, which means that the model needs to be either approximated (model-based reinforcement learning), for instance using statistical techniques, or having its value function or policy directly estimated (model-free reinforcement learning), where learners attempt to directly approximate a control policy through environmental interactions [20, 33, 46, 53].

MDP models can be applied to HEMS, however dynamic programming techniques can only be used in complete models. If there is a lack of information about the environment, for instance, rewards or transitions probabilities

missing, a model-free method such as Q-learning [43] can be used to generate optimal policies. Q-learning is part of the temporal difference methods, having the capability of being able to make predictions incrementally and in an online fashion. The update rule for Q-learning is defined in Equation (4.3) and calculated each time a state is reached which is non-terminal.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)] \quad (4.3)$$

Actions are selected according to the policy π and approximations of $Q^\pi(s_t, a_t)$ are calculated after each time interval. Different action selection policies can be chosen, such as *ϵ -greedy* and *softmax*, aiming to balance exploration and exploitation and achieve the best results. Over time the actions selected should converge to the optimal, where the agents consistently choose actions which give the greatest amount of cumulative reward. For instance, the *ϵ -greedy* strategy chooses the best action from the policy most of the time, however the agent explores a certain amount of the time governed by ϵ . Let $A'(s) \subseteq A(s)$, be the set of all non-greedy actions. The probability of selection for each non-greedy action is reduced to $\frac{\epsilon}{|A'(s)|}$, resulting in a probability of $1 - \epsilon$ for the greedy strategy. The discount factor γ , ($0 < \gamma < 1$), determines the importance of future rewards, where a value close to 1 assigns a greater weight to it, while a value close to 0 considers only the most recent rewards. Finally, α is a value lower than 1 that represents the learning rate of value estimates over the learning process.

Depending on the algorithm chosen, the set of estimated Q-values, actions, and states can be represented in tabular form (Q-table) or as part of a function approximator. Although tabular methods present good accuracy, they require continuous updating of the value estimates through repeatedly revisiting the states and choosing actions in the environment. As the size of the state-action space grows, the learning time to converge to an optimal policy also increases. This is known as the curse of dimensionality, where each additional state or action variable added increases the problem size exponentially.

One of the possible solutions is to replace the Q-table, using an artificial neural network to estimate the Q-values of Equation 4.3, which can now be

simplified. The learning rate α is no longer needed in Equation 4.3, as this method has a back-propagating optimiser that already has this function. After removing α , the two $Q(s_t, a_t)$ terms cancel each other. The Equation 4.4 shows the new update rule after the changes. This technique, known as deep reinforcement learning, can manage larger state-action spaces, thus bringing more possibilities when working with a huge number of variables.

$$Q(s_t, a_t) \leftarrow r_{t+1} + \gamma \max_a Q(s_{t+1}, a) \quad (4.4)$$

As the proposed HEMS will manage a variety of variables with a high range of values, and also thinking about future upgrades and additional features we can possibly add to it, we chose an online DRL approach as the core of our solution. The details about the neural network design are described in the following sections.

4.4.2 HEMS Operation and MDP Formulation

The proposed HEMS is composed of the elements described in Sub-Section 4.3.1. The goals are to optimise energy usage and to ensure user's comfort, by controlling the heating and DHW loads. The optimisation process is made by performing actions when PV production is higher and considering the variable setpoint from Sub-Section 4.3.2. The timestep is discretized to 5-min periods, where the system analyses the current state, performs an action, and moves to another state. The states are independent, as they only rely on weather conditions and building dynamics at the current time, thus following the Markov property. The state-space S comprises the following:

- *ot*: Outdoor temperature (°C).
- *it*: Indoor temperature (°C).
- *tk*: DHW tank temperature (°C).
- *pv*: PV production (kW).
- *hr*: Hour of the day.

The DRL algorithm target is to minimise energy consumption, whilst keeping *it* and *tk* temperatures within their respective thresholds, as presented in Sub-Section 3.3. The set of actions A is related to the heat pump

operation modes, which are space heating on ($HEAT_{ON}$), domestic hot water on (DHW_{ON}), and system on hold ($SYSTEM_{OFF}$). After performing an action, the environment will move to a new state and a reward r_t is calculated. The main target can be divided in three minor tasks, where individual rewards were assigned: indoor temperature control ($r1_t$: Equation (4.5)), DHW temperature control ($r2_t$: Equation (4.6)), and energy savings ($r3_t$: Equation (4.7)).

$$r1_t = \begin{cases} 0, & \text{if } it_{min} < it < it_{max} \\ |it - it_{set}|, & \text{otherwise} \end{cases} \quad (4.5)$$

$$r2_t = \begin{cases} 0, & \text{if } tk_{min} < tk < tk_{max} \\ |tk - tk_{set}|, & \text{otherwise} \end{cases} \quad (4.6)$$

$$r3_t = \begin{cases} 0, & \text{if } pv > a_{power} \\ a_{power} - pv, & \text{otherwise} \end{cases} \quad (4.7)$$

Regarding $r1_t$ and $r2_t$, the reward is 0 if the current temperatures are in between their respective minimum and maximum threshold. The it limits can be seen in Figure 4.2, and the tk limits are defined as DHW max. and the maximum temperature drop allowed (Figure 4.1). To increase the system flexibility, the limits chosen are from 40° to 55° . This way, the algorithm can either create a buffer of hot water in case of high PV production, or delay a DHW_{ON} action if production is low at the current state. On the other hand, if $r1_t$ or $r2_t$ is out of the boundaries, the rewards are calculated as the distance of the current temperature compared to their central setpoint. Moving to $r3_t$, the reward is 0 when PV production is higher than the energy spent to perform a specific action, which is on average 1.5 kW for heating and 2.0 kW for DHW, considering our modelled environment. If the PV production is low, the reward is defined as the remaining amount necessary to meet the action power. Finally, the three partial rewards are aggregated following Equation (4.8) along with the constants y , z , and w , added to balance the targets of comfort (indoor and DHW temperatures) and energy savings.

$$r_t = y \times (r1_t + r2_t \times w) + r3_t \times z \quad (4.8)$$

As mentioned previously, the state transitions follow the principles of Equation (4.4), where the values are approximated by a neural network. The steps involved in the simulation and DRL process is depicted by Algorithm 4.1.

ALGORITHM 4.1: HEMS simulation process

```

Initialise Building parameters
Initialise  $Q(s, a)$  arbitrarily
  Repeat (for each episode)
    Initialise  $s$ 
    repeat
      Choose  $a$  from  $s$  using policy from  $Q(\epsilon\text{-greedy})$ 
      Take action ( $a$ )
      Update building states ( $s_{t+1}$ )
      Calculate reward ( $r$ )
       $Q(s_t, a_t) \leftarrow r_{t+1} + \gamma \max_a Q(s_{t+1}, a)$ 
       $s \leftarrow s_{t+1}$ 
    until  $s$  is terminal
  end

```

4.4.3 Neural Network Design

The neural network designed to estimate the Q-values is similar to the one presented by Wei et al. [31]. It contains an input layer, two hidden layers, and an output layer, as can be seen in Fig. 4.3. First, each of the variables received from the current state is pre-processed and their values are normalised on a scale from 0 to 1 through the min-max method (Equation (4.9)). This practice helps the algorithm to achieve a more stable learning process, as it does not have to deal with numbers with different ranges. For instance, in this experiment outside temperatures range from approximately 4°C to 20°C, while PV production ranges from 0 to 2 kW.

$$value' = \frac{value - min}{max - min} \quad (4.9)$$

Next, there are two hidden layers, where the Rectified Linear Unit (ReLU) is the activation function chosen, followed by the output layer with possible

actions. The losses are calculated by the mean squared difference between the current output and the ideal target values. Moreover, we use a *Gradient Descent Optimiser*, which adjusts weights in the neural network to minimise these losses. The proposed algorithm also incorporates an ϵ -greedy policy to improve efficiency [20, 31, 53].

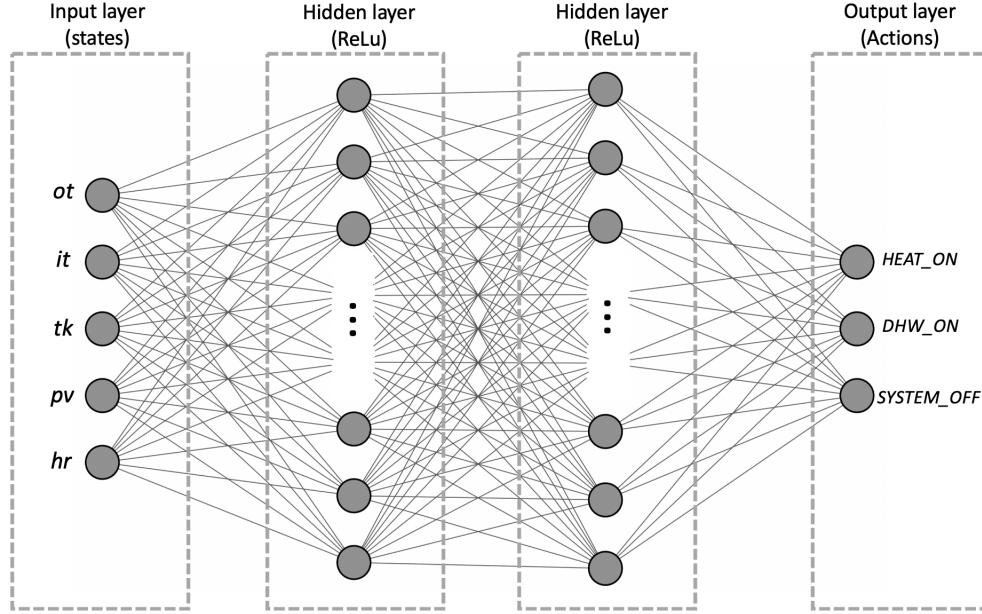


Figure 4.3: Neural network design.

4.5 Results

4.5.1 Experiments and Methodology

To validate the proposed DRL algorithm we simulated our test environment using weather data from Weatherbit.io [47], considering information from 1 May 2020 to 31 December 2020, and historical data about PV production from the mentioned Irish residence in the same period. The first step was to create a baseline, so the Mitsubishi’s rule-based operational mode from Sub-Section 4.3.1 was replicated, where the controller has to keep indoor temperature between 19°C and 22°C, and DHW temperature between 40°C and 55°C. This baseline is used in later stages for evaluating the efficiency of our method. The second step consisted of adding the dynamic indoor setpoint from Sub-Section 4.3.2 to the rule-based method. Finally, the DRL-based algorithm was deployed, aiming to get benefits from the dynamic setpoint and PV self-consumption optimisation.

Over the DRL deployment, a number of different parameters have been tested to determine the best configuration for handling comfort and energy savings. The final NN architecture is composed of 16 neurons in each of the hidden layers, with $\alpha = 0.0001$, $\gamma = 0.95$ and $\epsilon = 0.05$. The y , z , and w reward balancing parameters were defined as 1000, 3.5, and 3.5 respectively. The learning process and algorithm stability happen over the first month of simulation, as can be seen in Fig. 4.4. At this stage, the performance achieved in terms of comfort is similar to the rule-based method, but it is also performing actions aiming to reduce energy imported from the grid.

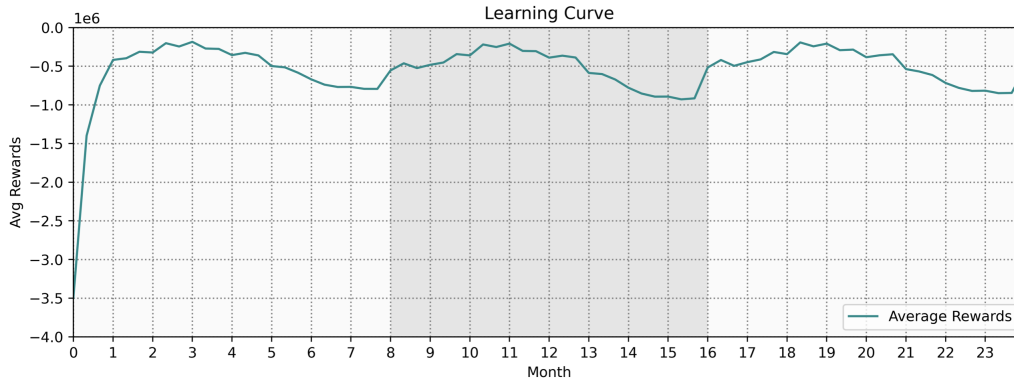


Figure 4.4: Learning curve.

After every 8 months of simulation (May to December), the episode is finished, and the environment is restarted. Fig. 4.4 shows the average accumulated rewards per month over 3 episodes. Note that the average rewards decrease from the middle to the end of each episode, and then start to increase in the subsequent months. This is related to the outside temperatures, because rewards depend on the number of actions executed. Colder temperatures found in winter months means that more heating actions are needed to keep user's comfort, hence lower rewards. On the other hand, warmer months need less heating actions and also have higher PV production, which makes rewards higher.

4.5.2 Comfort Analysis

This section shows the overall comfort performance after the full implementation of the DRL algorithm with a dynamic setpoint. Results show that

from June to December both indoor and DHW temperatures are kept in between their respective setpoints most of the time, with the indoor setpoint being violated less than 40 timesteps over the seven months, and less than 420 for DHW. The number is higher for DHW because the heating dynamic during the actions is different. Indoor heating is slow, which makes precision easier to be achieved. On the other hand, DHW heating is very fast, it needs less than 20 min to exceed the maximum setpoint limit, hence if the action of heating water is kept for one more timestep than necessary, there is a higher probability of surpassing the limit. But even in this condition, this temperature deviation represents less than 1% of the timesteps, and it does not exceed 2°C above the maximum allowed. Fig. 4.5 shows the seven months of temperature behaviour accumulated in an hourly granularity. Note that from the indoor temperature chart it is possible to see bigger bars from 11 am to 6 pm, which means the algorithm is exploring more of the setpoint limits due to a higher PV production activity. This is not possible to infer from the DHW chart, because its action occurs on average two times a day only, as the tank temperature decrease is slow.

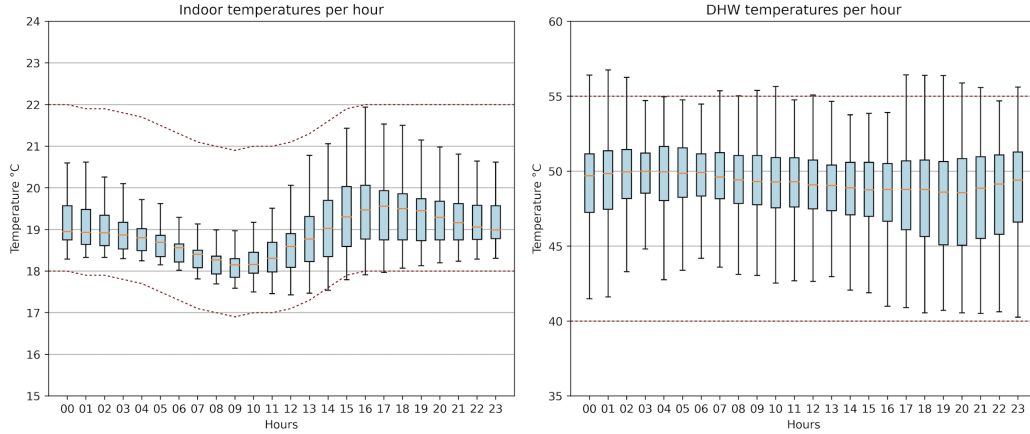


Figure 4.5: DRL-based control: Temperature behaviour.

4.5.3 Energy Analysis

Having achieved comfort for both indoor and DHW, the second goal of our proposed DRL-based HEMS is to optimise PV self-consumption, by prioritising performing actions when production is higher. The dynamic setpoint brings additional flexibility during the decision making process. For

instance, the algorithm can opt to anticipate the action and heat the space or tank creating a heating buffer, or it may postpone the action and save energy. There are two main benefits of this optimisation: consumers can reduce their energy bills and also help utilities to balance the grid, through a demand-response program of load shifting.

To assess energy savings, we first compared the rule-based methods with and without a dynamic setpoint by estimating the net energy consumption (action power minus PV production), which was assumed to be zero if production is higher than consumption. Although PV production is part of the calculus, both methods do not consider any optimisation in this regard, which means the actions are taken exclusively to keep comfort. As a result, the dynamic setpoint approach achieved up to 11% of savings over the months, with a 5.7% reduction average.

In the second analysis, we compared the DRL-based algorithm with a dynamic setpoint against the baseline. The average savings increased to 8%, reaching up to 16% of reductions over the summer months. There is a small energy reduction in November and December, mainly because PV production is low in those months. Also, outdoor temperatures are lower, hence more actions over the day are needed to keep indoor comfort, which reduces flexibility. Table 4.2 summarises energy saving across the different algorithms.

Table 4.2: Algorithms comparison.

	Rule-based Static	Rule-based Dynamic	DRL-based Dynamic
Month	kWh (Baseline)	kWh (Reduction %)	kWh (Reduction %)
June	146.15	130.01 (-11.5%)	126.55 (-13.4%)
July	105.08	94.10 (-10.4%)	87.85 (-16.4%)
August	111.80	100.47 (-10.1%)	93.28 (-16.6%)
September	140.76	127.79 (-9.2%)	126.07 (-10.4%)
October	207.06	196.18 (-10.1%)	190.28 (-8.1%)
November	247.96	245.88 (-0.8%)	241.06 (-2.8%)
December	257.95	253.34 (-1.7%)	253.99 (-1.5%)
Total	1,216.78	1,147.79 (-5.7%)	1,119.12 (-8.0%)

To evaluate the DRL-based algorithm capability of adapting its control actions according to PV production, we compared the accumulated number of actions $HEAT_{ON}$ and DHW_{ON} performed over the seven months, grouped by hour, as can be seen in Figure 4.6. The rule-based algorithm

(dotted-red line) has actions well distributed across the day. After applying DRL with PV optimisation (blue line), a number of actions that were originally in ranges from 2 AM to 10 AM and from 5 PM to 9 PM moved to the period from 10 AM to 5 PM. This load shifting represents 10.2% of more activity where PV production is higher.

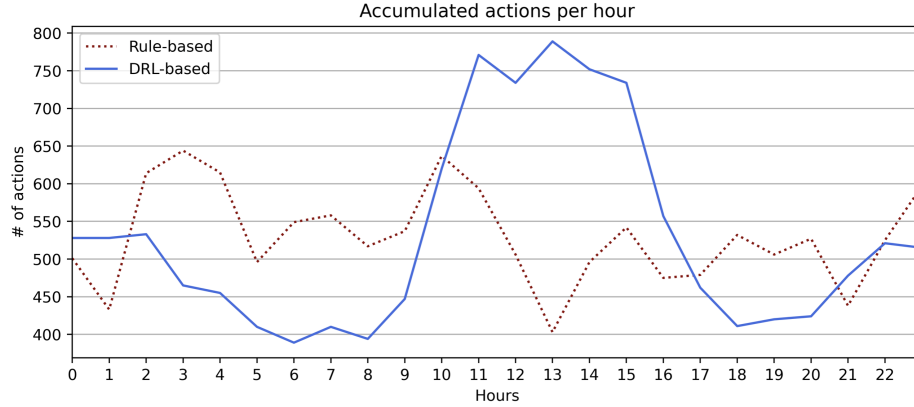


Figure 4.6: Actions distribution.

PV self-consumption optimisation is important to apply in situations where the policy is not favourable to the consumer. For instance, in this use case the PV system is connected to the utility grid and, if the energy is not consumed, the excess amount goes to the grid, but the user does not receive credits or incentives for that. In this regard, the proposed DRL-based algorithm was able to use up to 9.5% more of PV energy over the months than the rule-based one, hence consuming less from the grid.

In summary, the proposed algorithm was able to optimise PV self-consumption by choosing actions when production was higher, as shown in Fig. 4.7. The grey bars represent the accumulated number of actions per hour over the 8 months of the experiment. The green line shows the average PV production and, similarly to the previous analysis, the higher concentration of actions occurs when production is high. Finally, the orange dotted line is the average energy consumed from the grid. Even having more actions from 10 AM to 5 PM, the imported energy is lower compared to the other periods. There is also a slight increase of actions very early in the morning due to the DHW cycle, which happens on average twice a day. For instance, if DHW is turned on around 3 PM, it will probably turn on again 2 AM to keep the temperature to the specified setpoint.

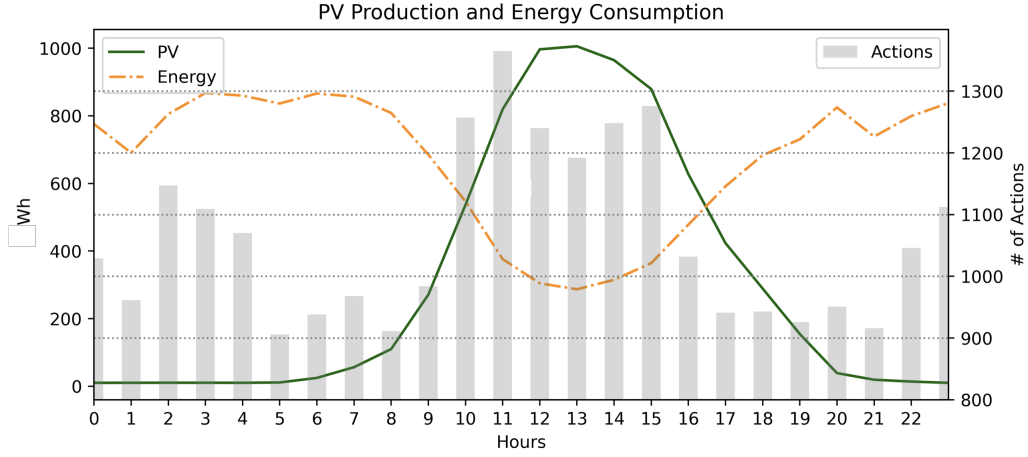


Figure 4.7: PV production and energy consumption.

4.5.4 Balancing Comfort and Energy Savings

An important role of the proposed DRL-based control is to balance users' comfort and energy savings, and this process is ruled by the z value of Equation 4.8. To define y , z , and w reward parameters, first we removed $r2t$, $r3t$, y , z and w . As a result, the algorithm only focuses on keeping the indoor temperature in between the setpoints. Then, we added $r2t$ and the parameter w . We started to change the value of w , verifying the impact on the indoor and tank temperatures, until both were in between their respective setpoint ranges. This was necessary because indoor and tank temperatures dynamics are different. Moreover, the setpoint operation zone is 4 degrees for indoor and 15 degrees for DHW, affecting directly the weight of the rewards. After this step, the algorithm achieved a similar performance to the rule-based method in terms of reaching temperature targets. Finally, $r3t$, y and z were added to the formula. The value of y only equalises the temperature rewards to a similar weight level of energy savings. Finally, we started to change z to find the best balance between temperature and energy savings targets.

A z number closer to zero represents an algorithm focused strictly on comfort, and in that case it will not try to optimise energy usage according to PV production. As can be seen in Figure 4.8, increasing z makes energy consumption to decrease, but after a certain limit the comfort matters start to be sacrificed. The purple line represents the percentage of timesteps out

of indoor and DHW setpoints for different z values, considering accumulated values from June to December. The red line shows the total energy consumption for the same period and specified parameters.

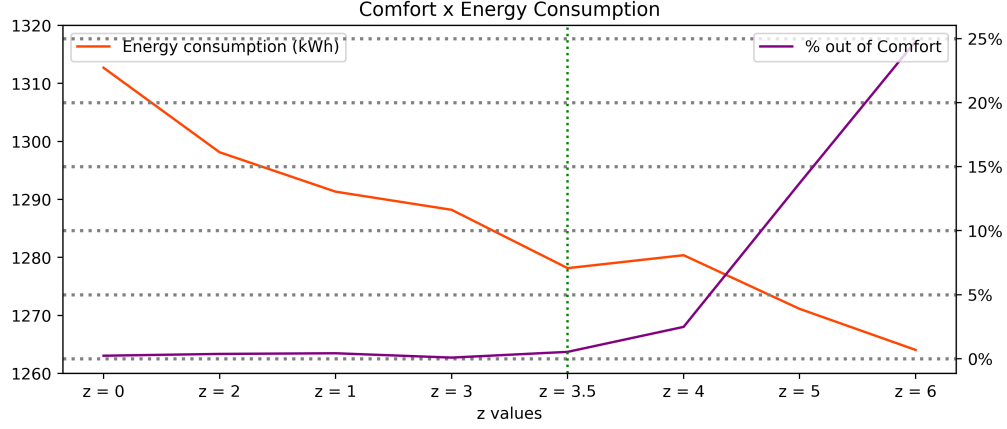


Figure 4.8: Balancing comfort and energy savings.

As our first target was to achieve a similar comfort performance of the rule-based control, a z value of 3.5 was chosen. The time out of comfort in this configuration is around just 0.5%, in addition to outperforming inferior z values in terms of energy savings performance. Note that a z greater than 3.5 achieves more energy consumption reduction, but the time out comfort starts to increase quickly, reaching 25% of discomfort rate for a z equal to 6. In a common sense with the user, another z could be selected targeting greater energy savings, such as 4.5, where time out comfort is around 5% and the energy consumption is lower when compared to a z of 3.5. Another option would be also changing the w parameter of Equation (4.8), which aims to balance the importance level between indoor and DHW temperatures.

4.6 Conclusions

This work presented a new DRL approach for a HEMS control, which includes a PV self-consumption optimisation and a dynamic setpoint definition for indoor temperature. The primary goal was to keep user's comfort, followed by energy savings and load shifting. Although our algorithm architecture and use case differs from previous works in many ways, the energy savings of up to 16% achieved is in line with values found by authors in [20, 33, 46, 49, 53]. Apart from bringing savings, PV self-consumption

optimisation also contributed to 10.2% of load shifting, by anticipating or delaying heating actions. Furthermore, right after the first month the level of comfort achieved by the proposed algorithm is nearly optimal, which means users would not experiment more than 1% of temperatures out of the specified setpoints range. The proposed DRL-based control also allows balancing energy savings and comfort according to the user's preferences.

Chapter 5

Transfer Learning Applied to DRL-Based Heat Pump Control to Leverage Microgrid Energy Efficiency

The work outlined in this chapter was published in:

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Abstract. Domestic hot water accounts for approximately 15% of the total residential energy consumption in Europe, and most of this usage happens during specific periods of the day, resulting in undesirable peak loads. The increase in energy production from renewables adds additional complexity in energy balancing. Machine learning techniques for heat pump control have demonstrated efficacy in this regard. However, reducing the amount of time and data required to train effective policies can be challenging. This chapter investigates the application of transfer learning applied to a deep reinforcement learning-based heat pump control to leverage energy efficiency in a microgrid. First, we propose an algorithm for domestic hot water temperature control and PV self-consumption optimisation. Secondly, we perform transfer learning to speed-up the convergence process. The experiments were

deployed in a simulated environment using real data from two residential demand response projects. The results show that the proposed algorithm achieved up to 10% of savings after transfer learning was applied, also contributing to load-shifting. Moreover, the learning time to train near-optimal control policies was reduced by more than a factor of 5.

5.1 Introduction

Domestic hot water accounts for approximately 15% of the total residential energy consumption in Europe according to a report from Eurostat [3]. It comprises water heating for uses other than space heating, such as showers or taps, and most of its usage happens during specific periods of the day, such as early in the morning and evenings, resulting in undesirable peak loads. Meanwhile, studies from the International Energy Agency [4] show that energy production from renewables are increasing all the time, representing almost 20% of the total generation in the EU, and it is expected to expand to 32% by 2030 as stated by the European Commission [48]. Energy from renewable sources, such as PV and wind, rely on weather conditions, and coordinating demand and production is one of the challenges arising from this change in the energy matrix. This new scenario is part of the smart energy system concept, which includes energy efficiency and better management of resources in an integrated manner. It is relevant in terms of achieving a strategy of 100% renewable envisaged for the future, as explored by Mathiesen et al. [103].

Demand Response solutions, which have been commonly applied in the industrial and commercial sectors, have started to involve residential homes, as can be seen in recent projects around the world, such as the EU Horizon 2020 projects RESPOND [9] and REACT [10]. DR solutions empower consumers to play a significant role in grid operation, by reducing or shifting loads during peak load, in response to any incentives or time-based rates [67]. The DR actions can be performed manually or automated, through the use of new technologies like the Internet-of-Things. Electric heat pumps are very suitable for DR because of their high electricity consumption and flexibility. In a common residential application, heat pumps can be used for HVAC and DHW applications. The first aims to provide indoor thermal comfort and air quality, while the second improves the user's life in specific aspects, such as

shower and hot water taps. From the control strategy perspective, electric heat pumps can be managed and turned on at different times of the day without affecting the needs of the end-user. They are an alternative option to fossil fuel based systems, thus contributing to the reduction of carbon emissions. Choosing the right time to use electric heat pumps is crucial in terms of cost savings, which should prioritise periods where electricity price is low or on-site energy production is high. Moreover, the water tank capacity in residences is limited and there are heat losses over the day, hence water heating has to be performed more than once to keep the temperature at the desired setpoint, increasing the need for a more complex control algorithm. For instance, the temperature drop can be around 0.075°C per 5-min interval for the operational range of $40\text{--}55^{\circ}\text{C}$, slightly reducing this value until reaching the indoor temperature level, depending on the type of tank insulation and water usage.

The utilisation of heat pump systems combined with renewables has been an object of study in the past years. Hedegaard et al. [101] stated that individual heat pumps and heat storage could play an important role in the integration of wind power. Their research shows that heat pumps can increase wind power utilisation, thus reducing the excess electricity production by up to 8% and 19%, for systems with and without heat storage, respectively. Protopapadaki and Saelens [102] also studied the interaction of heat pumps and renewables, but with a focus on solar generation. Results from their simulations confirmed that thermal storage offers the ability to shift electricity demand, leveraging the consumption of PV energy generation.

Rule and schedule-based approaches are the most common heat pump control methods, which allow users to set a specific temperature setpoint for the water tank or pre-program the time to activate the system, respectively. Nonetheless, the use of adaptative control solutions is gaining popularity due to their capacity of improving energy and cost savings while also allowing for load shifting capabilities, for instance considering optimisation by local renewable self-consumption and dynamic tariffs. Machine learning adaptative methods have demonstrated efficacy in this domain, as proven by the authors in [68–73], but there are some issues such as the lack of data available or the amount of time needed to train optimal policies.

Some DHW controllers are based on reinforcement learning, which is a sub-field of machine learning. Here the controller agent learns by interacting

directly with the environment in a trial and error manner, thus improving and learning near-optimal policies according to the previous experiences after a large number of trials, where no previous knowledge is needed. For a particular state, the agent receives a reward after performing an action, which indicates whether the chosen action was good or not. For instance, in DHW control the agent would be penalised if the tank temperature drops to less than a specified setpoint, or it would be rewarded if water was heated when the energy price at that specific time is low. The goal of the RL agent is to achieve a cost minimisation while ensuring the specified parameters of temperature over time.

Achieving optimal control policies may require a huge number of trials, where the learning agent will be exposed to a number of conditions and verify the consequences after performing different actions. Transfer learning techniques in contrast have been showing efficiency in reducing learning time, by reusing knowledge from another agent. For example, if an agent already has experience in a particular state, the hypothesis is that this can be shared with a second agent in a similar environment, hence avoiding the need to visit the all states a number of times to be able to act optimally. This was proved by authors in [53], where a significant speed-up in learning times was achieved in an HVAC control system. Apart from the building efficiency domain, TF has also demonstrated efficacy in other domains, such as computer vision [21–23], games [24, 25] and 3D environments [26, 27].

Considering that some houses from a specific community are equipped with similar heat pumps and PV systems, hence sharing common environmental and heating dynamics, this chapter investigates the use of transfer learning applied to a deep reinforcement learning-based heat pump control system to leverage energy efficiency in a microgrid, focusing on DHW. The thermal properties of the hot water tank will not differ much between the houses, as they are built using the same material and insulation levels. Also, external temperatures will not significantly vary among the houses, and PV production will be also similar if the houses are geographically located close to one another, although some small variations can happen depending on the position of the installation. The main contributions of this chapter include:

- The development of a model-free DRL algorithm for a DHW tank temperature control for a number of houses in a microgrid, aiming to reduce energy consumption from the grid by optimising the usage

of PV energy production and enhancing financial savings through a time-of-use tariff scheme.

- The utilisation of transfer learning techniques for a DRL-based DHW control, allowing for much faster learning rates, and comparison of its convergence speed with models without transfer learning.

The rest of this chapter is organised as follows: *Related Work*, which shows the related works regarding energy management systems and relevant transfer learning approaches. *Problem Formulation* provides information about the environment structure and the principles of ML applied in this research. *Experimental Setup* explains the variables and agent’s definition for the test cases, including also the algorithms. The *Results* section presents all the relevant outputs of our experiments, showing the performance of the algorithm and comparing the methods. Finally, the *Conclusions* section recaps the main points of the chapter.

5.2 Related Work

This section provides an overview of residential smart control for demand response, showing the current control methods and also detailing recent works in this subject, with a focus on water heating and reinforcement learning. Moreover, some examples of transfer learning applied to HVAC control and other domains are presented, aiming to show the advantages that this technique can bring when applied to the DHW domain.

5.2.1 Residential Smart Control for Demand Response

Demand response solutions can be applied in different types of residential assets, such as HVAC, DHW, smart appliances (e.g., dishwasher and washing machines), and electric vehicles. There is a variety of control methods for comfort and energy management, which can sometimes use renewable energy production, batteries, or dynamic prices as part of the optimisation process. The focus of this research is on RL model-free methods, where the agent controller does not need complete information about the environment, as it attempts to directly approximate a control policy through environmental interactions. Model-based methods can face significant bias, as unknown dynamics may sometimes not be complete modelled, which will impact the

policy effectiveness [66]. On the other hand, model-free methods are more adaptive to variations in the environment, likely to perform better by approximating the missing model in a complex task. Other methods out of the reinforcement learning domain include fuzzy control [35,36], and model predictive control [37,59,74], that can be very accurate, but sometimes costly and time-consuming as detailed models of the building are required.

Vázquez-Canteli and Nagy [29], in their review about reinforcement learning for demand response, identified more than one hundred studies across different applications in which only approximately 15% were about water heating (DHW, thermal storage, heat pump, and district heating). Their research shows that Q-learning is the most popular learning method in the energy and buildings context, which can also be seen in the reviews from Mason and Grijalva [49] and Han et al. [52]. Their studies also found that the utilisation of deep Q-learning has been increasing since 2015, mainly because its advantage of handling large state-action spaces with more efficiency, hence reducing learning time. Considering that in this research we will be handling multiple devices (heat pumps and PVs) from a number of houses and also thinking about future expansion of this study, we chose a deep RL approach combined with transfer learning techniques to enhance learning speeds.

The efficacy of reinforcement learning applied to DHW and residential PV production self-consumption can be seen in research carried out by De Somer et al. [70] and later on by Soares et al. [75], where the fitted Q-iteration control algorithm achieved on average 14% and more than a 20% increase in self-consumption, respectively, compared to a thermostat control method. Both works used a model-based reinforcement learning approach, where the transition function of the control strategy was learned from historical data. In complex environments, approximating the missing model by learning the transition function from the historical data will not be adaptive to changes beyond the scope. For instance, weather conditions may vary from year to year, so the control strategy learnt from a data set from the previous year may not represent the new reality. Al-jabery et al. [71] applied Q-learning for a number of different domestic electric water heaters models, targeted with reducing energy consumption by using the time-of-use tariff scheme as part of the optimisation process. Similarly, Ruelens et al. [69] reached a 15% reduction in the cost of energy using fitted Q-learning and

dynamic prices. In another work, Ruelens et al. [83] applied a model-free batch RL algorithm using a Monte Carlo and fitted Q-iteration approach in different experiments involving an electric water heater and heat-pump thermostat. The results showed that RL can be an option to model-based controllers, with a decrease in the total electricity cost of up to 19%. Kazmi et al. [68] applied deep RL to a set of 32 houses, reducing energy consumption used for water heating by approximately 20%. Kazmi et al. [84] applied a multi-agent RL approach to accelerate learning of a large-scale pilot consisting of 50 thermostatically controlled loads from different houses, where the proposed solution can potentially achieve savings up to 300 kWh annually per household (30% of reduction). Finally, Chen et al. [85] proposed an RL control method for DHW and space heating, using a differential Model Predictive Control policy, encoding domain knowledge on planning and system dynamics. The agent was pre-trained on historical data and once the controller behaviour was learnt, a policy gradient algorithm was applied, reaching 6.6% of energy savings and 16.7% of cooling demand reduction.

Recent works, also about deep reinforcement learning applied to heat pumps, but not with a focus on DHW, also showed the efficiency of such algorithms in this domain. For instance, research articles [32,86–96], which are mostly from the HVAC and space heating domain, use deep RL approaches with different targets. For instance, Vázquez-Canteli et al. [91] considered a multi-agent approach for a number of connected buildings, which reduced daily peak load by 15% by coordinating different loads. Kurte et al. [89] and Christensen et al. [95] used a price-responsive model as part of the optimisation, where the agent had to prioritise control actions when the cost of electricity was low.

Regarding convergence time, Yang et al. [76] proposed an RL approach with tabular Q-learning and batch Q-learning, applied to a heat pump control combined with a photovoltaic-thermal solution, outperforming the rule-based model by 10% after the third year of simulation. Patyn et al. [77] compared different neural architectures models for reinforcement learning applied to heat pump control, which outperformed standard thermostat controller and shift loads after 20-25 days. This is one of the motivations of our research: we aim to apply TF to reduce learning time, thus achieving near-optimal policies faster and bringing additional savings to users.

5.2.2 Transfer Learning for Reinforcement Learning

Transfer learning refers to the use of experience learnt by performing one task to improve another related, but different, task [38]. The objective is to increase convergence speed, as instead of learning from the beginning without prior knowledge, the agent integrates value function estimates from other agents to perform the tasks, achieving better results faster as a result of the knowledge sharing. Taylor and Stone [38] describes in their review five different types of transfer learning approaches, where our research is classified as part of the “allowed tasks differences group”, which is applied to methods that have the same state variables and actions. For instance, in our experiments, the agent’s representation of the world, which includes PV energy production metering, temperature sensing, and heat pump actuation all remain the same, while state values and/or actions can change. Furthermore, according to da Silva and Costa [78], our TF method is categorised as Inter-Agent transfer, in which knowledge is shared with another agent with a very similar characteristic, and this agent is able to merge the transferred knowledge with its own experience.

Although TF studies have been carried out and have demonstrated success in other domains, to the best of our knowledge this is the first time it has been applied to a model-free DRL-based heat pump control in a microgrid, leveraging PV self-consumption and reducing energy through ToU tariffs. In recent work from a close domain, Lissa et al. [53] proposed an RL algorithm which can transfer HVAC control policies between agents from different geographic locations. The results showed that the learning time to train near-optimal control policies was reduced by more than a factor of 6, compared to models without TF. Also in the HVAC field, Xu et al. [79] applied TF to a DRL-based HVAC control, transferring knowledge from different zones, equipment and locations, thus reducing the training time, energy cost, and temperature violations. Zhang et al. [96] proposed an RL approach with TF to reduce the training cost of an RL control policy for smart homes with different appliances and user preferences, showing that TL can effectively reduce the training time of a new policy if the new home is similar to the benchmark home. Finally, authors in [41, 80] also explored TF, but with focus on building predictive models, predicting energy consumption, temperature and humidity.

5.3 Problem Formulation

5.3.1 Test Environment

The experiments were deployed in a simulated environment using real data from two residential demand response projects, RESPOND [9] and REACT [10], in which one of their case studies was realised in the Aran Islands, Galway, Ireland. Energy supply in small islands can be challenging due to its geographic characteristics, hence the community relies on energy imported through submarine cables or fossil fuels. Both can be very costly, the first is expensive to deploy or upgrade if needed, and the second is not easy for logistical reasons and it is also not environmentally friendly, due to its carbon emissions. Some of the houses in the island have individual PV energy systems, which helps to reduce the grid load. However, a smart controller is needed to coordinate demand and production, which is the aim of RESPOND [9] and also the subject of study of REACT [10] in a number of islands across Europe.

Our study considers as reference five houses that have been recently upgraded in the Aran Islands, all equipped with an 8.5 kW Mitsubishi Electric Ecodan heat pump along with a PV panel array consisting of 8 panels, with a total nominal power of 2 kWp. For each house, the heat pump is connected to a 200-litre cylinder (tank) which is used to store hot water. The main characteristics of the cylinder can be seen in Table 5.1, extracted from the manufacturer’s specification [81]. These features rule the dynamics of the simulations in terms of water heating and temperature losses.

Table 5.1: DHW tank characteristics.

Domestic Hot Water tank	
Volume	200 L
Material	Duplex 2304 stainless (EN10088)
Time to raise DHW tank temp 15 – 65°C	22.75 min
Time to reheat 70% of DHW tank to 65°C	17.17 min
Heat loss	1.91 kWh/24h

The standard DHW mode operates automatically based on the upper and lower limits set by the user. For instance, if the user sets the maximum

temperature to 50°C and a maximum temperature drop of 10°C the hot water is automatically heated once the tank temperature drops to 40°C, until it achieves 50°C again, where this cycle restarts [60]. This rule-based method will be used as a comparison during the experiments.

Our final test environment is a python application deployed from a detailed building model. Sensors were installed in the houses to measure indoor temperature °C, total electricity consumption (kWh), total PV production (kWh) and heat pump electricity consumption (kWh). The construction characteristics obtained from a site survey, such as dimensions and u -values, along with data collected from each of the sensors, were used as a reference for developing a detailed and calibrated white-box model, using the IESVE software. Further details on each of the u -values and measurements can be found in [94]. The tank dynamics of Table 5.1 are also included as part of the environment. Finally, we added historical data from the PV sensors of the residences with a 15-min resolution, acquired in May 2020. In summary, the test environment reads the current state every 15 minutes and estimates the next DHW tank temperature values, considering the action taken by the RL algorithm, developed with TensorFlow, and the dynamics established previously.

5.3.2 Markov Decision Process and Reinforcement Learning

The main goal of a reinforcement learning algorithm is the solution of the Markov Decision Process. Its foremost property states that given the present, the future is independent of the past. It means that the current state concisely represents all the history, hence there is no need to keep previous records. Board games, such as chess or checkers are a good example of this property, in which the current configuration of pieces on the board at any stage of the match is sufficient for a player to make a decision on the next move. The MDP is composed of 4 components:

1. Environment state space (S).
2. Total action space (A).
3. Probability distributions of state transitions ($p(s_{t+1}|s_t, a_t)$).
4. Probability distribution governing the rewards received ($q(s_{t+1}|s_t, a_t)$).

Overall, an MDP task can be discretized into time periods, where at each period t the agent occupies a state $s_t \in S$, and then chooses an action a_t from the set of all possible actions within the current state. The execution of the selected action induces an environmental state s_{t+1} and results in the allocation of a reward $R(s_t, a_t)$. The state transition probability $p(s_{t+1}|s_t, a_t)$ governs the probability that the agent will transition to state s_{t+1} as a result of choosing a_t in s_t . $q(s_{t+1}|s_t, a_t)$ represents the expected reward received by the agent after transitioning from state s_t to s_{t+1} by executing action a_t . Solving an MDP results in the output of a policy, which is a mapping from states to actions guiding the agent’s decisions over the entire learning period.

Specific problems where the model is complete and fully observable can be solved through traditional techniques such as dynamic programming. However, the vast majority of real-world problems cannot be fully observed, as part of the environment may be unknown. To solve this issue, the possible solutions are to estimate the model by using statistical methods (model-based reinforcement learning) or by directly estimating its value function or policy (model-free reinforcement learning), where learners attempt to directly approximate a control policy through environmental interactions [53, 68–73]. Buildings and heat pumps can be modelled as an MDP if the complete information about the environment is available. However, in the case of missing information, dynamic programming cannot be applied to generate an optimal or near-optimal policy. As an option, the missing model may be either approximated by a model-based reinforcement learning algorithm, or directly have its policy or value function estimated by a model-free reinforcement learning technique, such as Q-learning [43]. This method is part of the temporal difference suite of methods and is capable of making predictions incrementally and in an online way, by bootstrapping estimates to estimates. For each time where a state is non-terminal, the update rule in Equation (5.1) is calculated.

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha[r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)] \quad (5.1)$$

Actions are selected according to the policy π and approximations of

$Q^\pi(s_t, a_t)$ are calculated after each time interval. To balance exploration and exploitation and achieve the best results, an action selection policy, for instance *ϵ -greedy*, is employed to choose actions. The *ϵ -greedy* strategy chooses the best action from the policy most of the time, however the agent explores and choose sub optimal actions a certain amount of the time ruled by ϵ . This is to ensure that the agent gets a wide amount of experience in order to make the best choices possible. After a number of trials, the agent should converge to a near-optimal operation, which means it will choose the actions that leads to the highest reward. In our experiments, a near-optimal policy means that the agent achieved the comfort standards of a rule-based controller and managed to do this at the lowest electricity cost possible, when PV production is high and/or the tariff is low. The importance of future rewards is governed by the discount factor γ , situated in a range from 0 to 1. Values close to 1 assigns a greater weight to it, while a value close to 0 considers only the most recent rewards. Finally, α is a value lower than 1 that represents the learning rate of value estimates over the learning process.

Overall, tabular Q-learning methods need to interact with the environment and revisit states a number of times in order to learn a good policy. The learning time to converge to an a near-optimal policy depends on the state-space size, where for each additional state or action included, the problem size can increase exponentially, which is known as the curse of dimensionality. The effect of this issue can be mitigated against by replacing the Q-table with a function approximator, using for example an artificial neural network to estimate the Q-values using Equation (5.1). Mnih et al. [100], followed by Wei et al. [31], used this method, where the Q-value estimates for all outputs can be calculated by performing one forward pass (inference) in the neural network. The loss function is calculated as the mean squared error between the target Q-value and the inferred output of the neural network, and the weights of the neural network are updated using a gradient descent method. As this method has a built-in back-propagating optimiser for the learning process, the learning rate α is no longer needed in Equation (5.1). Hence, both $Q(s_t, a_t)$ terms cancel each other, resulting in the simplified Equation (5.2).

$$Q(s_t, a_t) \leftarrow r_{t+1} + \gamma \max_a Q(s_{t+1}, a) \quad (5.2)$$

For the purposes of this work we utilise this method, called deep Q-learning, as it can manage larger state-action spaces and bring more possibilities of increasing learning speed by estimating values.

5.4 Experimental Setup

This section describes the 3 experiments proposed in our research, showing their algorithms and reward structure. Overall, all of them have the same common target, which is to achieve financial savings by using the heat pump to heat the hot water tank when the cost of electricity is low, which happens either when PV production is high, or the tariff applied at that specific time is reduced. As the controller has also to keep the tank temperature between the specified setpoints set by the user, this is a multi-criteria learning RL problem. Starting with the electricity cost (Ec), we define it as the cost to perform an action at a time t , calculated using Equation (5.3). In summary, it is the difference between the energy consumed by the heat pump and PV production, multiplied by the tariff at a specific time.

$$E_c = (HP_{power} - PV_{prod}) * Tariff \quad (5.3)$$

For instance, let's assume that a heat pump consumes on average 2 kWh when heating the water and the tariff cost is 0.0915 EUR/kWh for off-peak (from midnight to 9 am) and 0.185 EUR/kWh for other periods [82]. Considering also the average PV production per hour of May 2020 from one of the houses, Figure 5.1 shows the average electricity cost per hour to activate the heat pump for DHW purposes. The line represents the energy to be imported from the grid, considering the heat pump is switched on (total HP consumption minus PV production) times the cost of electricity at the specific hour. Note that in this example the best time to use the heat pump is from midnight to 9 am (reduced tariff) and from 1 pm to 6 pm (high PV production). In theory, the aim of the control algorithm in this regard is to operate within these ranges. This example shows the monthly average;

however, the near-optimal operation times will vary across the days, hence the algorithm will need to adapt itself according to the circumstances.

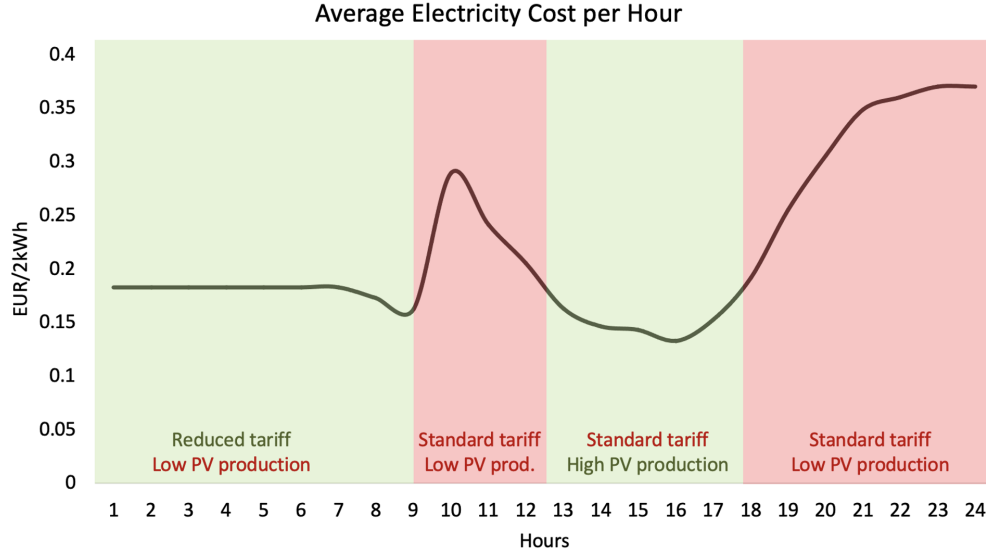


Figure 5.1: Average electricity cost (May 2020).

Moving to tank temperature control, the agent’s performance is measured according to the difference of the actual tank temperature and the lower temperature or maximum temperature in the action zone, as can be seen in Equation (5.4). The action zone ranges from the minimum temperature set by the user and the average of the upper and lower setpoints. For example, if the minimum and maximum tank temperatures are defined as 40°C and 50°C, respectively, the action zone range will be from 40°C to 45°C. The reason for limiting the action zone is to avoid exceeding the maximum temperature setpoint, as the heating process happens considerably fast (see Table 5.1). On the other hand, as the tank is well-insulated, heating losses are slow, hence an action zone of 5°C gives the algorithm enough flexibility to perform actions.

$$\Delta T = \begin{cases} |T_{actual} - T_{zone.min}|, & \text{if } T_{actual} < T_{zone.min} \\ |T_{actual} - T_{zone.max}|, & \text{if } T_{actual} > T_{zone.max} \\ 0, & \text{otherwise} \end{cases} \quad (5.4)$$

The timestep is discretized to 15-min periods, where the system analyses the current state, performs an action, and moves to another state. The states are independent, as they only rely on the heat pump and PV production conditions at the current time, hence following the Markov property. The

state-space S comprises the PV production, tank temperature and hour of the day. After performing an action, the environment will move to a new state and receive a reward r_t , which is 0 if the action is water heating “on” and the temperature is below T_{zone_min} or water heating is “off” and the temperature is above T_{zone_max} . For all other possibilities, r_t is calculated based on Equation (5.3) and Equation (5.4). A constant σ was added to balance the targets of temperature setpoints and economic savings, resulting in Equation (5.5).

$$r_t = -(E_c + \Delta T * \sigma) \quad (5.5)$$

The number of tank temperatures and PV production inputs may vary depending on the algorithm and the number of houses included, as detailed in the next subsections. In a simplified way, the proposed DRL control approaches follow the steps of Algorithm 5.1. The difference among them is basically the environmental parameters, for instance, if houses are treated individually or as a group, and the way agents access the policy and previous experiences.

ALGORITHM 5.1: Generic DRL Heat Pump Control

Initialise Building parameters

Initialise $Q(s, a)$ arbitrarily

Repeat (for each episode)

Initialise s

repeat

Choose a from s using policy from $Q(\epsilon\text{-greedy})$

Take action (a)

Update building states (s_{t+1})

Calculate reward (r)

$Q(s_t, a_t) \leftarrow r_{t+1} + \gamma \max_a Q(s_{t+1}, a)$

$s \leftarrow s_{t+1}$

until s is terminal

end

5.4.1 Independent Learners

The aim of this first experiment is to deploy the individual heat pump control DRL agent, which will be used as a base for comparisons in later stages. It considers that each of the houses has its own PV, heat pump and control system, operating without having any communication with other houses in the community, as can be seen in Figure 4.2. The neural network designed to estimate the Q-values is similar to the one presented by Wei et al. [31], from the HVAC domain. It is composed of an input layer, two hidden layers, and an output layer. The input values are tank temperature, PV production, and hour of the day, but first, they are normalised on a scale from 0 to 1 (min-max normalisation). This helps the algorithm to achieve a more stable learning process, as it does not have to work with numbers with different dimensions. As an example, tank temperature values are lower than one hundred degrees, while PV production can go up to two thousand W. Then, there are two hidden layers, where softmax is the activation function chosen, followed by the output layer with possible actions (turn on or off DHW). The losses are calculated by the mean squared difference between the current output and the ideal target values. Furthermore, we use a Gradient Descent Optimiser, which adjusts weights in the neural network to minimise these losses. The proposed algorithm uses an ϵ -greedy policy to improve efficiency. Finally, rewards are calculated individually for each of the houses following Equation (5.5).



Figure 5.2: Independent learners topology.

5.4.2 Independent Learners with Transfer Learning

The second experiment consists of having individual learners in each of the houses, similarly to the independent learners' experiment, but now the agents can share their knowledge among the other houses in the community, as illustrated in Figure 5.3. The NN structure, normalisation process and

rewards follow the same approach as 5.4.1 and Algorithm 5.1. As the equipment available in the houses is the same, the agents update their respective policy in each timestep but also take into account the neighbours' policy. For instance, if an agent from a specific house faces the same state that was previously visited by any other house, it will either choose the same action performed (exploit) or explore, if the agent decides to take another action. Similar to the previous experiment, the possible actions for each agent are to turn on or turn off DHW. Next, it will receive a reward and update its own policy according to the new experience learnt. Similarly, the agents from other houses will follow this process. The idea is to reduce the number of trials a single agent has to perform to learn near-optimal policies by using the knowledge obtained from the agent that already visited some particular state. This concept is called parallel transfer learning [97,98], where the knowledge to be shared between agents does not need to wait until the end of the process to be available. Similarly, da Silva and Costa [99] proposed that agents can advise each other in a multi-agent system composed of simultaneously learning agents.

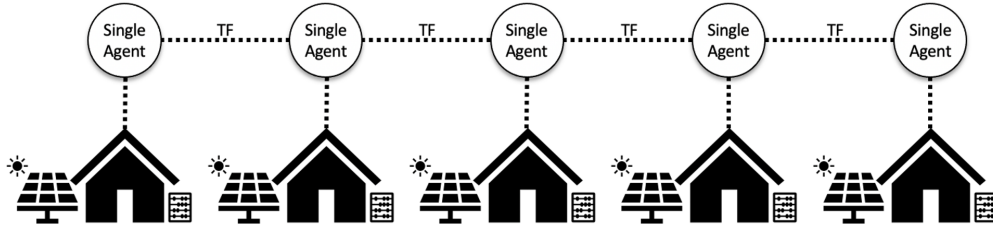


Figure 5.3: Independent learners with transfer learning topology.

5.4.3 Global Learner

The third experiment explores a different solution, where a global agent is used to control all heat pumps in the community, as presented in Figure 5.4. The general idea follows Algorithm 5.1, but now there is only one NN that receives all tank temperatures. Also, as the energy production is very similar among the houses (PV systems have the same capacity and are very close geographically), we considered the average of the PV production as input instead of having individual inputs for each of the PV systems. In this new model, for each time step the agent reads all tank temperatures and the PV average at once, and takes an action that can be turning on only one of the heat pumps or keeping them all off. The rewards are now calculated

considering the summation of temperature difference values (Equation (5.4)) and the electricity cost (Equation (5.3)). These changes impact on the setpoint and savings balance, hence σ will have a different value to keep a good performance in both. In summary, the global agent has a higher number of inputs and outputs compared to the previous models, but control actions cannot be done in parallel. In theory, as the tank temperature drop is slow, the agent can manage the time slots of activation and keep the performance without violating the lower setpoint temperature.

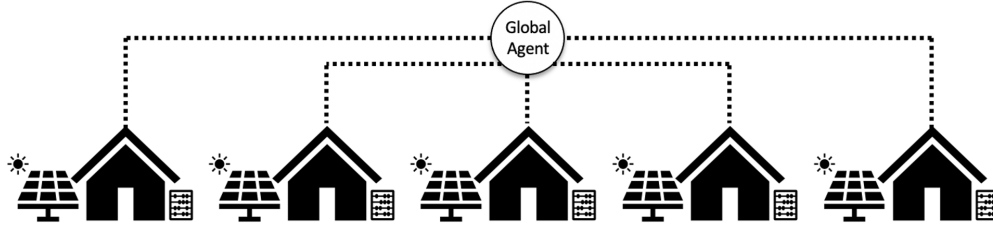


Figure 5.4: Global learner topology.

5.5 Results

Several tests with different parameters were performed to determine the best configuration for balancing energy savings and tank temperature matters. The final NN architecture is composed of 8 neurons in each of the hidden layers, with $\alpha = 0.0001$, $\gamma = 0.95$ and $\epsilon = 0.05$, where these values are fixed during the whole experiment for a fair comparison of learning speed. First, we removed Ec and selected $\sigma = 1$ in Equation (5.5), to certify that the algorithm was achieving a similar performance to the rule-based method. Next, we added Ec and started to tune the value of σ , in which the final value was defined as 100. Values close to zero mean low savings, while high values affect the tank temperature as the algorithm will start to focus only on savings and possibly will not turn on the equipment even if the temperature is lower than the expected setpoint. The final control algorithms were analysed in terms of control efficiency, energy savings, and convergence speed.

5.5.1 Control’s Efficiency

After selecting and balancing the algorithm parameters, simulations were performed for the independent and global learner. The agent’s goal was to achieve a near-optimal policy by interacting with the environment, the details about the number of episodes needed and convergence speed is found in subsection 5.5.3. The first metric to evaluate the control’s efficiency was to certify the agent’s ability to keep the tank temperature in an acceptable range, so we selected one random month (episode) after the learning process achieved the near-optimal policy. After achieving stability, both independent learners’ methods (with and without transfer learning) have the same result, hence this analysis considers only one independent and one global agent, both with 5 houses. Figure 5.5 summarises the tank temperatures for each of the methods.

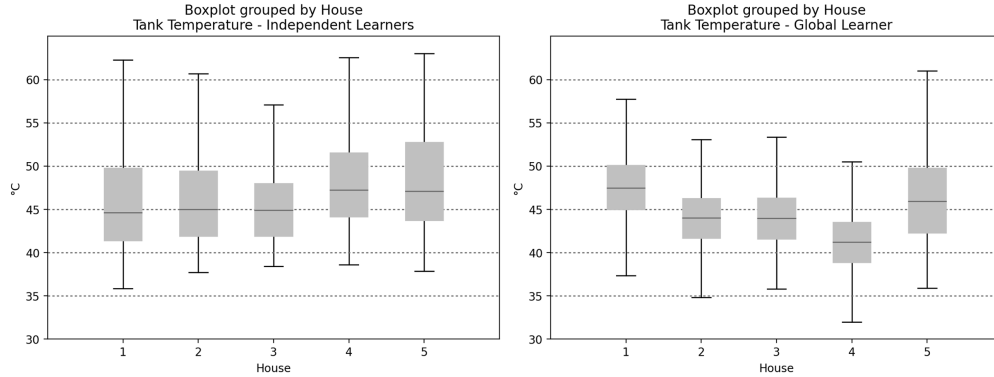


Figure 5.5: Tank temperatures demonstration.

The independent learners presented a slightly more uniform temperature distribution, as each agent had to handle only their own tank temperature. Their average temperatures were around 45°C, with a few exceptions where it achieved up to 60°C. This can represent, for instance, a moment when PV production is high and the agent decides to heat the water creating a buffer, which will consequently postpone later actions. The overall performance of the global agent was similar, as managing 5 tank temperatures seems to be a more challenging task. In this control strategy, rewards of the five houses are accumulated, and sometimes the agent can prioritise one house over another, as long as the net rewards are low, and this can result in more temperatures out of the expected range.

The second criteria for a control’s efficiency analysis consists of evaluating

if the control algorithm can optimise energy usage by performing actions when the electricity cost is lower, as presented in Figure 5.1. Similarly to the first test, we selected a month after a near-optimal policy was achieved, then we verified the accumulated distribution of the loads over the hours for the two DRL algorithms (independent and global) and also compared them with a rule-based controller, as can be seen in Figure 5.6. The average PV production of May 2020 is also plotted in the chart as a reference for a quick understanding of the best control periods.

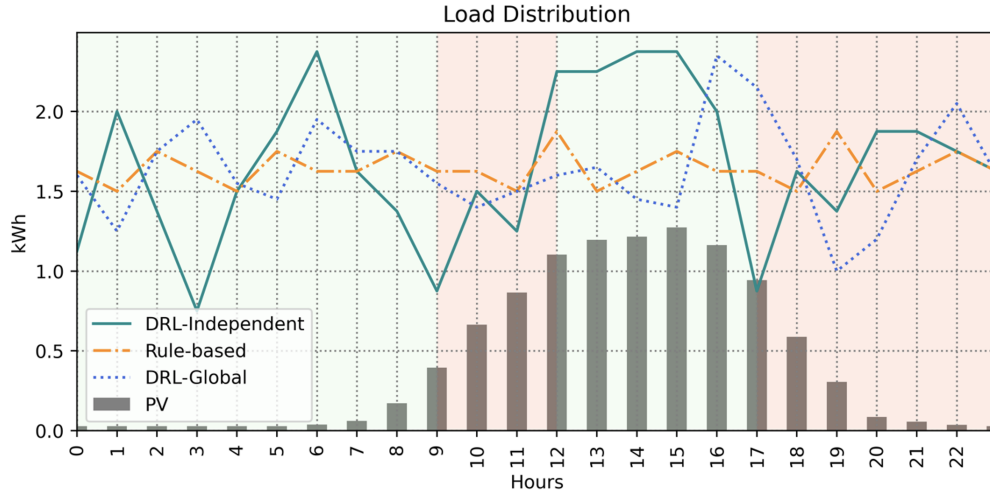


Figure 5.6: Load distribution per control type.

The results show that the independent learner (green line) was able to perform water heating actions in the periods of low energy price (midnight to 9 am) or high PV production (noon to 5 pm), and that brings benefits for both end-users and utilities. In a simple manner, the first one gets a reduction in the electricity bill, while the second will have a load reduction during the peak hours. According to EirGrid [104], the Irish national grid operator, the energy peak in Ireland happens at 5 pm, which matches with one of the periods where energy imported from the grid is lower in the independent learner experiment. On a large scale, it could help even more the utility, by avoiding unnecessary upgrades in the only connection point with the main island [10].

Due to the tank dynamics, on average, the heat pump needs to perform a control action more than once a day to keep the temperature in between the setpoints. The agent must coordinate actions in such a way the next action will also be in periods of low energy cost. For instance, although hours 2

to 4 am have a low cost, performing an action in that period may result in a next action to be done in the period of high cost 9 am to noon due to a temperature drop below the setpoint. In that case, it would be better to anticipate the action, making the next action fall before 9 am (first low-cost period), or delay it to 5 to 7 am, so the next action would probably be performed in the next low-cost period (high PV production). The global agent has a similar distribution of the rule-based control, having the loads well-distributed over the hours, and one of the reasons is the difference in the process flow. In the independent learners' approach, actions can be performed in parallel in each 15-min timestep, while the global agent only allows one heat pump activation per timestep.

5.5.2 Energy Savings

In the energy savings analysis, we compared the amount of energy consumed from the grid by the independent and global learners against the rule-based method, our reference model. Overall, the independent learner performed better in all the houses, achieving up to 9.55% of reduction. The global learner reached almost 5% of reduction in the best case, but the average of this model was just around 2%. The individual information for each model and house can be found in Table 5.2. Without considering reductions from the PV production, the gross average energy consumption presented across the DRL-based models was reduced by 2% for most of the houses. Only house 2 presented an increase of 1%, but this fact does not mean a real increase in terms of final savings, as the actions of this specific house could have happened in periods of low-cost tariff or high PV production, resulting in a net value that could be lower. In summary, as the comfort element was the first control target, the average gross energy consumption was very similar to the rule-based model, but the difference is the time energy is consumed.

Table 5.2: Algorithms comparison.

House	Rule-based		Independent DRL-based		Global DRL-based	
	Ref.	Avg. (kWh)	Total (kWh)	Reduction	Total (kWh)	Reduction
1			29.86	-5.12%	31.15	-1.02%
2			28.57	-9.23%	31.03	-1.40%
3		31.47	29.86	-5.13%	30.31	-3.69%
4			28.46	-9.55%	29.97	-4.77%
5			28.93	-8.06%	32.12	2.06%

Regarding financial savings, we verified the energy bought from the grid for each of the periods, low-cost and high-cost tariffs. Both DRL-based models achieved an average of 5% compared to the rule-based, with a maximum reduction of 7.46%. This maximum is lower than the 9.55% found for energy savings and the main reason is the ToU tariff chosen, which has a long period of low-price energy (9 hours a day). Considering that heat pumps have to turn on more than once a day for water heating, there is a big probability that one of the cycles will be within the low-cost tariff, even operating through a rule-based model.

5.5.3 Convergence Speed

The convergence speed test aims to verify if transfer learning helped the algorithm to achieve a near-optimal policy faster. First, we evaluated the performance of an independent learner (Subsection 5.4.1), where we identified the number of episodes necessary to learn the control strategy, as can be seen in Figure 5.7. Each episode represents the whole month of May 2020, and the agent took around 11 episodes to reach the plateau, mainly because the tank heating losses are slow. Overall, it is necessary to execute only two actions on average for water heating in an entire day, hence visiting the same set of states can take a large number of trials even using estimates from the NN. Secondly, we performed an experiment to assess the independent learner with transfer learning applied (Subsection 5.4.2), considering a different number of houses sharing experiences in each of the tests ($H = 1, 2, 3, 4, 5$). The idea is to compare if rewards are reduced even from early episodes, thus achieving an optimal control policy faster than letting a single agent learn by itself.

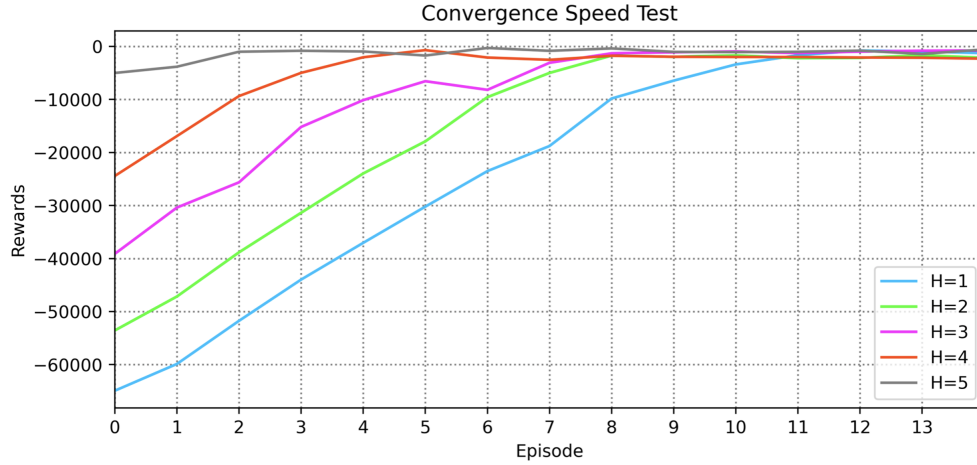


Figure 5.7: Convergence speed.

In summary, the algorithm convergence speed increases gradually according to the number of houses sharing experiences. Thus, the best result can be found in $H = 5$, where the total reward is considerably lower since episode 0 compared to the other experiments. In this scenario, the learning time to train near-optimal control policies was reduced by more than a factor of 5. This improvement is achieved mainly because agents are most of the time at different states, hence their knowledge is complementary to each other. Considering 4 houses ($H = 4$), the time was reduced by a third. Although the experiment with three houses ($H = 3$) presented a lower initial total reward compared to $H = 2$, both achieved the optimal policy in episode 8.

Besides improving convergence speed, reducing total rewards in the initial episodes also helps the agent to present an initial better behaviour. For instance, it will partially perform the right water heating actions following its policy instead of just guessing randomly. The difference between the total rewards of the agent without TF (reference) and the agent with TF is known as Jumpstart [38, 78]. In our experiment, the Jumpstart also increased gradually following the number of houses included.

5.6 Conclusions

This work presented a new DRL algorithm with a transfer learning approach for a DRL-based heat pump control to leverage energy efficiency in a microgrid. The main objective was to control the heat pump tank temperature at the lowest energy cost possible, by optimising PV self-consumption and

taking advantage of the time-of-use tariff. Moreover, transfer learning was applied to speed up the learning process, hence achieving a satisfactory performance since early trials. A total of three different DRL topologies were proposed and tested, having their results compared to a rule-based reference model. Energy savings of almost 10% were achieved by the TF DRL algorithm, is similar to the values found by [76] and [70]. In addition to that, our TF approach reduced the time to learn near-optimal policies by more than a factor of 5, considering the whole cluster of houses. This performance is similar to the one achieved by the Q-learning model with TF proposed by [53] in the HVAC domain.

Chapter 6

Conclusion

6.1 Summary of Achievements

This research aimed to demonstrate how reinforcement learning algorithms can improve residential energy management control. Three main sets of experiments, presented in Chapters 3 to 5, were conducted in a simulated environment involving different technologies, such as HVAC systems, PV generation and IoT devices for smart homes, in which two of them were based on real demand response projects. Reinforcement learning algorithms for autonomous control were deployed, exploring tabular Q-learning and deep Q-learning as a solution, also using transfer learning to speed up the learning process and improve performance from early trials. Overall, the algorithms' objectives were to operate the residential EMS components autonomously without affecting the user's comfort and to achieve additional targets depending on the experiment, such as energy and financial savings, load shifting, or improved learning rate. The main results are summarised as follows:

- Chapter 3:
 - A reinforcement algorithm (Q-learning) with transfer learning capabilities for HVAC control was deployed.
 - Learning time to train near-optimal control policies after transfer learning was reduced by more than a factor of 6 when comparing to experiments without it.
 - After applying transfer learning on the geographic changes experiment, the proposed method demonstrated higher performance in

countries where the temperature variation is lower, reducing time out comfort by one-third.

- The two proposed control methods, static and dynamic, presented very low ToC (2.5%) for both static and dynamic HVAC control when the new room was not greater than 50%. For rooms with variation greater than 50%, the dynamic approach had to be used to keep ToC low.

- Chapter 4:

- A deep reinforcement learning algorithm was deployed for indoor and DHW tank temperature control, leveraging PV self-consumption in a dwelling.
- Energy imported from the grid was reduced by up to 16% over the months.
- Apart from bringing savings, PV self-consumption optimisation also contributed to 10.2% of load shifting, by anticipating or delaying heating actions.
- Users would not experiment more than 1% of temperatures out of the specified setpoints range.
- Renewable energy consumption was 9.5% higher for the DRL-based model compared to the rule-based.

- Chapter 5:

- A deep reinforcement learning algorithm with TF capabilities was deployed for DHW temperature control of a number of houses in a microgrid.
- Energy imported from the grid was reduced by almost 10% across the houses.
- The DRL-based algorithm presented a similar control performance as the rule-based in terms of keeping the temperatures within the setpoints, hence the user would not feel discomfort.
- The transfer learning approach reduced the time to learn near-optimal policies by more than a factor of 5.

- Indirect load shifting was observed, as energy consumption moved to periods of high PV production and out of peak hours.
- Considering the optimisation based on leveraging PV generation self-consumption and the use of ToU tariff, the proposed algorithm achieved up to 7.5% of financial savings.

6.2 Lessons Learnt

The main algorithms used in the three experiments, tabular Q-learning and deep Q-learning, were able to understand the environment and perform near-optimal control actions after a number of trials. The first is considered more accurate as it stores all visited states and actions, while the second tends to generalise better as it uses estimations instead. The algorithm chosen was determined according to the characteristics of the problem to be solved. For small and discrete state-spaces, tabular Q-learning is recommended. The downside of the tabular method is that it can face issues once the state-space increases, for instance when increasing the number of houses or adding other devices to be controlled under the same policy. This could result in a slower learning process, which can possibly be avoided by using a deep Q-network to replace the Q-table, or even using transfer learning, if training information in a similar environment is available. A different algorithm has to be chosen if the action space is continuous (e.g. HVAC fan speed or setpoints), such as soft actor-critic.

As one of the main constraints of this research was to maintain user comfort, all the algorithms developed were first parameterised to ensure this was achieved, and then savings targets were included. End-users may be reluctant to new technologies that affect their comfort, especially automated devices with autonomous control, as pointed out by the RESPOND project [9]. Across the experiments, the time out comfort was around 3% or lower, with one exception in Chapter 3, where the Spanish test case had a 9.49% of ToC due to the huge weather variations over the trial months. The energy imported from the grid was reduced by up to 16% in experiments from Chapter 4 (August 2019), and by 10% in Chapter 5 (April 2020). These reductions could have been even higher if the reward coefficients that balance comfort and savings were configured focusing on savings, but then users comfort would start to be affected.

As leveraging PV self-consumption was part of the optimisation process, the weather conditions impacted considerably in the results. The summer period has the highest PV production, so the flexibility to perform control actions is higher compared to colder months. This can be observed in the experiment of Chapter 4, which had a more diverse weather dataset, running from May to December 2020.

Transfer learning proved to be a good addition to both reinforcement learning algorithms (tabular Q-learning and deep Q-network), by reducing the time to learn near-optimal policies by more than a factor of 5. Consequently, the benefits of an RL-based controller can be seen from the start of its application, avoiding discomfort or extra expending that could happen in a real test case during the learning process. Although this method has been applied in several domains, there are just a few works in the HVAC and residential energy management system domain. The studies carried out so far can enhance future applications, such as using synthetic trained data from simulations in real environments, thus skipping part of the learning process that may affect the end-user.

6.3 Limitations

Residential energy management systems rely on end-user engagement, hence the social aspect is an important point to take into account to explore the best it can offer. The application has to be as transparent as possible, so users can trust they will not feel discomfort and will get some additional benefit, such as financial savings. Although most of the simulations of this research are based on real test cases, additional tests in a controlled real environment should be performed before scaling the solution to other homes. This is necessary because human behaviour can be unpredictable, as people can suddenly change their daily routine, or randomly keep windows open, which will affect the heating dynamics of the building. The additional tests would help to verify how well the algorithm can act and adjust the control actions in these kinds of adverse situations.

Another important topic for the success of a residential energy management is the technology available. The proposed algorithms take into account a specific set of equipment. Thus, the learning time and control actions can be different if a house has a different heat pump or PV system. A similar

issue may be found in terms of building construction (materials and geometry). Moreover, the experiments performed utilised the central temperature sensor close to the heat pump controller as reference, located in the main corridor, and not individual sensors for each of the rooms. Random manual actions performed by users in heaters installed in other rooms may effect the building dynamics and the results.

Finally, comparing reinforcement learning algorithms is not an easy task, as each of the environments found in the literature are mostly unique. Buildings have specific particularities in terms of construction, equipment, and pattern of utilisation by the user. Hence, an algorithm developed for a building may not be effective in another. New initiatives have arisen to tackle this issue, such as the CityLearn challenge [19]. From the point of view of an algorithm’s performance, it allows a direct comparison among the reinforcement learning models, but the generalisation to other topologies would still be a challenge in different environments.

6.4 Future Works

Reinforcement learning applied to residential energy management control is becoming more popular over the past years, especially models using deep Q-networks [29, 49]. Furthermore, considering that ML algorithms are in constant evolution across several domains, future works would include the application of those novel methods. This also applies to transfer learning, which has only very recently been applied in the residential EMS domain. As presented in Chapter 2, there are other types of TF that could potentially be explored as a solution to enhance the learning process in future research, such as task mapping and knowledge transfer.

Another important improvement would be moving from the simulated environment to a real test case experiment. This would allow for a better understanding of the algorithm capacity of dealing with unexpected events caused by the user’s interaction with the environment. To avoid affecting user’s comfort, a study about transferring the control’s policy knowledge from the simulated environment to the real one would also be considered. Transfer learning using synthetic data from simulations is already being applied in fields such as computer vision, where a virtual representation of an object can be used during the training process. This reduces the need of

gathering real pictures, which can be costly and time consuming depending on the nature of the object.

The experiments can be further expanded by adding more complexity to the environment. First, instead of a single point of indoor temperature verification, a multi-zone topology could be used, which is interesting from the point of view of a fine-tuning of the experiment in terms of comfort. This could be further expanded with a multi-agent control approach. Moreover, some heat pumps allow other types of control actions that may bring additional savings and comfort, such as changing the setpoint boundaries or fan speed. Additional studies can also be carried out including battery storage systems and electric vehicles, or considering demand curve of the neighbourhood as part of the optimisation. These new assets, along with PV and ToU tariffs, would bring more flexibility for the residential energy management system, as the algorithm could be configured to find the best time to store, use or sell energy, for instance.

In chapter 4, a different approach to get the user’s setpoint preferences could be applied. For instance, a smartphone app could be available for users, where they can evaluate if the temperature is good or not at a specific time. Another option, if the house already has temperature sensors installed with historical data available, a data-driven statistical or machine learning models could be used to infer the setpoints. This could information could be used as input for the RL model.

Finally, the application of reinforcement learning in a microgrid presented in Chapter 5 can be improved by increasing the number of houses with a more diverse configuration and building construction materials. For example, the Aran islands, used as a reference in the experiments, has around 450 dwellings with similar characteristics that could integrate a bigger project. A study considering transfer learning for houses out of the microgrid in different geographic locations may also be positive in terms of generalisation of the models. The core of this research was the individual houses, with a positive side effect on the grid. New research could be conducted with targets from the utility and grid perspective. For instance, if there is a critical need for demand reduction in a certain period of the day to avoid shortages, the RL algorithm could perform actions to reduce or coordinate the utilisation of certain loads in a community to keep the grid stable and operational for all residents.

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Appendix A

Demand Response for Residential Buildings: Case Studies and DR Program Design by the RESPOND Project

The work outlined in this appendix was published in:

Lissa, P., Seri, F. and Keane, M. Demand Response for Residential Buildings: Case Studies and DR Program Design by the RESPOND Project. In Proceedings of the 4th Annual APEEN Conference 2019 - Energy Demand-Side Management and Electricity Markets.

Abstract. This appendix presents an analysis of the contemporary residential demand response (DR) programs and solutions, already applied at the European and Worldwide level, and provides recommendations for the design and implementation of DR programs for three different pilot sites: private apartments buildings in Madrid (Spain), social housing community in Aarhus (Denmark) and dwellings on the Aran Islands (Ireland). Throughout the world, residential energy consumption ranges from 30% and 40% of the total energy demand, it shows how representative residential DR can be in the future, helping to distribute the energy usage, hence reducing peak loads. DR programs can bring savings to end-users also additional bene-

fits to utilities, which can better manage their energy assets. This research also explores the main DR drivers and barriers, showing the current development status in the pilot countries. Results from the analysis show that there is still work to be done mainly regarding the policies, which varies across Europe. For instance, in Ireland end-consumer cannot send its own production of energy directly to the utility grid, hence DR should focus on individual energy management. On the other side, Denmark and Spain have some aggregation and prices issues to be improved to better disseminate DR programs.

A.1 Introduction

Throughout the world, residential energy consumption ranges from 30% and 40% of the total energy demand. Users' consumption patterns contribute directly to sporadic peaks of demand and, to support this variation, utility companies need to increase their generation in order to avoid interruptions in power supply [1]. Demand response programs help both consumers and utilities to reduce peak demand and price volatility. DR provides responsive and interactive consumers with a wide range of potential benefits on system development, operations, and market efficiency. Through DR programs, consumers can play a significant role in grid operation by reducing or shifting loads during peak load, in response to any incentives or time-based rates [2]. DR is more performed in residential sectors than commercial or industrial because the consumers are more susceptible and responsive to the price signal from the utility, adjusting or shifting the time of use of their appliances, such as washing machines, tumble dryers, water pumps, and heaters [3].

Nowadays, there is no critical need for residential demand flexibility in most of the European countries because they have sufficient generation to supply the energy even during peak times. There is also some provision from industrial customers that can be used if necessary [4]. This scenario is starting to change. The use of renewable sources is increasing affecting directly the energy matrix. Renewables represented almost two-thirds of new net world electricity capacity additions in 2016, with almost 165 gigawatts (GW) coming online [5]. Between 2017 and 2022, we expect global renewable electricity capacity to expand by over 920 GW, an increase of 43% [5]. In

order to reach the goal set by the European Union (EU) of 20% of renewable energy in 2020 [6] and at least 27% in 2030 [7], this growth has to continue and some challenges regarding renewable electricity must be resolved. The intermittent capacity of those resources will tend to increase the variability of the overall electricity supply challenging their integration into the grid [8]. Depending on weather conditions, solar and wind generation are considered more reliable intermittent resources and, consequently, residential demand flexibility will play an important role in balancing the network [4].

Contemporary solutions which achieve the matching between energy generation and demand, are based on increasing the generation capacity. Backup power plants are a costly solution, which should be used for short periods. Building a traditional power plant will provide the necessary energy back up to match demand during peak events, but also increasing greenhouse gas production [8]. An alternative could be found on storage technologies. However, they are currently expensive and not efficient [9]. Demand response can be a possible solution, aiming to manage the demand in order to meet the available energy [10].

Demand response is defined as “changes in electric usage by end-use customers from their normal consumption patterns in response to changes in the price of electricity over time, or to incentive payments designed to induce lower electricity use at times of high wholesale market prices or when system reliability is jeopardised” [11]. Demand response initiatives have become quite popular recently as a new tool for, both, supply guarantee and energy efficiency and cost-saving measures. But until now these initiatives have always been focused on large consumers.

This appendix presents an analysis of the contemporary residential demand response (DR) programs and solutions, already applied at the European and Worldwide level, and provides recommendations for the design and implementation of DR programs for three different pilot sites. The document is divided as follows: *Demand Response* introduces the different classification of DR from the point of view of the most relevant aspects found in the literature. *Drivers and Barriers* shows what can leverage or can be obstacles for DR programs. *Case Studies* presents the current status of DR policies in the selected pilot projects. *Discussion* section presents all the relevant outputs of our analysis, showing the suggested design for each one of the pilot sites. Finally, *Conclusions and Future Work* recaps the main

points of the appendix, introducing ideas for future work.

A.2 Demand Response

The interest in DR has been growing over the last few years. A number of review studies have been carried out related to DR. All of these publications focused on particular parts of a comprehensive framework of DR. For instance, [12] carried out a review on DR challenges, issues and barriers, which is crucial for the DR penetration into the energy market. [14] presents an overview of the DR enabling control, metering and communication technologies; the same study also explores different DR programs and consumers types and the current status of development at the level. [15] reviews and make a comparison of local DR deployment methods for better use of renewable energy. [1] presents an overview of the literature on residential DR systems, load-scheduling techniques, and the latest ICT that supports residential DR applications. [16] provides a review on DR programs and schemes, based on the motivation of the consumers to participate in the program. [17] provides a review of existent DR models, identifying gaps and recommendations for future model developments. Reviews studies on DR are focused on different aspects of it, a comprehensive approach to guide the design of a DR program is missing.

The authors in [17] classified DR models on criteria based on: thematic, methodological, temporal, spatial, technological and practical properties. Another classification is proposed in [18], dividing DR types between potential impact of efficiency and service to the customers. This work considers the classification proposed by the US Department of Energy [11], because it is the most commonly used, in which DR programs can be categorised in two main types: incentive-based and price-based, as shown in Fig. A.1.

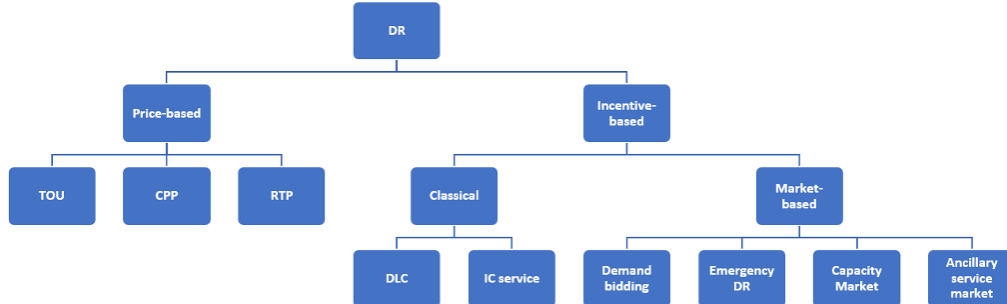


Figure A.1: DR programs classification.

Regarding incentive-based programs, end-users get a recompense to reduce their electric loads on request or for giving permissions to regulate their electric loads [13]. This typology of programs began with industrial and large commercial customers. Recently, residential customers have been included in those programs as well. Following the categorisation proposed by [11], there are six different incentive-based DR programs. They can be grouped into two main categories: classical and market-based. Classical ones are direct-load control and interruptible/curtailable service. Demand bidding, Emergency DR, Capacity Market and Ancillary service belong to the market-based programs. Price-based demand response are programs where the electricity price is not fixed, it varies in time. This category is divided into three major groups: time-of-use rates, critical peak pricing and real-time pricing. The common goal is to reduce electricity consumption over a certain period.

A.3 Drivers and Barriers

Research carried out extensively by [12] aimed to identify the main barriers to demand response programs, besides classifying them and thus presenting some possible solutions. The fundamental classes used in the analysis were economics, technological and social, which relate to intrinsic human nature (social/economic barriers). From political and regulatory perspective, it shows that regulation may cause distortion in markets if goods that are practicably substitutable are precluded from competing with each other. A further a political barrier may result from uncertainty derived from unclear policy and role definitions. In order to achieve the European's energy goals and meet the political promises, both price-based and incentive-based demand response will be strongly required for all groups of consumers (industrial, commercial and residential) through different approaches. This will give energy resource flexibility to the utilities and price benefits to the consumers. SEDC [19] has developed a set of regulatory requirements to enable Demand Response, that can be structured around four main criteria:

- Enable market access for demand response.
- Enable different service providers to access the market.
- Create viable products.

- Develop measurement and verification requirements and ensure fair payments and penalties.

The penetration of distributed generators, the prevalence of competitive electricity markets, the advancement of end-use technologies and control systems, and the advent of smart grids also result in the reform of conventional demand-side management theoretical frame [20]. Moreover, low energy prices are a natural demand response driver and regions with high electricity costs would benefit the most from DR and should therefore pursue more programs and resources. Another important driver for DR development is the market structure. In many cases an organised wholesale power market facilitates DR and the rules and conditions of the particular market define the opportunity for customers, enhancing or hindering participation [20].

A.4 Case Studies

Demand response programs have been playing an important role throughout the European Union in recent years, either better integrating the network due to new renewable technologies or reducing consumption in residences at peak times. Some countries, such as Belgium, Finland, France, Ireland, Switzerland and the UK present a more advanced framework, making them an attractive environment for new projects and investments [19]. However, even for these countries, DR is still a new concept and there are areas for improvement, such as a standardisation of the responsibilities of all participants involved (consumers, aggregators, suppliers), new regulations and interoperability issues. It is also important to highlight the development of demand response in countries that historically had little or no openness to these types of programs, such as Estonia, Italy and Spain, which are working on new regulations and studying their potential for future applications through public and private partnerships [19].

The case studies chosen are three different pilot sites where RESPOND (integrated demand response solution towards energy positive neighbourhoods) [13] is being deployed. Each one of them is located in a country at different stages of DR development, a positive fact that will give a good view of its applicability throughout the European Union. The next subsections aim to explore the individual situation of each pilot.

A.4.1 Spain

Despite presenting the most critical picture when compared to other countries, Spain on the other hand presents good initial opportunities in DR, being encouraged mainly by government initiatives. Currently, the main barrier to be overcome is relating to demand-side flexibility and aggregators, which are not allowed under current legislation, similar situation is happening in countries such as Portugal and Estonia [19].

Spain is the first country in the world where the default price for households is based on hourly spot prices, which is an important driver for price-based demand response programs [19]. Consumers are already encouraged to participate in time-of-use contracts, that give them the possibility to shift consumption patterns in response to signals given, thus saving energy and network resources. On the other hand, even though some pilot projects are being deployed in this area, the development of incentive-based demand response is still limited to industrial consumers [19]. Since 2016, decentralised and renewable energy resources (especially wind generators) have been able to pre-qualify and participate in the balancing market (tertiary reserve), this could signify a good step in new regulations for aggregators [19].

In the Spanish pilot, the utility is conducting a demonstration project in a community of 3 private apartment buildings with 24 dwellings. Fenie Energia [21] is the electricity and gas provider of the pilot site, with access to data collected by the network operator. There is no self-generation installation or monitoring system for household devices, apart from fiscal meters for electricity and gas. RESPOND is being deployed to receive consumption inputs, perform optimisation and take control actions on the devices, as well as installing smart meters and home automation devices provided by consortium members.

A.4.2 Denmark

Denmark has several initiatives and discussions about DR progress and some modalities are already allowed, although an improvement in the definition of rules and roles of those involved in the system is needed. Along with Germany, Austria and other Nordic countries, Denmark is part of the group of countries that have started the process of standardisation for independent aggregators, which can bring integration and flexibility opportunities

between them as a single energy market [19].

Consumers are allowed to participate in all the ancillary services in Denmark. However, due to a weak business case as well as a regulatory environment which makes it difficult for independent aggregators to develop innovative demand response businesses in the market, demand response participation within the markets remains limited. The lack in the regulatory environment have affected the aggregation, which is legal in Denmark, but the responsibilities are not well-defined, creating a barrier to independent aggregators [19].

The Danish pilot consists of a public housing district with 30 residential buildings, where 4 of which have been selected for the RESPOND demonstration pilot. Although the estate is equipped with solar panels, there is no demand response program deployed, so part of the RESPOND demonstration project is to deploy a program reflecting DR possibilities. The pilot dwellings are also being equipped with home automation solutions in order to give full control to the RESPOND system. The RESPOND platform will aim to integrate the metering points, including smart meters, for optimal data monitoring, analysis and control.

A.4.3 Ireland

Ireland currently has presence in some DR sectors. In order to foster greater applicability and also to standardise the roles of participating agents, the new "Integrated Single Electricity Market" implemented in 2018 together with the DS3 (Delivering a Secure, Sustainable Electricity System) program are bringing improvements to balancing market, wholesale market, and as well as a newly designed Capacity Mechanism [19]. As a comparison, UK has a similar framework with an active open DR market, which is already working, however some regulation and role definition are still needed for a better dissemination.

Regarding aggregators, they are allowed in Ireland but the minimum size of demand side unit is 4 MW. The reduction of this value to 1 MW would encourage greater participation and competition in this area, since currently many loads shall be grouped to achieve this minimum of 4 MW. Furthermore, DR programs can participate in the wholesale electricity market from the point of view of bidding and dispatch, however DR providers do not earn an energy payment for this [19]. The tariff scheme in Ireland is close to flat

and does not reflect congestion or real-time need, even industrial tariffs do not have a significant effect on demand-side response activities [19].

Ireland’s pilot site is being deployed at dwellings on the Aran Islands, which are located 10 nautical miles west of Galway. These 3 islands comprise a total population of 1,225 inhabitants. There are 448 total dwellings, 12 of which have been selected for the RESPOND demonstration project. The islanders used to rely on imported fossil fuels for gas and heating, but have since switched to electrical heating with heat pumps, PV and solar-thermal arrays. Smart metering exists in the form of temperature sensors and power meters, while a number of heating devices can be controlled wirelessly. The RESPOND project is aiming to deploy additional home automation and smart metering devices for the purposes of the pilot demonstration.

A.5 Discussion

After having analysed each one of the pilot countries and considered the elements from the literature review, it is possible to notice that in fact, they are not at the same level of DR development, hence individual actions should be taken in order to improve the participation of residential DR on the electricity market. According to [22], the DR potential in Ireland and Denmark is around 0.45 GW, while Spain has 4.80 GW. It means, in relative terms, a percentage of peak load ranging from 7.5% to 10% across the countries assessed. Table A.1 summarises the main findings, taking into account the DR enablers, barriers, accesses to markets and service provider access.

Table A.1: DR programs characterisation in the pilot countries.

Country	Denmark	Ireland	Spain
DR Potencial	0.50 GW	0.40 GW	4.80 GW
% peak load	7.5%	8.0%	10%
Main Enablers	1) Ancillary services are open to Demand Response; 2) Prequalification is made at the aggregated pool level; 3) Demand Response can participate in the wholesale market.	1) Aggregation is allowed and the minimum bid size is 4 MW for DSUs. 2) The new “Integrated Single Electricity Market” implemented in 2018, together with the DS3 programme, bring a range of markets for demand-side response, specifically the balancing market, and the wholesale market.	1) Spain is the first country in the world where default price for households is based hourly spot prices, which is an important driver for price-based demand response programs. 2) All consumers are able to take advantage of Time-of-Use (ToU) contracts.
Main Barriers	1) Payments in the wholesale market are too low to make a positive business case; 2) Some markets have a volume demand of 10 MW which makes it impossible for smaller pools of flexibility to enter the market.	1) An ex-post pricing mechanism prevents the involvement of price-based demand-side measures given the lack of actionable price signals. Lack of different market products limits the level of participation.	1) Aggregation is not legal; 2) There is only one scheme allowing incentive-based demand response: The interruption load program, reserved only for large industrial consumers.
Demand Response Access to Markets	Consumers can trade their flexibility into the common Nordic wholesale markets. However, the traded volume is very limited, mainly due to low prices.	Individual Demand Response sites may be aggregated in order to be operated as a single demand side unity (DSU). Demand Response participates in the wholesale electricity market from the point of view of bidding and dispatch.	While some of the markets are open for Demand Response in principle, in practice this applies only for large industrial consumers. Aggregated Demand Response is allowed only for Tertiary Control.
Service Provider Access	The current market definition requires that independent / third-party aggregators must bilaterally contract with the consumer’s BRP and retailer (if they only wish to sell flexibility and not energy) to provide Demand Response services.	In the balancing market, a medium to large electricity users (> 4 MW) can participate in a Demand Side Unit (DSU) or an Aggregated Generating Unit (AGU). The aggregator does not require BRP’s agreement prior to load management.	Overall, there is no possibility for aggregated demand-side resources to take part in the Spanish electricity market. Only consumers with contracted power above 5 MW have access to interruptible demand service managed by the TSO.

Regarding the pilot projects, the Danish and Spanish ones fit better the utility-based business model, where demand response products are negotiated directly to regulated utilities, which contracts with DR providers to procure firm capacity, operating reserves and mitigation of network constraints [23]. For the Irish pilot, the market-based is more appropriated. In this business model the customer receives an energy management system to control energy consumption and production and also to participate in an ISO-based DR program, manual response is also allowed if the EMS is missing [23]. The main reason for that is because in Ireland consumers cannot sell energy directly to the utility, so in that case the EMS can help customers, autonomously or not, to choose the best time to use the appliances or even deciding when to consume energy from their own generation (e.g. PV panels) or from the utility grid. On top of the aforementioned, a successful DR program to be applied to the pilot sites should take into account the peculiarities from Table A.1.

A.6 Conclusions and Future Work

This work has demonstrated how distinct DR programs can be when comparing countries at different regulation stages and there is no common sense in Europe yet about the best model to be followed. Apart from legislation, the DR program decision can be more complex when adding other variables, such as user behaviour or the type of technology and equipment to be controlled, which can affect the program effectiveness even within the same country. The DR literature is quite fragmented since many contemporary DR programs do not reveal completely the design of the DR solution and the final results, which does not allow a proper definition of best practice to be used for further DR program design and application. For future work, a similar analysis can be carried out considering a bottom-up perspective: starting from user needs, then moving to house and infrastructure, the utility grid and the regulation of the country. This approach could facilitate the DR program design phase for residential customers, allowing the development of a general DR characterisation methodology, which will focus on applicability and early-stage user DR assessment.

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Appendix B

Machine Learning Methods Applied to Building Energy Production and Consumption Prediction

The work outlined in this appendix was published in:

Lissa, P., Peretti, D., Schukat, M., Barrett, E., Seri, F. and Keane, M. Machine Learning Methods Applied to Building Energy Production and Consumption Prediction. In Proceedings of the 27th AIAI Irish Conference on Artificial Intelligence and Cognitive Science. AICS 2019. <http://ceur-ws.org/Vol-2563/>.

Abstract. The utilisation of renewable sources of energy is growing all over the world due to pressure for sustainable solutions. It brings benefits to the environment, but also adds complexity to the electricity grid, which faces energy balancing challenges caused by intermittent production from this kind of generation. Having a good energy prediction is essential to avoid losses and improve the quality and efficiency of the energy systems. There are many machine learning methods that can be used in these predictions; however, every consumer is different and will behave in a distinct way. Therefore, the objective of this article is to compare the application of different ML methods, aiming to predict PV energy production and energy consumption for residential users. Four different ML methods were applied

in a real dataset from the RESPOND project: Linear Regression, Decision Forest Regression, Boosted Decision Tree Regression and Neural Network. After the simulation, the predicted values were compared against the real data, considering 150 days of measurement from two Irish houses. Overall, all the algorithms applied achieved mean errors below 14%, but the Boosted Decision Tree overperformed, with mean errors of 2.68% and 10% for energy consumption and energy production prediction, respectively.

B.1 Introduction

Energy generation through renewable sources is becoming popular in recent years. The European Union targets to achieve about 20% of renewable energy production in 2020 [1] and at least 27% in 2030 [2]. Along with this expected growth, other challenges start to arise, mainly because the energy produced from wind or solar sources depends on weather conditions and presents an intermittent capacity. This will tend to increase the variability of the overall electricity supply, thus making its integration to the grid a complex process [3].

Understanding and predicting how the electricity network works, including distributed generation from renewables, is essential in this new framework, as it can bring benefits to the utilities. Contemporary solutions for energy balance, such as backup fossil power plants [3] and storage [9], are costly and sometimes not efficient. With a proper energy management system (EMS) utilities can provide new services. These include Demand Response (DR) solutions, where utilities can give benefits to users that change their consumption behaviour according to the network load, hence reducing total energy demand during peak times [19]. Moreover, having information about the grid status can help utilities to plan their own energy production, thus avoiding unnecessary costs with new assets otherwise required to match peak demand over small periods.

To support this new trend, some data-driven methods for energy production and consumption prediction have been arising, ranging from statistical models to complex machine learning algorithms. These aim to find correlations and meaning among variables in large datasets. Although more than 80% of the previous studies about energy consumption prediction have been carried on non-residential customers [4], research from [25] shows that in the

EU residential applications correspond to 42% of the total energy flexibility potential, whereas 31% comes from industry and 27% in the tertiary sector.

In summary, the main contribution of this work is to assess machine learning techniques for energy prediction and to deploy a simulation environment, aiming to provide the following predictions for a hypothetical EMS:

- PV energy generation.
- Residential energy consumption for a small group of houses.

The rest of this appendix is organised as follows: *Background Research*, which shows the related work regarding energy production and consumption prediction. *Machine Learning* section provides information about the principles and techniques applied in this research. *Environment Setup* describes how the environment monitoring has been structured and explains the dataset preparation stage. *Results* section presents all the relevant outputs of our experiments, comparing the prediction methods and real data. Finally, *Conclusions and Future Work* recaps the main points of the appendix, introducing ideas for future work.

B.2 Background Research

Electricity is a development indicator, it boosts the country's economy and brings comfort to our homes, improving quality of life in most of the daily tasks. However, it is also strongly associated with CO₂ emissions, where buildings represent 36% of the total produced gas in the EU [5]. As energy production using fossil power plants is one of the CO₂ emissions reasons, the use of renewable sources is increasing, therefore affecting directly the energy matrix. Renewables represented almost two-thirds of new net world electricity capacity extensions in 2016, with almost 165 gigawatts (GW) coming online. Between 2017 and 2022, it is expected that the global renewable electricity capacity is to expand by over 920 GW, an increase of 43% [6].

Amasyali and El-Gohary (2018) carried out an extensive review [4] on data-driven building energy consumption prediction, having categorised more than 60 previous studies across five categories: type of building, temporal granularity, type of energy consumption, type of data and ML algorithm.

As a result, they identified that 19% of models belong to residential buildings and the granularity chosen was mostly hourly (57%) followed by daily (15%). Most of the datasets considered only the overall energy consumption (47%) from the electricity meter and 67% of the models used real data instead of simulated or public benchmark data. Finally, the most frequent ML algorithms applied were artificial neural networks and support vector machine (SVM), with 47% and 25% respectively.

In a different approach, Naji et al. [7] proposed the application of an extreme machine learning algorithm for estimating energy consumption based on a building envelope's parameters, district heating and cooling loads, achieving an accuracy improvement when comparing the results against genetic programming and artificial neural network. Authors in [8] and [10] utilised a genetic algorithm applied for building performance, the first predicting energy consumption and the second one predicting heating/cooling. Besides ML methods, there is also a physical modelling approach, known as engineering methods or white-box models, but they rely on thermodynamic rules for a detailed energy modelling and analysis [4] and thus are not part of the present work.

Regarding energy production prediction, Das et al. [11] assessed more than 20 recent works about forecasting of PV generation, from physical models to ML, and compared their performance across different factors, such as accuracy, reliability, computational cost and complexity. According to the study, ANN and SVM-based forecasting models performed well under rapid and varying environmental conditions. Voyant et al. [12] presented another list with almost 50 works where ML was applied through different methods, with ANN the most popular followed by SVM, regression trees and others. There is no common agreement with regard to the evaluation criteria, but as a reference, the root-mean-square error (RMSE) of some of them ranged from 5% to 24%. [13] compared three different prediction models for a PV plant in south Italy: phenomenological detailed model, Multi-Layer Perceptron (MLP) neural network and a regression approach. The results demonstrated that more accurate predictions can be reached by statistical machine learning approaches.

The authors in [14] applied ML algorithms, such as SVM and Forest regression, in order to predict solar radiation values for seven different places in Spain. Our proposed research is about PV generation, but there are other

important studies that show the application of ML for a different renewable energy source. For instance, [15] and [16] presented a review of current methods for wind power generation forecasting.

B.3 Machine Learning

Machine learning is a sub-field of computer science that is classified as an artificial intelligence method. It can be used in several domains and one of the advantages is the capability of solving problems that are impossible to be represented by explicit algorithms. Some of the ML methods are regression-based, which can be widely used to create projections about the future, with the objective to predict a numeric target.

The best ML model will rely on the equilibrium between predicted error and complexity of the system. Depending on the database particularities, a complex model may result in a greater error than using a simple model, as shown in Fig. B.1.

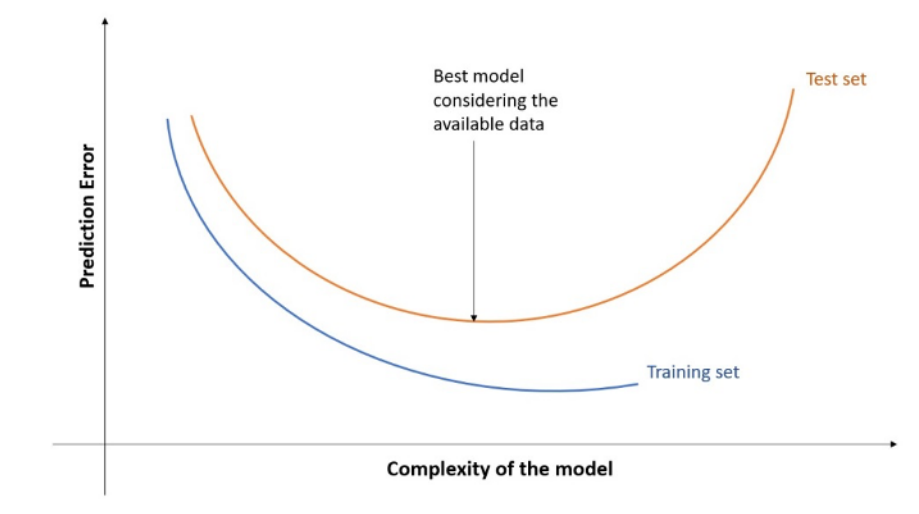


Figure B.1: Model complexity versus prediction error.

All methods to be presented here are regression-based. The following subsections describe the methods applied to our proposed prediction model.

B.3.1 Linear

Linear regression is a statistical method, which has been adopted for use in ML. In spite of being one of the simplest models for a basic predictive task,

this method also tends to work well on high-dimensional sparse datasets [21]. The classic regression problem involves a single independent variable and a dependent variable, this is called simple regression. Multiple linear regression involves two or more independent variables that contribute to a single dependent variable. Problems in which multiple inputs are used to predict a single numeric outcome are also called multivariate linear regression.

B.3.2 Decision Forest

Decision trees are non-parametric models that perform a sequence of simple tests for each instance, traversing a binary tree data structure until a leaf node (decision) is reached. The advantage of decision trees is that this method is efficient in both computation and memory usage during training and prediction. The Decision Forest model consists of an ensemble of decision trees. Each tree in a regression decision forest outputs a Gaussian distribution as a prediction. Aggregation is performed over the ensemble of trees to find a Gaussian distribution closest to the combined distribution for all trees in the model [22].

B.3.3 Boosted Decision Tree Regression

Boosting is one of several classic methods for creating ensemble models, along with bagging, random forests, and so forth. In Azure machine learning studio [18], boosted decision trees use an efficient implementation of the MART gradient boosting algorithm, which is an ML technique for regression problems. It builds each regression tree in a stepwise fashion, using a predefined loss function to measure the error in each step and correct for it in the next. Thus, the prediction model is an ensemble of weaker prediction models [20].

B.3.4 Neural Network Regression

Although neural networks are widely known for applications in deep learning and modelling complex problems, such as image recognition, they are easily adapted to regression problems. Any class of statistical models can be termed a neural network if they use adaptive weights and can approximate non-linear functions of their inputs. Thus, neural network regression

is suited to problems where a more traditional regression model cannot fit a solution [23].

The layers of a neural network are made of nodes, the place where computation happens. A node combines input from the data with a set of coefficients, or weights, that either amplify or dampen that input, thereby assigning significance to inputs with regard to the task the algorithm is trying to learn [24]. These input-weight products are summed and then the sum is passed through a node’s so-called activation function, to determine whether and to what extent that signal should progress further through the network to affect the ultimate outcome. If the signals pass through, the neuron has been “activated.” Fig. B.2 shows a diagram of what one node might look like [24].

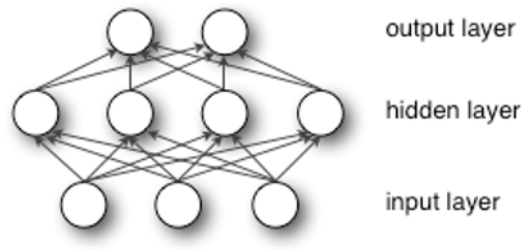


Figure B.2: Neural network layers.

A node layer is a row of those neuron-like switches that turn on or off as the input is fed through the net. Each layer’s output is simultaneously the subsequent layer’s input, starting from an initial input layer receiving your data. Pairing the model’s adjustable weights with input features is how we assign significance to those features regarding how the neural network classifies and clusters input [24].

B.4 Environment Setup

The case study chosen is part of the Irish pilot from RESPOND project [17]. It consists of data collected from two houses in the Aran Islands over 150 days, from May to September 2019, both equipped with PV panels. In this experiment, the data were grouped and houses were considered as a cluster because of the goal to analyse energy generation and consumption in the whole grid. The dataset was then uploaded to Microsoft Azure machine learning studio [18], where the information was processed following our pro-

posed architecture showed in Fig. B.3.

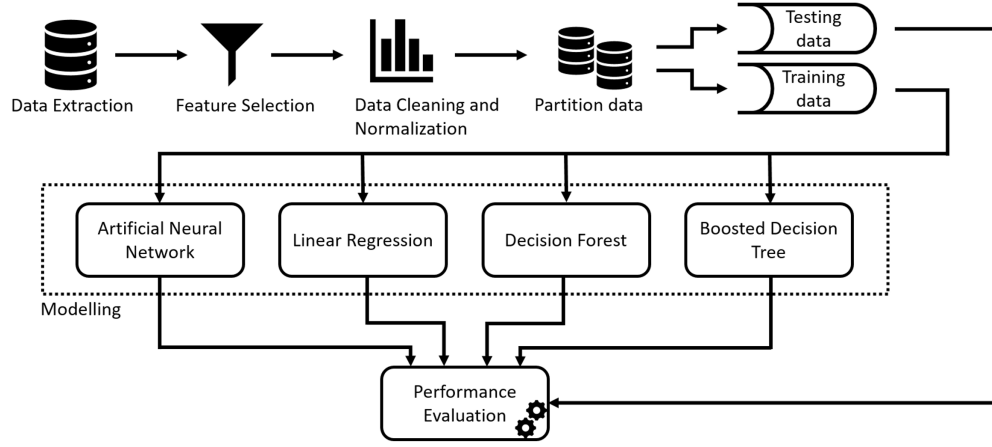


Figure B.3: Process diagram.

B.4.1 Data Extraction

Data extraction is the first step of the process. It is a collection of all available data from sensors and weather information from external sources. Furthermore, additional classification features for day classification has been created to improve the algorithm decisions in later stages. The initial dataset has hourly resolution and is composed of the following features:

- Complete timestamp (day, month, year, hour).
- PV generation.
- Energy consumption (electricity meter).
- Weather (temperature, precipitation, humidity, wind speed, solar radiation).
- Day classification (weekend or working days).

The energy consumption data comes from the utility’s electricity meter, which is the most common source of this kind of measurement. It does not consider social aspects, such as a number of family members, because it would be hard to track changes considering large groups, thus making the generalisation process complex.

B.4.2 Feature Selection

This stage is where the selection of features for energy production or consumer consumption prediction model happens. For instance, the production prediction model uses almost all available variables, only energy consumption is removed from the dataset. On the other hand, consumption prediction does not depend on PV production or some weather features, such as wind speed or solar radiation, so the final dataset is reduced in that case. This practice helps the algorithms to converge faster and allows a better generalisation.

B.4.3 Data Cleaning and Normalisation

This process aims to make the dataset as homogeneous as possible. Data cleaning works identifying incomplete, incorrect, inaccurate or irrelevant parts of data and then replaces, modifies or deletes the dirty or coarse data. The normalisation process aims to change the values of numeric columns in the dataset to a common scale, without distorting differences among variables values. For example, temperature values range from 0 to 25, while solar radiation can achieve values greater than 500. The normalisation method applied to the proposed experiment is the Z-score, where the values in the specified columns are transformed using Equation (B.1).

$$Z = \frac{x - \text{mean}(x)}{\text{stdev}(x)} \quad (\text{B.1})$$

where mean and standard deviation are computed for each column separately.

B.4.4 Test and Training Dataset

Once data preparation is done and ready for processing, the dataset is randomly divided and follows two different paths:

- 70% of data is sent to training models.
- 30% of data is separated and used for testing purposes, to be compared against trained models later.

B.4.5 Modelling

The training dataset received from the previous stage is trained across four different regression methods: Linear, Decision Forest, Boosted decision tree and Neural Network. Detailed descriptions about each one can be found in Section B.3.

B.4.6 Performance Evaluation

The predicted results from the different models are compared against the real data. Our experiment considers a performance evaluation of daily and hourly predictions for both, PV generation and energy consumption. Azure ML Studio provides outputs about overall accuracy, but we have also added a Python script to plot and calculate additional outputs allowing an intuitive visual analysis.

B.5 Results

Over the performance evaluation stage, the four algorithms have been parameterised in different ways targeting to minimise errors. Azure ML Studio allows us to input a range of parameters to be trained. For instance, instead of using a static value for the number of decision trees in the Decision Forest method, you can set a range of numbers and the algorithm will try all of them, choosing the best combination of parameters. Immediately below you can find the final parameterisation of each method:

- Linear Regression: Method: Ordinary Least Squares. Regularisation weight: 0.001.
- Decision Forest Regression: Re-sampling method: Bagging. Number of trees: 8. Maximum depth of the decision trees: 32. Number of random splits per node: 128.
- Boosted Decision Tree Regression: Maximum leaves per tree: 20. Minimum number of training instances: 10. Learning rate: 0.1. Total number of trees constructed: 100.
- Neural Network regression: Hidden layer specification: Fully connected case. Number of hidden nodes: 100. Learning rate: 0.0001.

Number of learning iterations: 100. Initial learning weights diameter: 0.1. Normaliser: Gaussian.

B.5.1 Daily Predictions

The objective of the model presented in this appendix is to provide an energy forecast to utilities, in order to help them plan their own energy production on a day-ahead basis. The weather inputs for PV production relate to one day before the actual day. The model could also use forecast data from two or three days before, but the accuracy will drop. Users' consumption prediction considers historical trends and also weather forecasts.

Table B.1: Prediction errors.

	Linear	Decision Forest	B. Decision Tree	Neural Network
PV Production				
MAE	35,24%	27,63%	21,84%	26,33%
Mean Error	14,10%	14,24%	9,99%	11,92%
Energy Consumption				
MAE	16,38%	11,93%	10,31%	12,70%
Mean Error	5,56%	4,31%	2,68%	4,98%

The results from our tests show a mean error below 14% for PV production prediction and below 6% for energy consumption. The mean absolute error (MAE) ranges from 21% to 35% for PV production and from 10% to 16% for energy consumption. The boosted decision tree presents the best performance across the methods, followed by the neural network. Table B.1 presents a compilation of results.

In some circumstances, the use of mean error can better represent reality. For instance, if the utility is analysing the forecast of a huge number of houses, some of them will present positive errors, predicting more energy than necessary, and others will have the opposite effect with negative errors, so this kind of measurement could result in a better balance than absolute values. Overall, the four methods follow the real data trend, as can be seen in Fig. B.4 (PV generation prediction) and Fig. B.5 (Energy consumption prediction).

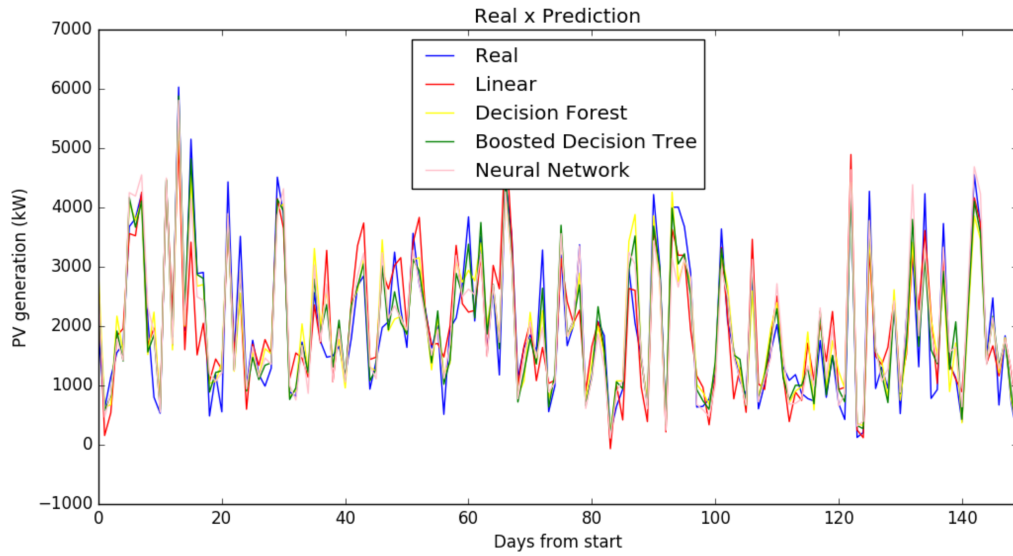


Figure B.4: PV generation prediction model comparison.

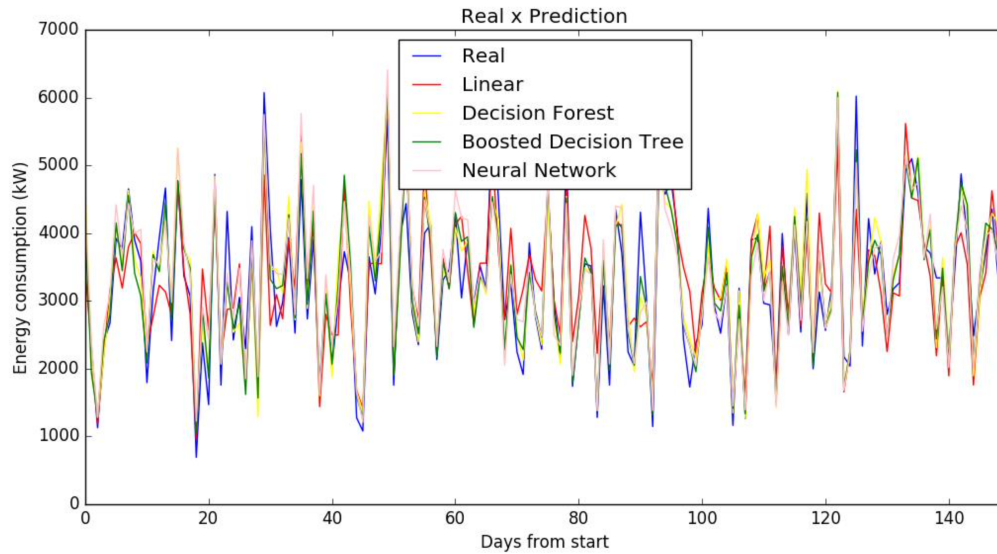


Figure B.5: Energy consumption prediction model comparison.

B.5.2 Hourly Predictions

Hourly energy consumption prediction can be hard to perform, mostly because consumers' behaviour can vary throughout the week with no defined pattern. PV production forecasting can also suffer variation and uncertainty, due to sudden weather changes. In order to show the accuracy across the

models, we selected the best performer (boosted decision tree) and the worst (linear regression), both presented in Fig. B.6 (consumption prediction) and Fig. B.7 (PV generation).

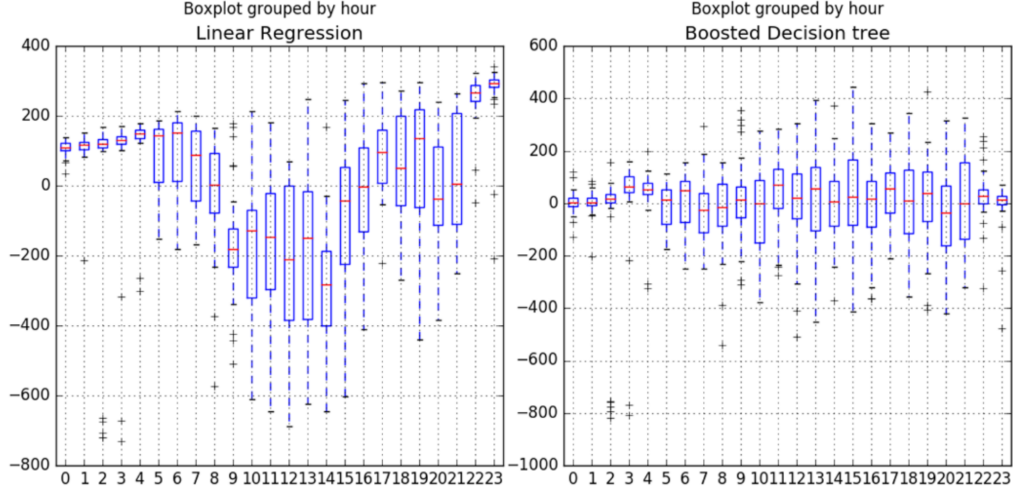


Figure B.6: Energy consumption boxplot of residuals grouped by hour.

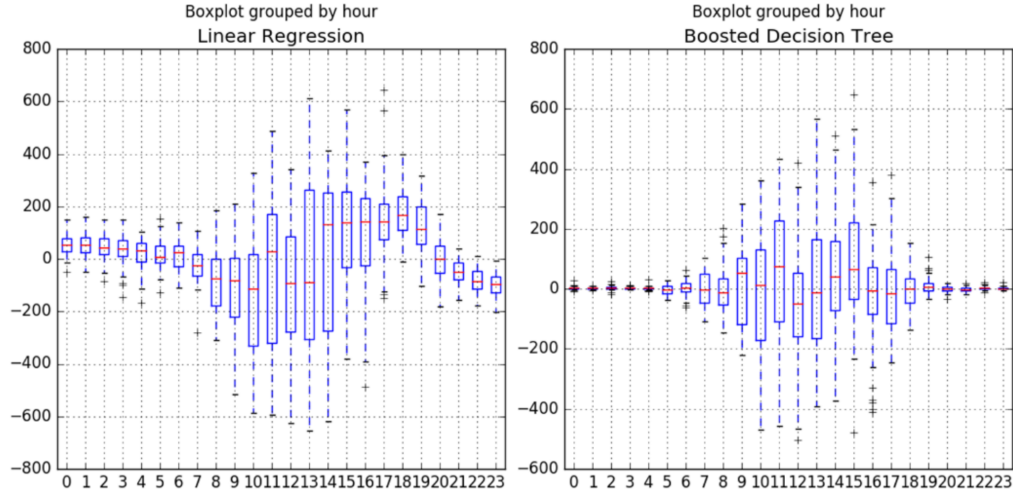


Figure B.7: PV generation boxplot of residuals grouped by hour.

The residuals are grouped by hours, across 150 days of data. As expected, it is easier to predict the end of the night and beginning of the morning, when the house's activity is lower and there is no solar radiation. Higher pattern changes mean higher errors.

B.6 Conclusions and Future Work

This work has demonstrated the application of distinct machine learning methods applied to PV energy production and energy consumption predictions, achieving 9.99% and 2.68% of mean error respectively, considering the best case (boosted decision tree). The dataset pattern is unique, so different ML methods should be applied in order to find the best one that suits each specific application.

Due to limitations of our dataset, only two houses in Ireland were analysed. For future work, the dataset will be improved by adding houses, hence more historical data. Furthermore, other ML techniques could be applied, examples include Support Vector Machine and Multi-Layer Perceptron neural network. Finally, energy production from other renewable sources and storage systems can be included, adding more complexity to the proposed model.

B. References

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